Science AAAS

Supplementary Materials for

Covalent organic framework–based porous ionomers for high-performance fuel cells

Qingnuan Zhang *et al*.

Corresponding authors: Xiao Feng, fengxiao86@bit.edu.cn; Bo Wang, bowang@bit.edu.cn

Science **378**, 181 (2022)
DOI: 10.1126/science.abm6304

The PDF file includes:

Materials and Methods
Figs. S1 to S77
Tables S1 to S6
References

Other Supplementary Material for this manuscript includes the following:

Movies S1 and S2

Materials and Methods

Synthesis of DhaTab-COF

DhaTab-COF particles were prepared by a reported solvothermal method *(26)*. Typically, aqueous 6 M acetic acid (0.2 mL) was added to a mixture of 2, 5-dihydroxyterephthalaldehyde (21.6 mg, 0.13 mmol) and 1, 3, 5-tris(4-aminophenyl)benzene (30.2 mg, 0.086 mmol) in n-butyl alcohol (1 mL) and ortho-dichlorobenzene (1 mL). The suspension was heated at 120 °C for 72 h. An orange-colored fluffy powder of DhaTab-COF was collected by centrifugation and washed with DMF, THF, and MeOH. Elemental analysis (EA) of activated DhaTab-COF sample was as follows: calcd C 79.12%, H 4.40%, N 7.69%, O 8.79%; found C 76.26%, H 5.09%, N 7.62%.

Synthesis of DbdTab-COF

Similar to DhaTab-COF, aqueous 6 M acetic acid (0.3 mL) was added to a mixture of 3,3'-dihydroxy-[1,1'-biphenyl]-4,4'-dicarbaldehyde (36.3 mg, 0.15 mmol) and 1,3,5-tris(4-aminophenyl)benzene (35.1 mg, 0.1 mmol) in *n*-butyl alcohol (1.5 mL) and ortho-dichlorobenzene (1.5 mL). The suspension was heated at 120 °C for 72 h. An orange-colored powder of DhaDbd-COF was collected by centrifugation and washed with DMF, THF, and MeOH. EA of activated DhaDbd-COF sample was as follows: calcd C 81.80%, H 4.58%, N 6.36%, O 7.26%; found C 80.52%, H 4.68%, N 6.19%.

Synthesis of DbdBtt-COF

DbdBtt-COF were prepared by a reported solvothermal method *(36)*. Typically, aqueous 6 M acetic acid (0.4 mL) was added to a mixture of 3,3'-dihydroxy-[1,1'-biphenyl]-4,4'-dicarbaldehyde (36.3 mg, 0.15 mmol) and 4,4',4''-(benzene-1,3,5-triyltris(ethyne-2,1-diyl))trianiline (42.4 mg, 0.1 mmol) in dimethylacetamide (DMAc, 1 mL) and ortho-dichlorobenzene (3 mL). The suspension was heated at 120 °C for 72 h. An orange-colored powder of DbdBtt-COF was collected by centrifugation and washed with DMF, THF, and MeOH. EA of activated DbdBtt-COF sample was as follows: calcd C 83.59%, H 4.13%, N 5.73%, O 6.55%; found C 81.68%, H 4.45%, N 5.61%.

Synthesis of LZU-1

LZU-1 was prepared by a reported solvothermal method *(37)*. Typically, aqueous 6 M acetic acid (0.2 mL) was added to the mixture of 1,3,5-triformylbenzene (32.4 mg, 0.2 mmol) and 1,4-diaminobenzene (21.6 mg, 0.2 mmol) in dioxane (2 mL). After being degassed through three freeze-pump-thaw cycles, the tube was flame sealed under vacuum and placed in an oven at 120 ºC for 72 h. Bright yellow powder of LZU-1 was collected by centrifugation and washed with DMF, THF, and hexane. EA of activated LZU-1 sample was as follows: calcd C 80.00%, H 4.44%, N 15.56%; found C 79.04%, H 5.61%, N 13.53%.

Synthesis of RDT-COF, RBT-COF, RBB-COF and RLZ-COF

RDT-COF was synthesized according to the literature with minor modifications *(28, 29)*. Glacial acetic acid (35 μL, 0.61 mmol) was added to the suspension of DhaTab-COF samples (10.0 mg, 0.051 mmol by imine) in 25 mL of absolute THF. The mixture was stirred for 10 min at room temperature, and then $NaBH_3CN$ (66.0 mg, 1.05 mmol) was added in small portions over 10 min. The mixture was further stirred at room temperature for 24 h. A light-yellow solid was collected by centrifugation and washed with DMF (20 mL × 3), THF (20 mL × 3), and H_2O (20 mL × 3). Finally, the as-obtained RDT-COF particles were collected by centrifugation and dried in a vacuum oven at 60 ºC overnight. EA of the activated RDT-COF samples was found: C 68.71%, H 6.28%, N 9.84%.

The reduction method of RBT-COF, RBB-COF and RLZ-COF is similar to that of RDT-COF. EA of the activated RBT-COF samples was found: C 78.45%, H 5.32%, N 6.30%; EA of the activated RBB-COF samples was found: C 76.59%, H 4.86%, N 5.25%; EA of the activated RLZ-COF samples was found: C 70.51%, H 6.58%, N 16.52%

Synthesis of SDT-COF, SBT-COF, SBB-COF and SLZ-COF

RDT-COF (100 mg, 0.457 mmol -NH-), 3-bromopropanesulphonic acid sodium salt (2.078 g, 6.857 mmol), KI (75.8 mg, 0.456 mmol), Na_2CO_3 (1.263 mg, 9.138 mmol) and 40 mL of dimethyl sulfoxide (DMSO) were placed into a 100 mL round-bottom flask equipped with a magnetic stirrer, a condenser, and a gas inlet/outlet. The mixture was ultrasonic dispersed for 30 min using a bath sonicator. Then, the suspension was heated to 120 °C for 48 h. After cooling to RT, the product was washed with methanol and water several times to remove residual other salts. Subsequently, the product was stirred and soaked in 1 M $HClO_4$ for 12 hours to fully ion-exchange (H^+ and Na^+), and then washed with deionized water until the pH was neutral. Finally, the as-obtained SDT-COF particles were collected by filtering with a nanofiltration membrane and dried in a vacuum oven at 60 ºC overnight, affording SDT-COF powder. EA of activated SDT-COF samples was found: C 52.25%, H 3.77%, N 4.28%, S 5.93%.

The grafting method of SBT-COF, SBB-COF and SLZ-COF is similar to that of SDT-COF. EA of activated SBT-COF samples was found: C 64.69%, H 5.30%, N 13.01%, S 2.52%. EA of activated SBB-COF samples was found: C 69.37%, H 5.24%, N 5.52%, S 3.41%. EA of activated SLZ-COF samples was found: C 51.06%, H 4.86%, N 6.29%, S 11.02%.

Synthesis of CRDT-COF

CRDT-COF was prepared according to literature (*29*).

Sample information of control group

Sulfonated graphene (SG) was purchased from Tanfeng Tech. Co., Ltd.. The purity of the sample is higher than 99%, and the number of layers is less than 5. EA of SG sample was found: C 43.69%, H 2.60%, S 6.76%.

Sulfonated polystyrene spheres (SPS) was purchased from Wuxi Rigor Tech. Co., Ltd.. The diameter of the microspheres was 100 nm.

Sulfonated activated carbon (SAC): 50 mg of activated carbon and 20 mL of 98% sulfuric acid were added into the flask, and further stirred for 6 hours at 80 °C in an oil bath. The product was obtained by filtration and washed with deionized water until the pH of the filtrate is neutral. The sample was dried in an oven at 105 °C. The IEC of SAC was 2.84 meq g^{-1}.

Structure optimization

Density functional theory (DFT) calculations were performed with B3LYP-D3 functional with 6-31G(d) basis set, where the D3 version of Grimme’s dispersion with Becke-Johnson damping was used to account for dispersion. The Gaussian 16 program package was used for all the calculations. For RDT-COF, we built a one-layer cluster model with four rings, where the truncated C-C bonds were compensated with H atoms (Fig. S2), and all atoms were relaxed during the optimization.

Materials characterization

Powder X-ray diffraction (PXRD) analyses were performed on a Rigaku MiniFlex 600 diffractometer using Cu-Kα X-ray radiation (λ = 0.154056 nm) with 40 kV voltage and 50 mA current. Samples were mounted onto silicon zero background holder. The PXRD patterns were recorded from 1.5 to 80º (2θ) with a step size of 0.02 degree and a scan rate of 10 º min^{-1}.

Fourier transform infrared attenuated total reflection (FTIR-ATR) spectra were recorded at the range 400 cm^{-1} – 4,000 cm^{-1} on a Bruker ALPHA spectrometer.

The elemental analysis (EA) was performed on an Elementar Vario MICRO analyzer.

The catalyst morphologies and microstructures were characterized by transmission electron microscopy (TEM) using FEI Talos TEM. Energy-dispersive X-ray spectroscopy (EDS) were taken on the same microscope.

The COFs microstructures were characterized by FEI Themis Z advanced probe aberration corrected analytical high-resolution TEM, which operates under the electron energy of 300 kV.

Field-emission scanning electron microscopy (FE-SEM) was performed on a JEOL model JSM-7500F operating at an accelerating voltage of 5.0 kV.

Focused ion beam-scanning electron microscope (FIB-SEM) was performed on ThermoFisher Helios G4 UC operating at an accelerating voltage of 5.0 kV.

X-ray photoelectron spectroscopy (XPS) was performed on a Thermo Scientific ESCALab 250Xi using 200 W monochromated Al-Ka radiation. The binding energies were referred to the C 1s peak (284.8 eV) from adventitious carbon.

The specific surface area of the catalyst was measured by the Brunauer-Emmett-Teller (BET) method at liquid nitrogen temperature using a Quantachrome Instrument ASiQMVH002-5 surface area analyzer. The samples were first evacuated under dynamic vacuum (< 10^{-3} torr) at 120 ºC overnight to remove the guest molecules and adsorbed moisture before re-measuring the accurate sample weight.

Thermogravimetric analysis (TGA) was performed on a PerkinElmer STA6000 simultaneous thermal analyzer with a constant heating rate of 10 ºC min^{-1} under air within the temperature range from 40 to 800 ºC.

Atomic force microscopy (AFM) was performed on Bruker Dimension Icon to characterize the thickness of the SDT-COF nanosheet.

Raman spectra were recorded by a micro-Raman system (Renishaw Invia), which with a charge-coupled device (CCD) to record backscattering signals dispersed by an 1,800 g/mm grating.

Ion exchange capacity (IEC)

The ideal IEC (element value) of ionomers were calculated using elemental analysis. A dehydrated sample was heated in an elemental analyzer to calculate the content of the sulfur element (recorded as S%). The IEC value was calculated using:

$$\text{IEC (meq g}^{-1}) = \frac{\text{S\%} \times 1000}{32} \quad \text{Eq 1}$$

The apparent IEC (titrimetry value) of ionomers was measured by titrimetry (*38*). The ionomers in their acidic form were immersed in 50 mL of NaCl (2.0 M) solution for 2 h and titrated with NaOH (0.025 M) to a phenolphthalein end point. The IEC (meq g^{-1}) was calculated from the mass of dry polymer (m_{dry}), the volume (V_{NaOH}), and the concentration of NaOH (c_{NaOH}) as follows:

$$\text{IEC} = \frac{V_{NaOH} \times c_{NaOH}}{m_{dry}} \quad \text{Eq 2}$$

Proton conductivity measurements

For the impedance measurement, SDT-COF was heated at 70 °C for 24 h in vacuum before measurement. SDT-COF was manually ground to obtain fine powders. An SDT-COF cylinder was prepared by applying a pressure of 4 MPa for 10 min, yielding an electrode with 3 cm × 2 cm × 0.1 cm in size. The in-plane proton conductivity of the thin block was measured, and the distance between the two gold electrodes was 3 cm. The whole cell assembly was maintained in a humidity chamber to control the temperature and humidity. Electrochemical impedance spectroscopy (EIS) was measured on an electrochemical workstation (CHI 760E: CH Instrumental Inc.) with an amplitude of 5 mV in the frequency range of 1 MHz to 100 Hz.

Proton conductivity equation:

$$\sigma = \frac{l}{A \cdot R} \qquad \text{Eq 3}$$

where l represents the thickness of the cuboid, A represents the electrode cross-sectional area, and R represents the bulk resistance of the material.

Arrhenius equation:

$$\log(\sigma T) = \log A - E_a/k_B T \qquad \text{Eq 4}$$

Here, σ is the conductivity, k_B is the Boltzmann constant, A is the pre-exponential factor, T is the temperature, and E_a is the activation energy for proton transport. The activation energy (Ea) of the pressed SDT-COF tablet is calculated to 0.46 eV, which indicates a Grotthus proton conduction mechanism.

The effect of temperature on SDT-COF was analyzed by keeping the relative humidity (RH) constant at 100% and changing temperature between 20 °C and 80 °C. The resistance was obtained from the semicircular arc in the Nyquist plot. Typically, if the sample has high conductivity, the semicircular arc is not clear, and, in this case, the intercept with the x-axis is considered the value of resistance.

Unlike the behavior in the pressed tablet, the boundary impendence between the COF nanosheets in the CL poses an obstacle for proton conduction, which requires the incorporation of Nafion with flexible chains to provide sufficient proton transport pathways to connect COF nanosheets. Thus, the proton conductivity of thin films was tested. We prepared Nafion film and SDT-COF/Nafion (1:1) films by drop coating on the gold interdigitaled electrodes. The film proton conductivity (k_f) was calculated from the resistance (R_f) using Eq. 5, where, d is the spacing between the IDA electrode teeth (100 μm), t is the thickness of the ionomer film (25 μm), l is the length of the teeth (0.6 cm) and N is the number of electrodes (20).

$$k_f = \frac{1}{R_f} \cdot \frac{d}{l(N-1)t} \qquad \text{Eq 5}$$

The EIS of films was measured on PARSTAT 4000A potentiostat galvanostat (AMETEK Scientific Instruments).

MEAs preparation

MEAs were fabricated using a sprayer to directly coat on the proton-exchange membrane (Nafion N212, 50 μm), which was catalyst coated membrane (CCM) process. Wetproof carbon paper with a microporous layer (TORAY YLS-30T) was used as a gas diffusion layer. Pt/C@Nafion CL was prepared from an ink containing 120 mg of catalyst (40% Pt/C (Pt/Vulcan, Johnson Matthey), 40% PtCo/C (PtCo/Vulcan XC-72, Premetek), or 40% Pt/KB (Pt/Ketjenblack EC-300J, Premetek)), and 57.6 mg (dry weight) of PFSA polymer (DuPont™ 20% Nafion DE2020 solution, dissolved in n-propanol/H_2O mixture, 57 wt%/43 wt%) to 50 mL of isopropanol. Pt/C@SDT-Nafion (1:1) CL was prepared using the same method, but only changed the ionomer component to 28.8 mg SDT-COF and 28.8 mg of Nafion® (dry weight). All the inks were ultrasonic dispersed for 3 h using a bath sonicator with ice water. The catalyst coated membrane was sandwiched between the gas diffusion layers by hot-pressing at 120 ºC and 0.4 MPa for 3 min to form an MEA.

The MEA with different Pt-loading (0.07 mg_{Pt} cm^{-2} - 0.3 mg_{Pt} cm^{-2}) used the same ink for spraying, and the subsequent treatment was also the same.

Performance tests for HEMFCs

Fuel cell testing was carried out in a fuel cell test station (Scribner 850e) with H_2 and O_2 using a single cell block with a serpentine flow channel. Before the polarization test, a “break-in” step was applied by measuring the voltage gradually decreasing from 0.8 V to 0.2 V and held for 3 min at each point and cycled 10 times, which took 6.5 hours in total. The performance test conditions were: cell temperature of 80 °C, the RH in the cell of 100%, the gas flow rate of 300 mL min^{-1} and 300 mL min^{-1} for pure H_2 and O_2, respectively, and back pressures both of 50 kPa (150 kPa abs). The performance of H_2/air test conditions were: cell temperature of 80 °C, the RH of 100%, the gas flow rate of 500 mL min^{-1} and 1,500 mL min^{-1} for H_2 and air, respectively; and back pressures both of 50 kPa (150 kPa abs). The polarization curves were recorded in voltage-control mode, from the open-circuit voltage to 0.2 V and holding each potential for 30 s.

Mass activity measurement

All the mass activity (MA) of catalyst in MEA is measured under 1,000 mL min^{-1} H_2 and 2,000 mL min^{-1} pure O_2 at 80 ºC and 100% RH, with the back pressure is 150 kPa. Such test conditions are to minimize the impact of mass transfer on the performance of MEA. The MA of ORR catalysts in MEA were evaluated by normalizing the iR-free currents at 0.9 V in the H_2-O_2 test divided by the cathode Pt mass.

Electrochemical surface area analysis

CV curves were obtained in a potential range of 0.1 V – 1.2 V *vs.* RHE at a scan rate of 50 mV s^{-1}. The CV test conditions were: the gas flow rate of both 500 mL min^{-1} for pure H_2 and N_2, and the back pressures of 50 kPa. The ECSA value was determined by integrating of the hydrogen (desorption) region integrated from 0.1 V to 0.45 V *vs.* RHE. The final cycle of a set of three cycles was used for data analysis. The ECSA (m^2 g_{Pt}^{-1}) was calculated using the following formula:

$$S_{ECSA} = \frac{Q_{Integraled\ area}\ (mC)}{0.21\ (mC \cdot cm^{-2}) \times M_{Pt}(mg)} \qquad \text{Eq 6}$$

where $Q_{\text{Intergated area}}$ (C) and M_{Pt} (mg) are the integrated charge and mass loading of Pt, respectively, and 0.21 (mC m^{-2}) corresponds to a monolayer hydrogen adsorption charge on polycrystalline Pt *(39)*.

The accelerated stress tests

The accelerated stress tests (AST) on the cathode catalyst in MEA were carried out with humidified H_2 to the anode and N_2 to the cathode under 80 ºC based on DOE’s AST protocol required for MEA catalyst durability evaluation *(40)*. An external potentiostat drove the fuel cell voltage in triangle sweep cycle at 50 mV s^{-1} between 0.6 V and 0.95 V. After the completion of designated voltage cycles, the cathodic gas was switched to O_2 and the whole fuel cell current-voltage polarization was taken to measure the change of the catalyst specific activity and durability. Fuel cell polarization, MA, and CV curves were measured after 10,000, 20,000, and 30,000 voltage cycles.

MEA chemical stability

The steady-state open circuit voltage (OCV) of MEA was measured under 90 ºC, the relative humidity of the cathode and anode were both set to 30%, and the outlet pressure was controlled at 150 kPa abs. The fuel gas was 500 mL min^{-1} H_2, and the oxidant was 1,500 mL min^{-1} air.

Fuel cell impedance analysis

The EIS was tested from 20 KHz to 1 Hz at 3 A with supplying H_2 (300 mL min^{-1}) and O_2 (300 mL min^{-1}) at 80 ºC and 100% RH or supplying H_2 (500 mL min^{-1}) and air (1,500 mL min^{-1}). high-frequency resistance (HFR) was *in-situ* measured during the polarization curve test and durability test. The perturbation amplitude for the AC impedance was 5% of the direct current, and the frequency was 10 kHz, which would not disturb the electrochemical reaction of fuel cell.

Mass transport resistance *(16)*

The mass transfer process in the CL is a main limiting factor for the limiting current density, which is an essential parameter for evaluating the electrochemical power generation capacity of fuel cells. The mass transport resistance was evaluated from limiting current measurements at 80°C and 100% RH under 500 mL min^{-1} of H_2 and 1,500 mL min^{-1} O_2/N_2 (X_{O2} = 0.5, 1, 2, and 4%.). Stay at the voltage of 0.1 V for 3 minutes to determine the maximum current density.

$$R_{tot} = \frac{C_{O_2}}{i_L/4F} \quad \text{Eq 7}$$

$$C_{O_2} = \frac{P - P_{H_2O}}{RT} \cdot x_{O_2} \quad \text{Eq 8}$$

where i_L is the limiting current density (A m^{-2}), C_{O2} is the oxygen concentration, P is the absolute gas pressure (Pa), P_{H2O} is the water vapor pressure (Pa), x_{O2} is the oxygen mole fraction, R is the gas constant (J mol^{-1} K^{-1}), T is the gas temperature (K), and F is Faraday's constant (C mol^{-1}). The saturated vapor pressure of water vapor at 80 ºC is 47334.8 Pa.

Dry proton accessibility measurement

The dry proton accessibility to the Pt sites was calculated by dividing the ECSA at 20% RH by the ECSA at 100% RH *(9)*.

Gas permeation measurement

For gas permeation measurement of COF/Nafion and pure Nafion membranes, the membranes were fixed into Wicke–Kallenbach cells sealed with O-rings tested, the gas separation measurement system was shown in Fig. S53, and the effective diameter of the membrane is 5 mm. In all cases, helium was used as the sweep gas. For O_2 test (0% RH and 100% RH), feed and sweep gas were all 50 mL min^{-1}, and held a pressure of 1 bar at both sides. The permeate stream was analyzed through online gas chromatography (Agilent 7890 B with thermal conductivity detector, and the detection limit is 400 pg Tridecane/mL with helium).

The permeability of membranes was calculated by the equation as follow:

$$P_{O2} = \frac{N_{O2}}{A \times \Delta P_{O2}}$$
$$1\ GPU = 3.3928 \times 10^{-10}\ mol \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1} \qquad \text{Eq 9}$$

$$Barrer = l \times GPU$$
$$1\ Barrer = 3.344 \times 10^{-16}\ mol \cdot m^{-1} \cdot s^{-1} \cdot Pa^{-1} \qquad \text{Eq 10}$$

in which P_{O2} is the permeability of O_2 (mol m^{-1} s^{-1} Pa^{-1}), N_{O2} is the permeating rate of O_2 (mol s^{-1}), A is the effective membrane area (m^2), ΔP_{O2} is the transmembrane pressure difference of O_2 (Pa), f is the O_2 content measured by gas chromatography (mL min^{-1}), d is the diameter of membrane (mm), and l is the thickness of the membrane (μm).

Rotating disk electrode

Four kinds of catalyst films (Pt/C, Pt/C@Nafion, Pt/C@SDT, and Pt/C@SDT-Nafion) were used to measure the ORR activity with RDE at room temperature in an O_2-saturated 0.1 M $HClO_4$ solution. LSV under the scan at 5 mV s^{-1} and 1600 r. p. m. The electrochemical measurements were performed with CHI 760E electrochemistry workstation connected in a three-electrode configuration at room temperature using a carbon and an Ag/AgCl wire as the counter and reference electrodes, respectively. A glassy carbon electrode with the diameter of 3 mm was used as the working electrode. To prepare a catalytic slurry, the catalyst composite (2.0 mg) was ultrasonically dispersed in 10.0 mL 80% (v/v) 2-propanol aqueous solution. A portion of ink was cast on the glassy carbon electrode to provide a uniform layer of CL. The Pt loading density was controlled to be 0.5 mg cm^{-2}.

Model construction and dynamics simulation

The cathode catalytic layer is composed of Pt particles, carbon-supports, ionomers (Nafion and SDT-COF or pure Nafion), water, and oxygen. The diameter of carbon-supports is about 3 nm; 4–6 Pt particles are dispersed on the carbon-support surface randomly. The chemical structure of the Nafion is demonstrated in Fig. S74. In this work, the SDT-COF single ring was used as a simplified simulation object. All simulations were conducted using Materials Studio 2017 molecular modeling software (Accelrys, San Diego, CA, USA), and the COMPASS force field is used, as recommended in the references *(41)*. The organic molecular system and the inorganic molecular system covering all the molecules used in the simulation could be uniformly calculated. The three-dimensional (3D) model was 89 Å in the Z direction and 89 × 89 $Å^2$ in the XY direction. The time step was set to 0.5 fs. First, geometry optimized the system to minimize energy. Then, molecular dynamics simulations of 1,000 ps were performed at 353 K in the NVT ensemble.

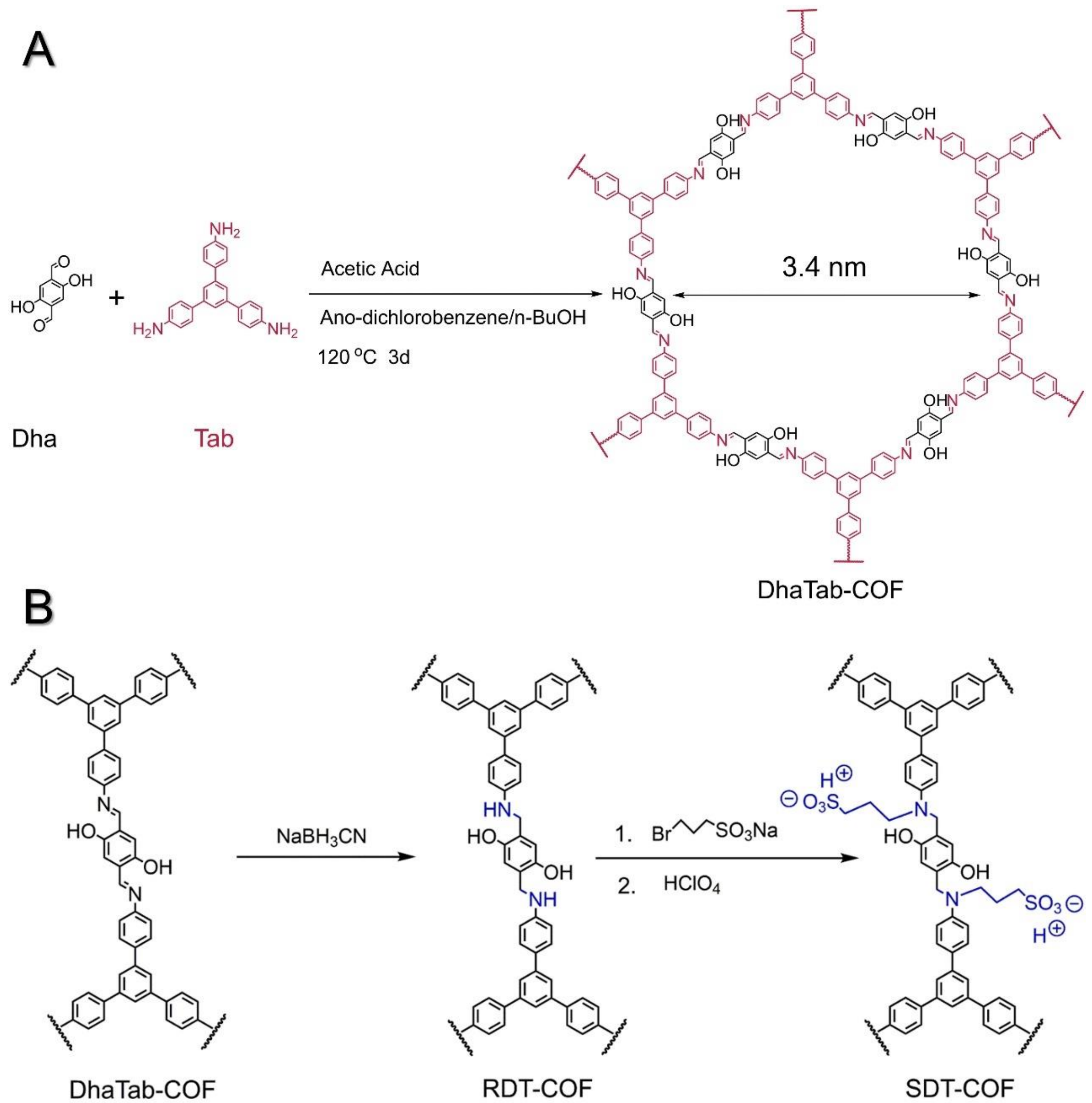


Fig. S1. (**A**) Reaction scheme of DhaTab-COF synthesis. (**B**) The synthetic route of SDT-COF.

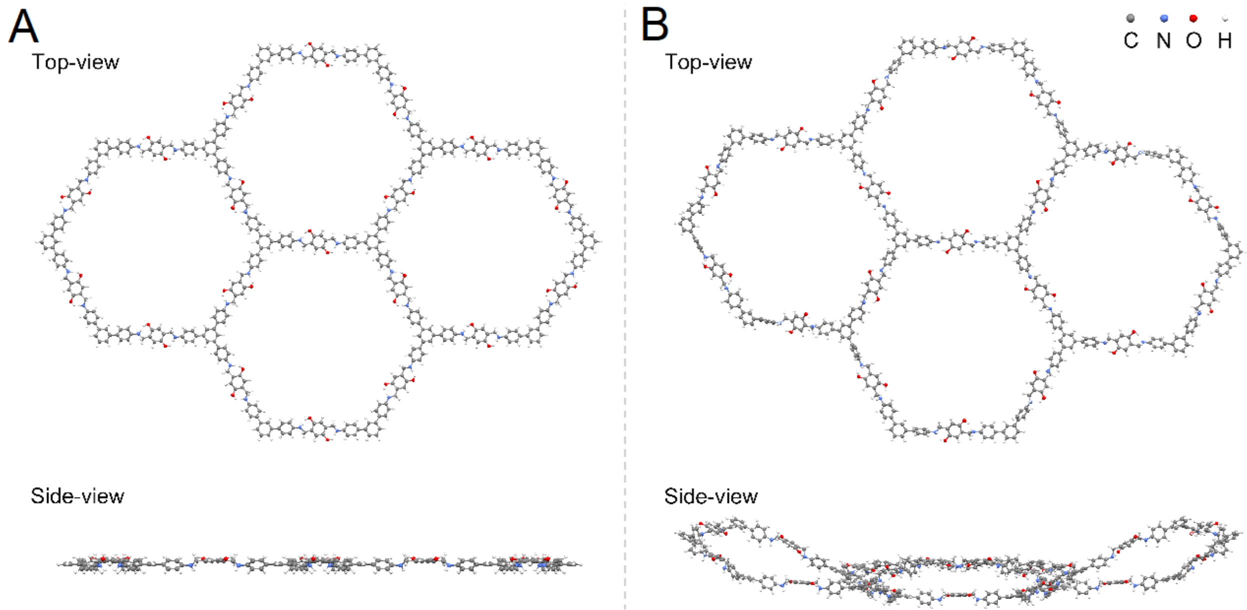


Fig. S2. One-layer model with four polygons for RDT-COF that (**A**) before and (**B**) after optimization using B3LYP-D3/6-31G method.

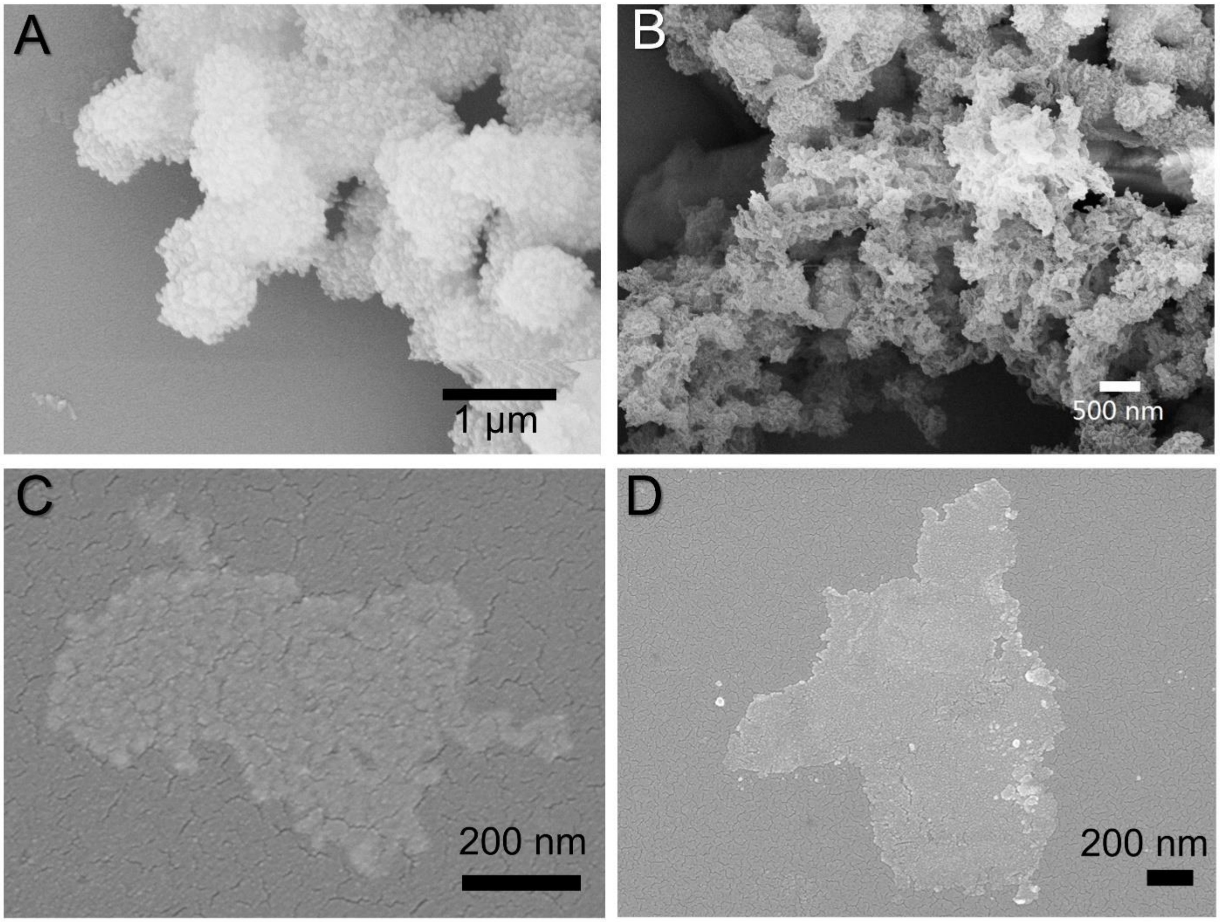


Fig. S3. The SEM images of (**A**) DhaTab-COF, (**B**) RDT-COF, and (**C, D**) SDT-COF.

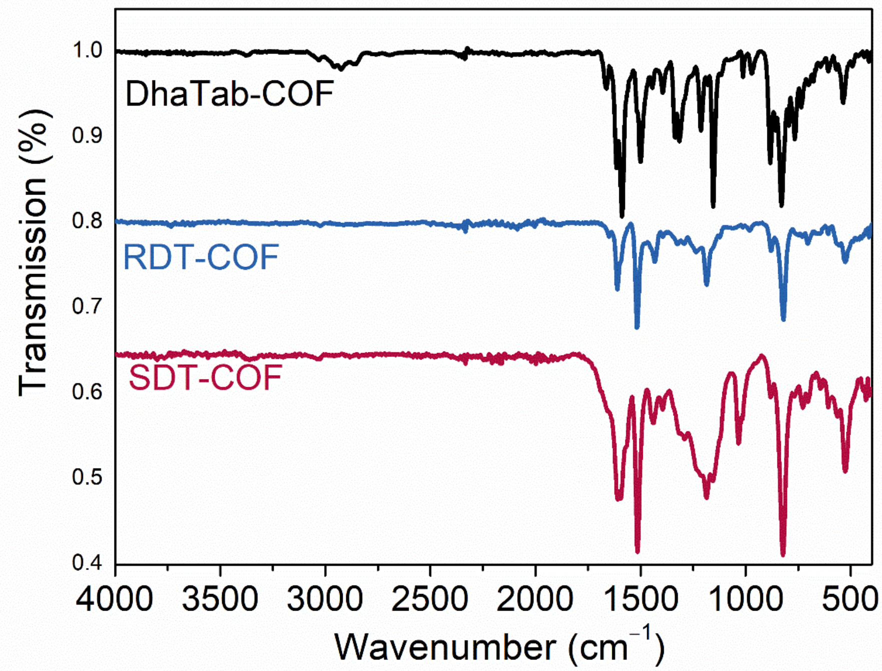


Fig. S4. The FTIR spectra of DhaTab-COF, RDT-COF, and SDT-COF.

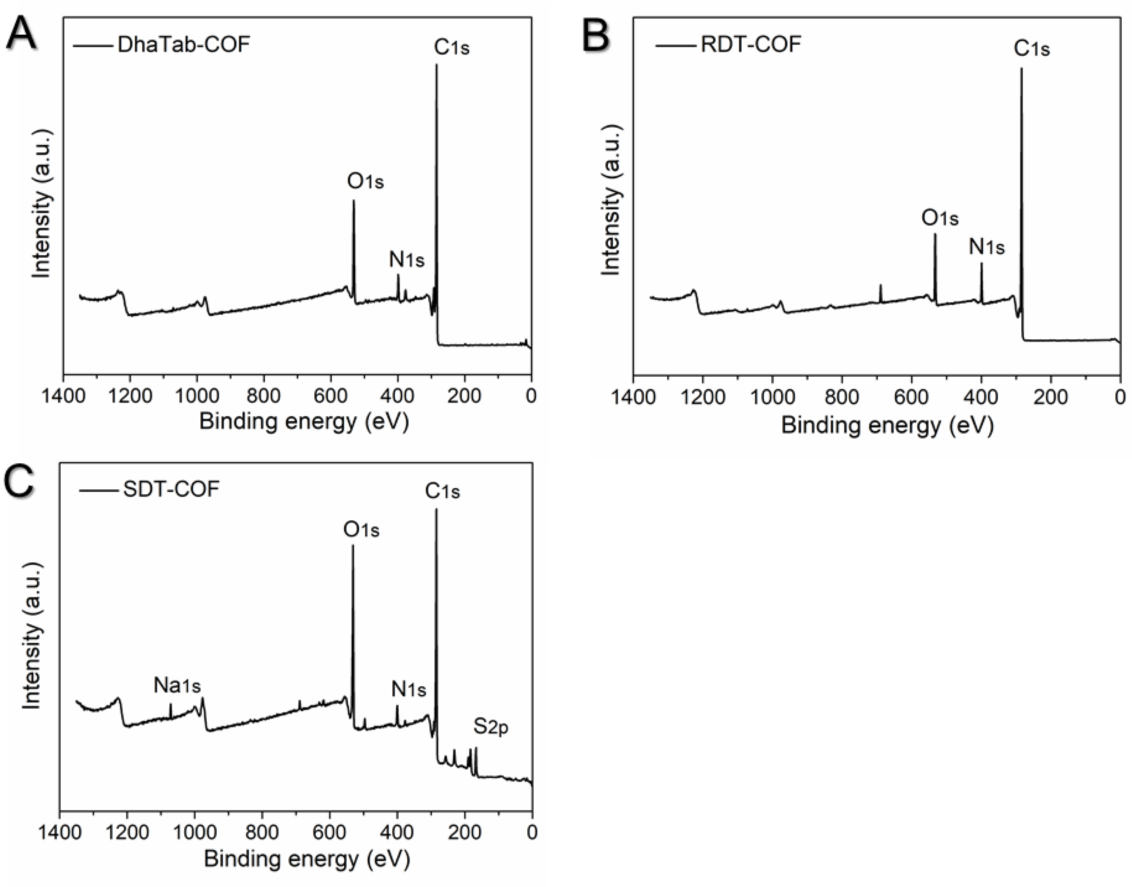


Fig. S5. The XPS spectra. (**A**) DhaTab-COF, (**B**) RDT-COF, and (**C**) SDT-COF.

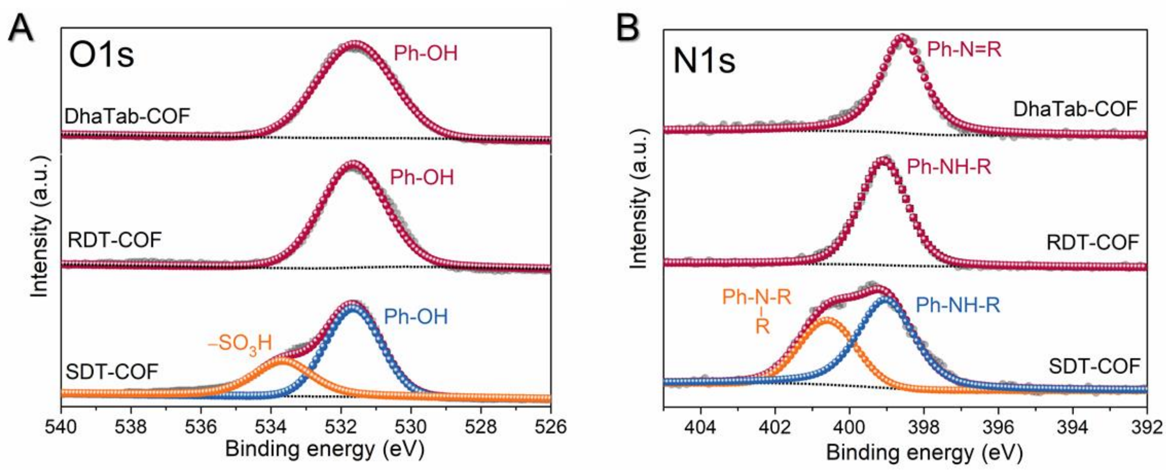


Fig. S6. (**A**) O spectra and (**B**) N spectra in XPS analysis of DhaTab-COF, RDT-COF, and SDT-COF. The XPS confirms that the grafting reaction occurred on the N rather than the phenolic hydroxyl group, as evidenced by the appearance of the characteristic peaks assigned to tertiary amines and no occurrence of the peak corresponding to ether.

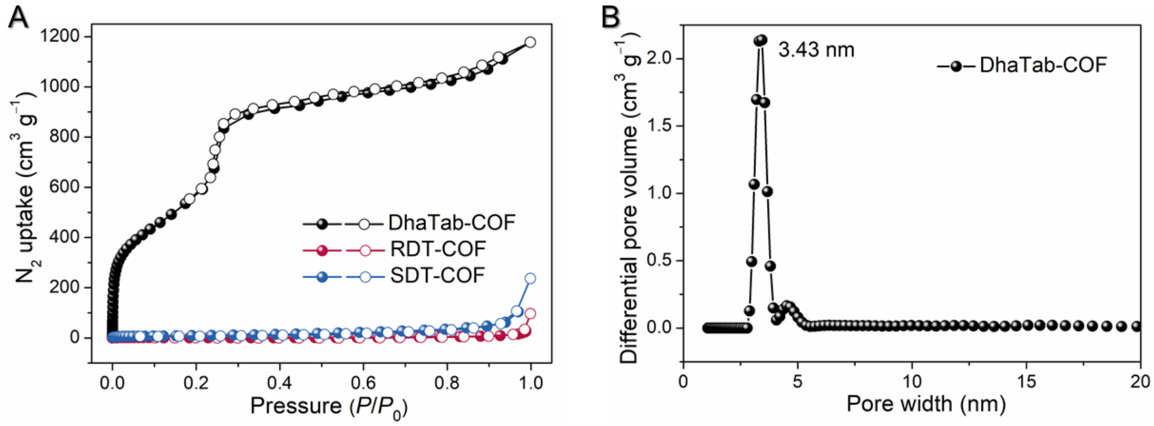


Fig. S7. (**A**) The nitrogen sorption isotherm profiles of DhaTab-COF, RDT-COF, and SDT-COF measured at 77 K. The BET specific surface areas are: DhaTab-COF: 2255 m^2 g^{-1}. (**B**) The pore-size distribution profiles for DhaTab-COF. The formation of sp^3 hybrid N and C atoms in RDT-COF and SDT-COF reduced the π-π interactions between the adjacent layers, leading to the randomly stacking of polymeric layers. It means that only sharp edges rather than wall surfaces were exposed. The well-stacked COF layers facilitated the binding of N_2, due to the extra interaction among layers of the COF. Therefore, the BET surface areas obtained by N_2 sorption at 77 K were quite low. However, when we used CO_2 as probe molecules to enhance the interaction between the adsorbate and the framework, the RDT-COF and SDT-COF both showed CO_2 uptake ability and proved the porous nature of the modified COFs

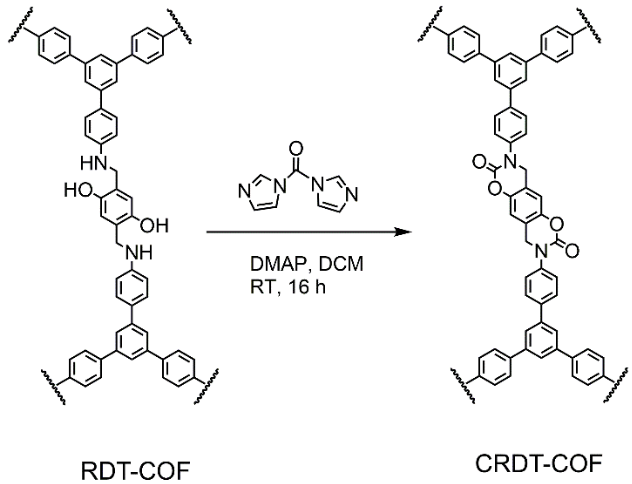


Fig. S8. The synthetic route of CRDT-COF.

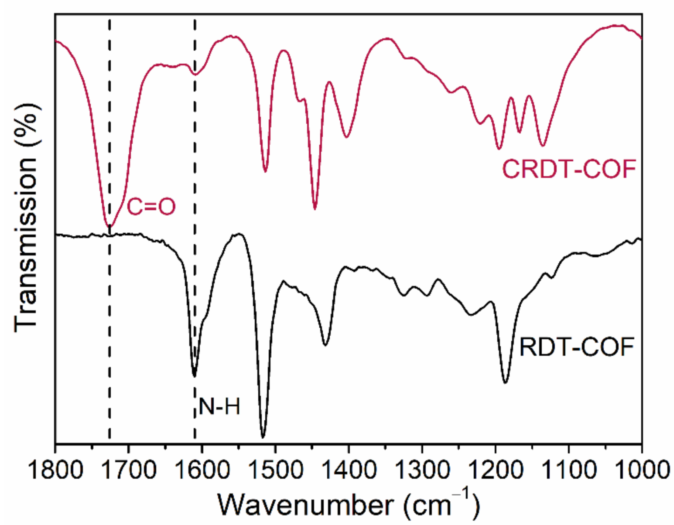


Fig. S9. The FTIR spectra of CRDT-COF and RDT-COF.

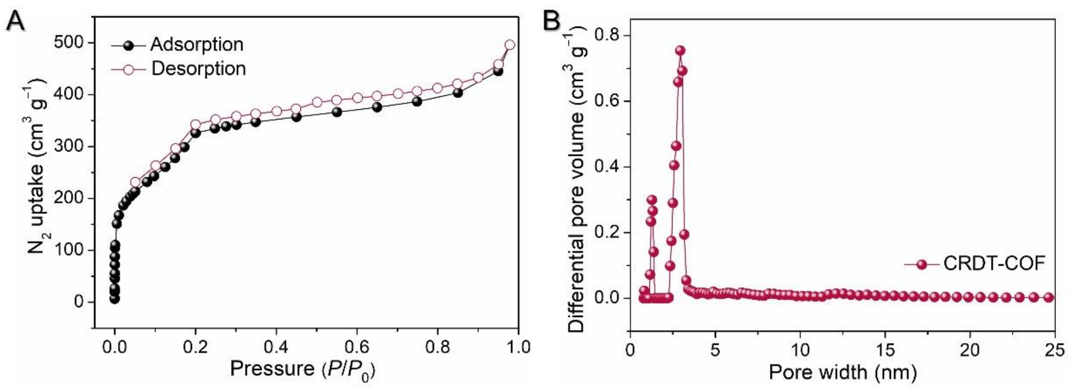


Fig. S10. (**A**) The nitrogen sorption isotherm profile of CRDT-COF measured at 77 K. (**B**) The pore-size distribution profile of CRDT-COF.

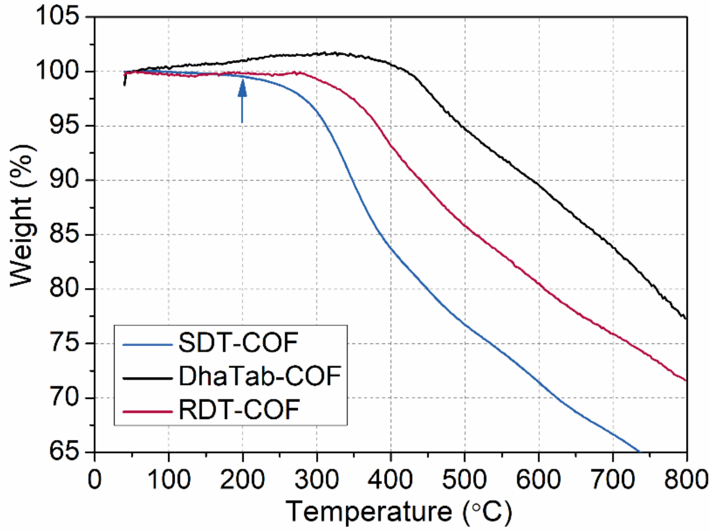


Fig. S11. TGA profiles of SDT-COF (blue), RDT-COF (red), and DhaTab-COF (black).

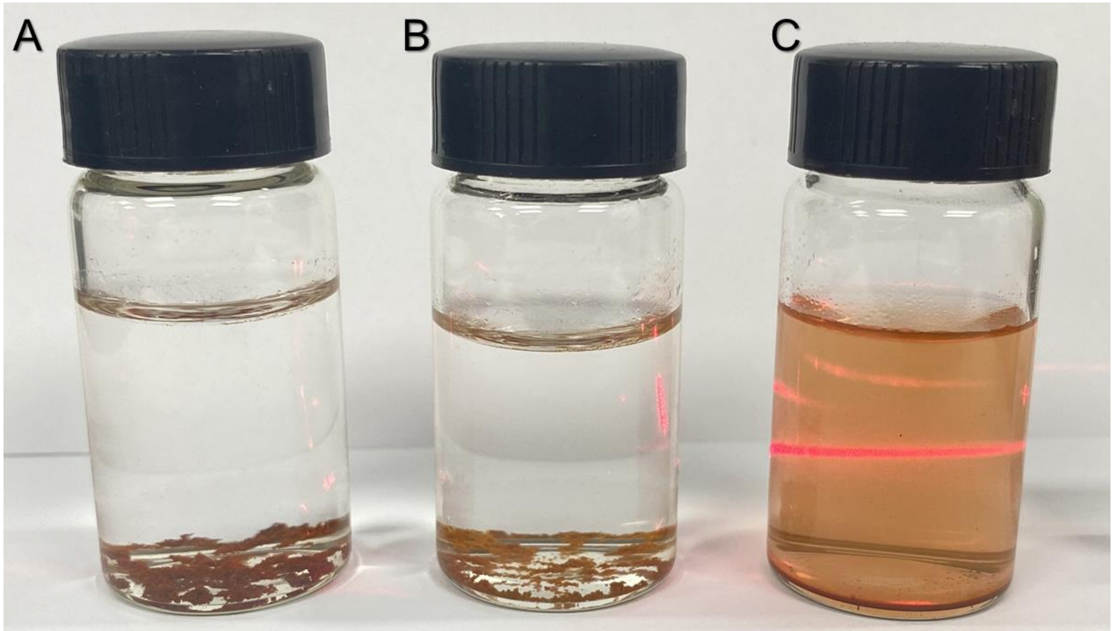


Fig. S12. The Tyndall phenomenon produced by (**A**) DhaTab-COF, (**B**) RDT-COF, and (**C**) SDT-COF aqueous solution. The Tyndall phenomena were tested after the solutions were kept standing for 24 hours.

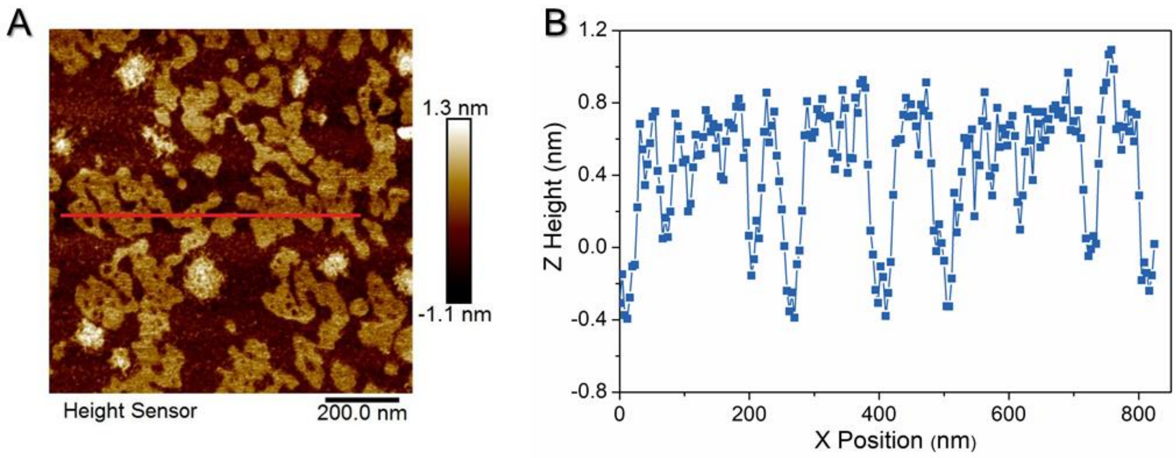


Fig. S13. High-resolution AFM tapping height images of SDT-COF. (**A**) 2D image; (**B**) The thickness of nanosheet evaluated from AFM micrographs.

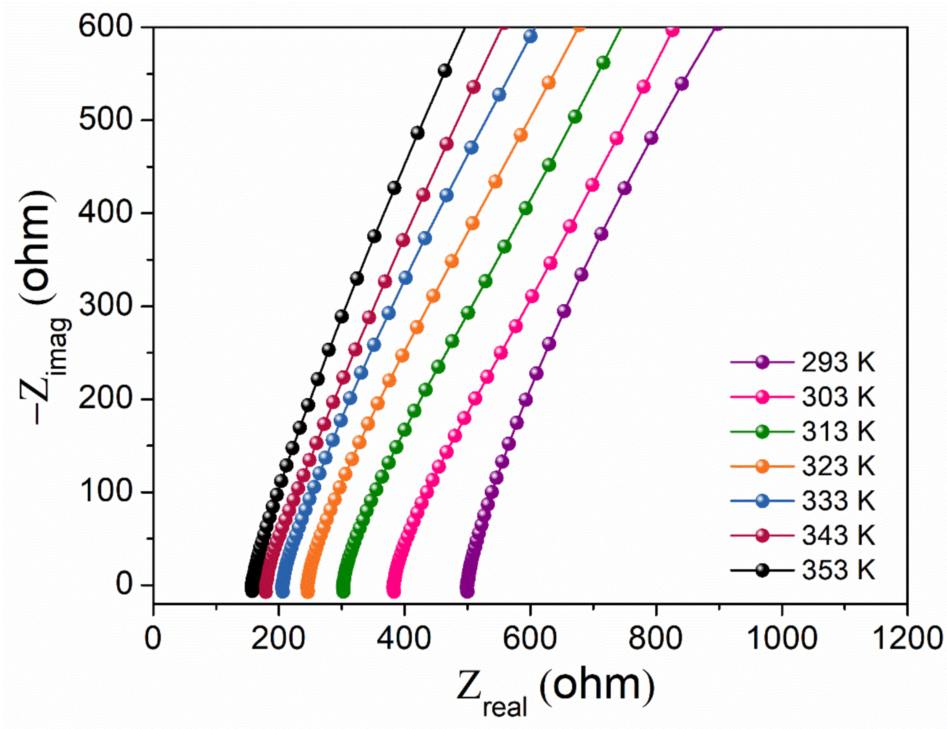


Fig. S14. Nyquist plots for SDT-COF tablet measured under 100% RH at different temperatures.

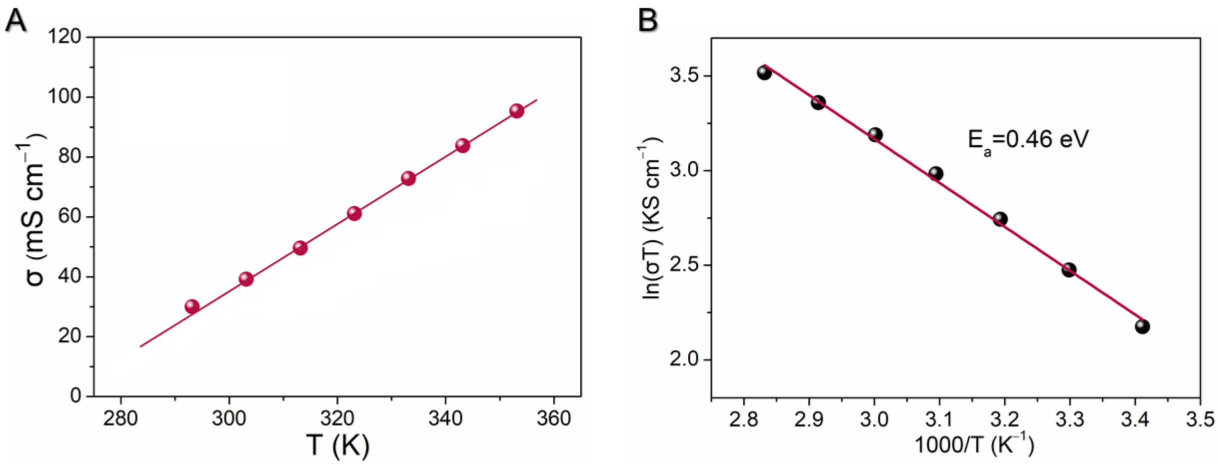


Fig. S15. (**A**) The proton conductivity for SDT-COF tablet at 100% RH and temperatures from 20 °C to 80 °C. (**B**) The Arrhenius plot used to calculate the activation energies for SDT-COF.

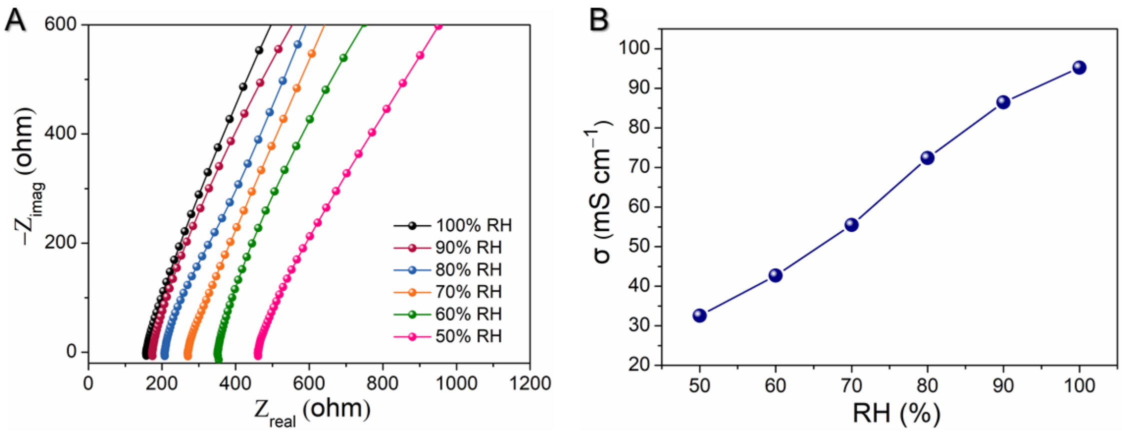


Fig. S16. (**A**) Nyquist plots for SDT-COF tablet measured under different RH at 80 ºC. (**B**) The proton conductivity of SDT-COF varies with relative humidity.

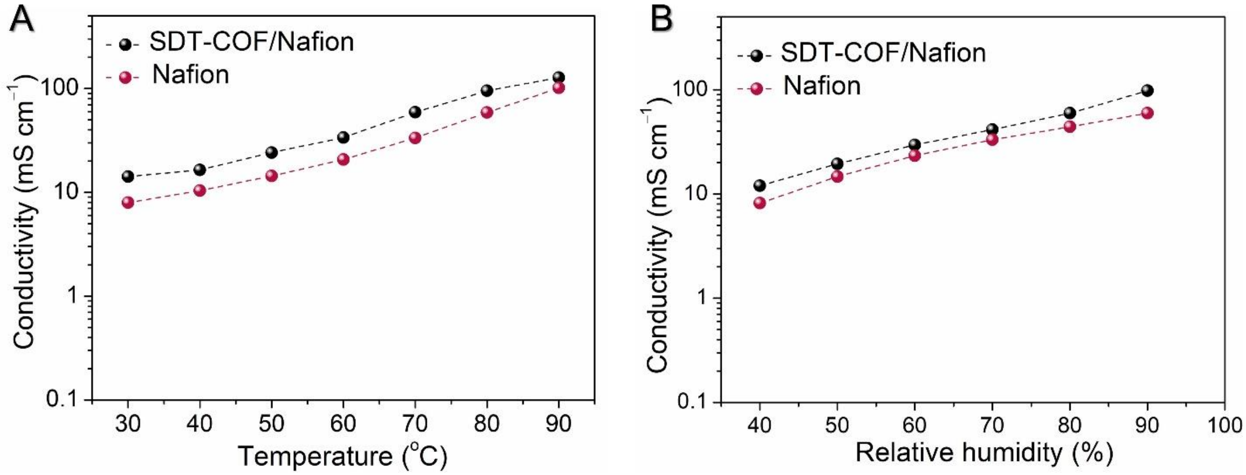


Fig. S17. (**A**) The proton conductivity of SDT-COF/Nafion (1:1) and Nafion films varies with temperatures under 90% RH. (**B**) The proton conductivity of SDT-COF/Nafion (1:1) and Nafion films varies with relative humidity under 80 °C.

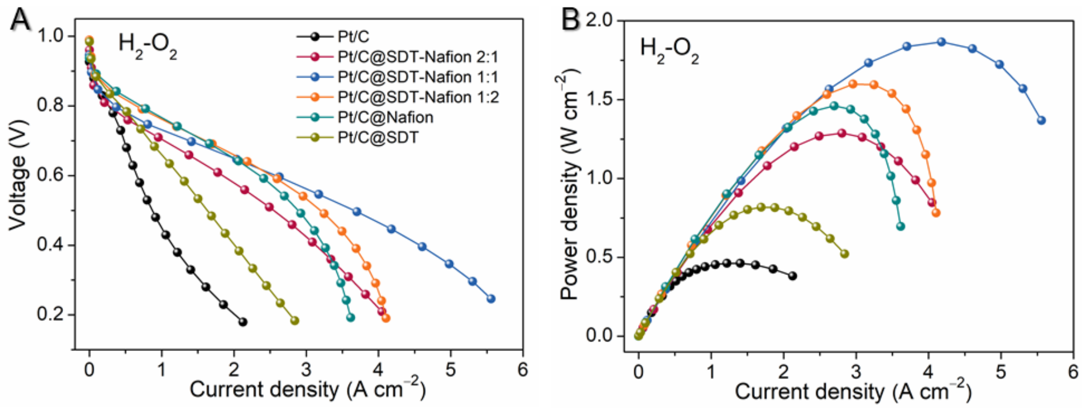


Fig. S18. Determination of the optimal ratio of ionomer. H_2-O_2 fuel cell: (**A**) *I*-*V* polarization without *iR* compensation; (**B**) power density plots of the cells with Pt/C without ionomer, Pt/C@Nafion, Pt/C@COF, and Pt/C@COF-Nafion (1:2, 1:1, 2:1) measured at 80 ^{o}C and 100% RH under 50 kPa back pressure.

Note:

Only the MEAs of this test used Hesen's carbon paper (Hesen HCP-120), and the other MEAs of this work all used Toray's carbon paper (TORAY YLS-30T).

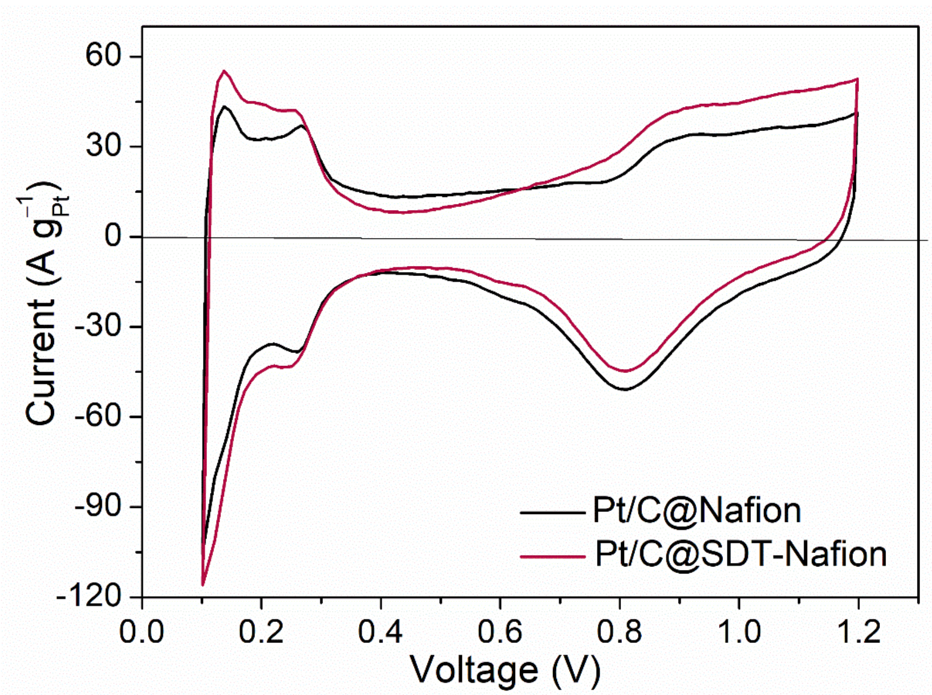


Fig. S19. The CV curves of the cells with a Pt/C@Nafion catalyst and a Pt/C@SDT-Nafion catalyst with nitrogen on the cathode side. The scan rate was 50 mV s^{-1}.

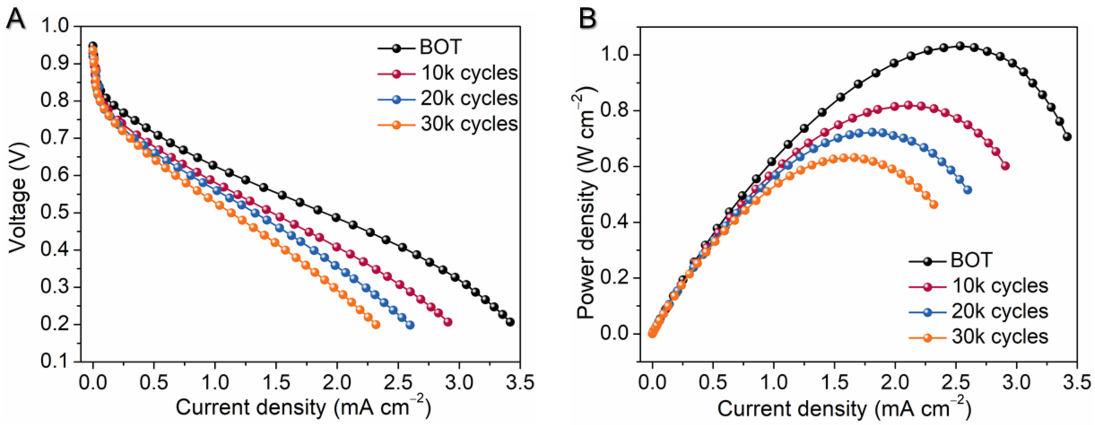


Fig. S20. The *I-V* polarization (without *iR* compensation) and power density variations in the AST of the H_2-air cell with Pt/C@SDT-Nafion, which measured at 80 ºC and 100% RH under 50 kPa back pressure.

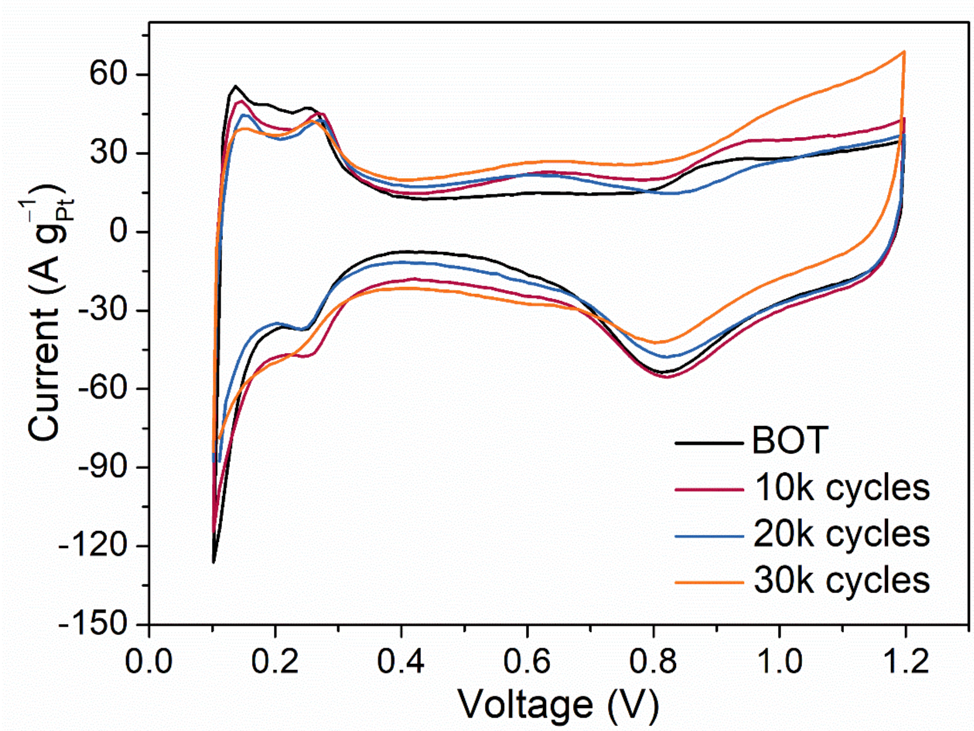


Fig. S21. The CV variations in the AST of the cell with a Pt/C@SDT-Nafion. The scan rate was 50 mV s^{-1}.

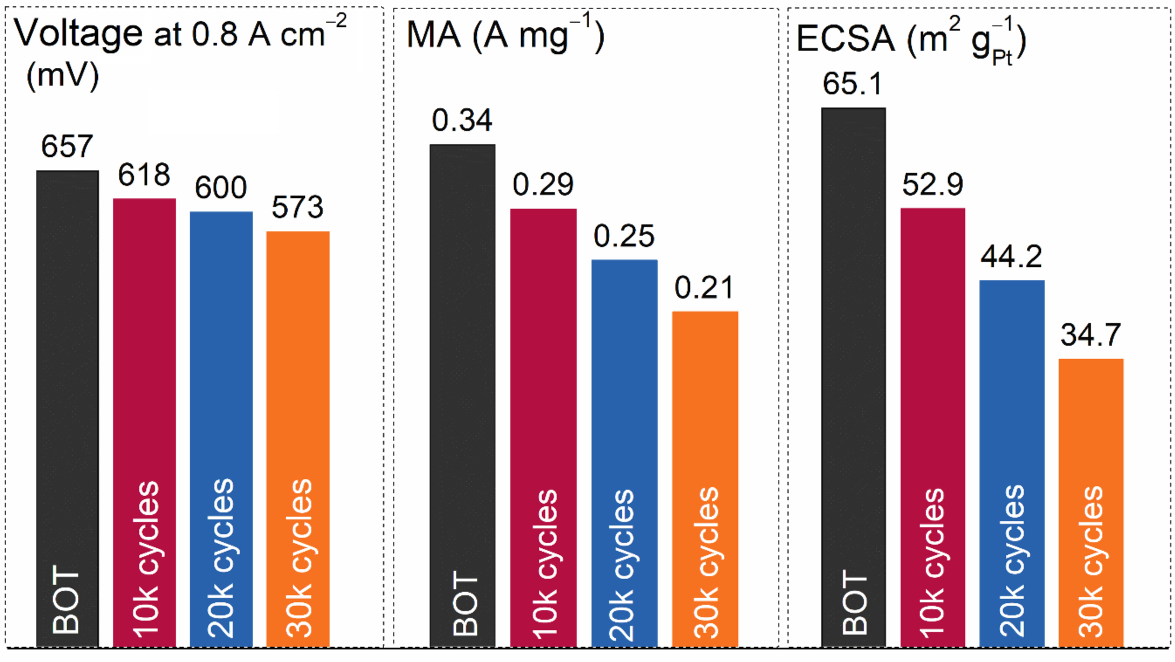


Fig. S22. The voltage changes at 0.8 A cm^{-2}, the change of mass activity, and the ECSA change after 10,000, 20,000 and 30,000 cycles AST test.

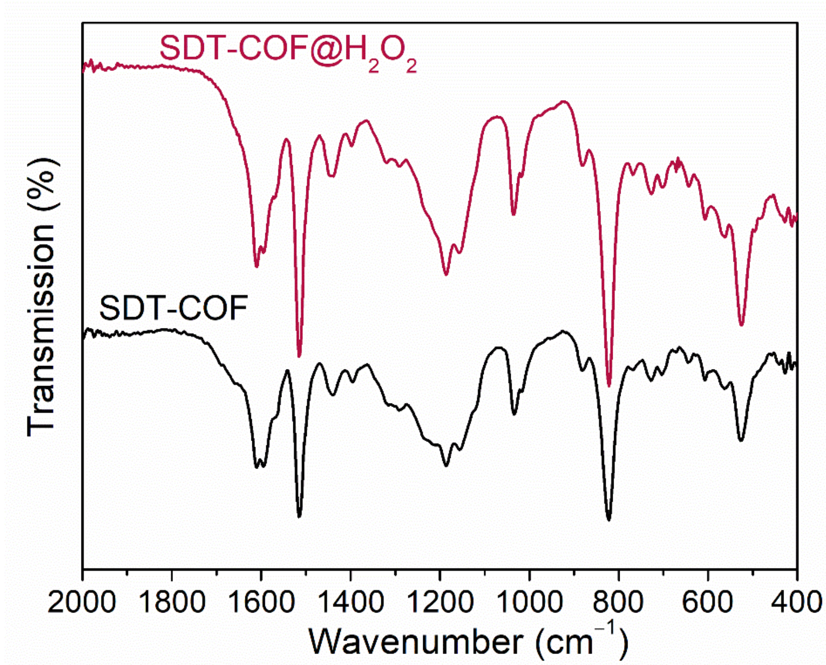


Fig. S23. Stability test of SDT-COF in hydrogen peroxide. The SDT-COF was directly soaked in 1% hydrogen peroxide for 5 hours. The IR spectrum does not show obvious changes after soaking.

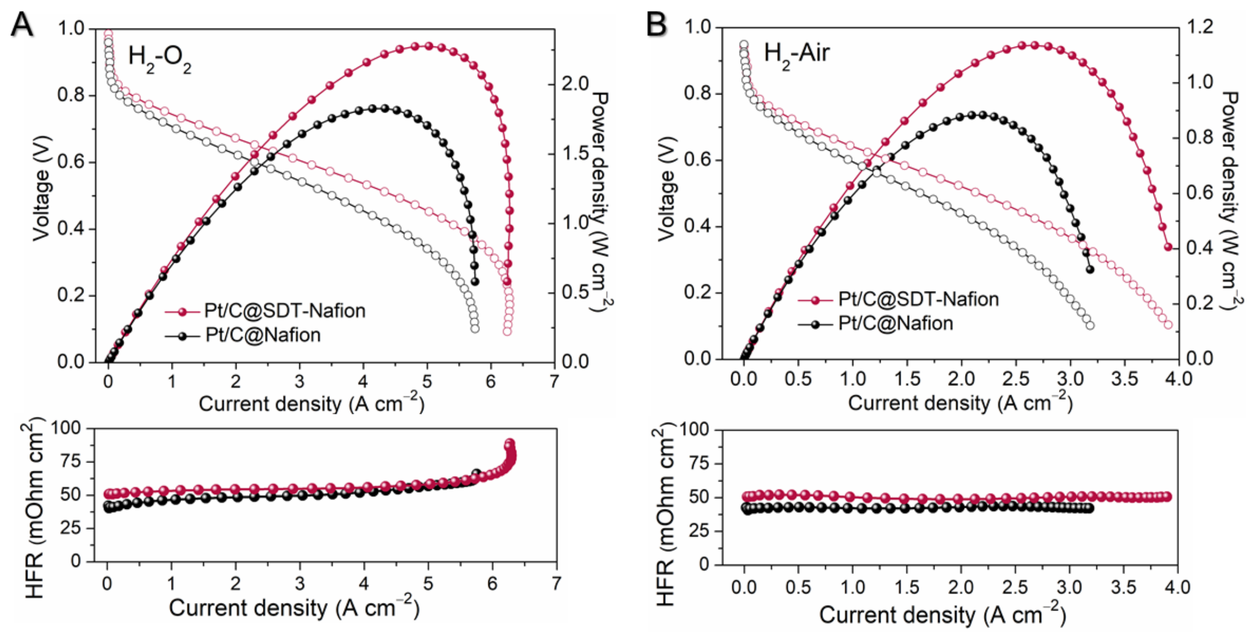


Fig. S24. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I-V* polarization (without *iR* compensation) and power density plots with Pt/C@Nafion and Pt/C@SDT-Nafion measured at 80 ºC and 100% RH under 50 kPa back pressure. Cathode, ~0.1 $mg_{Pt}\ cm^{-2}$; anode, ~0.1 $mg_{Pt}\ cm^{-2}$; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is high-frequency resistance (HFR).

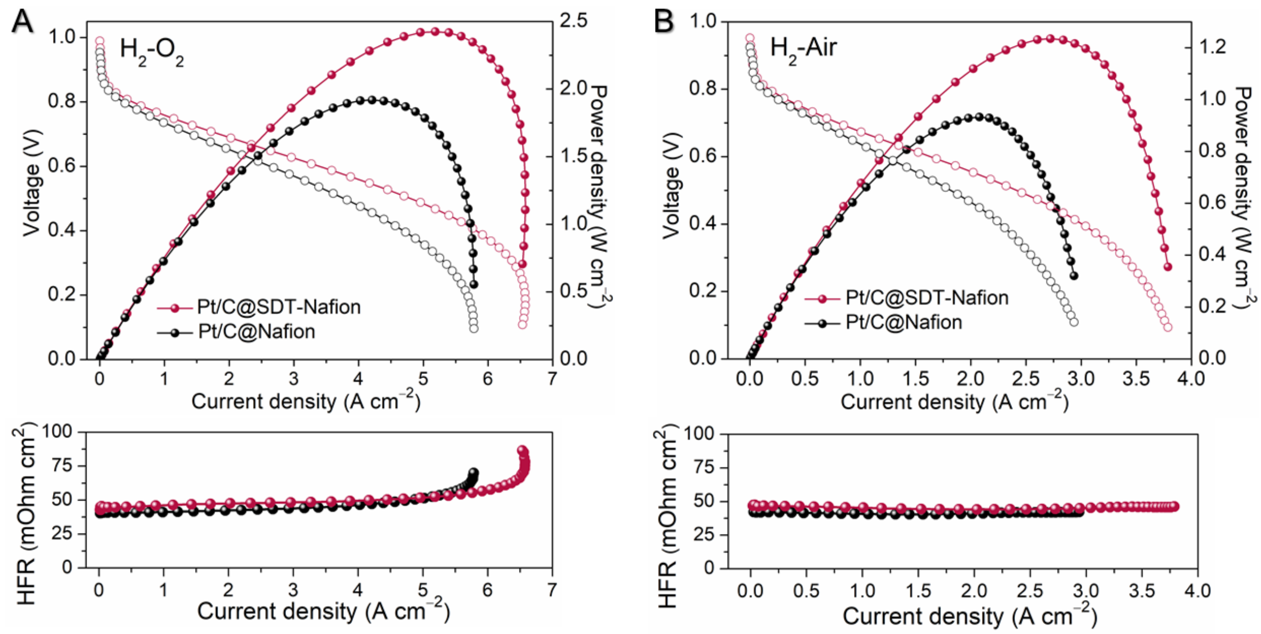


Fig. S25. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@Nafion and Pt/C@SDT-Nafion measured at 80 ºC and 100% RH under 50 kPa back pressure. Cathode, ~0.2 mg_{Pt} cm^{-2}; anode, ~0.2 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

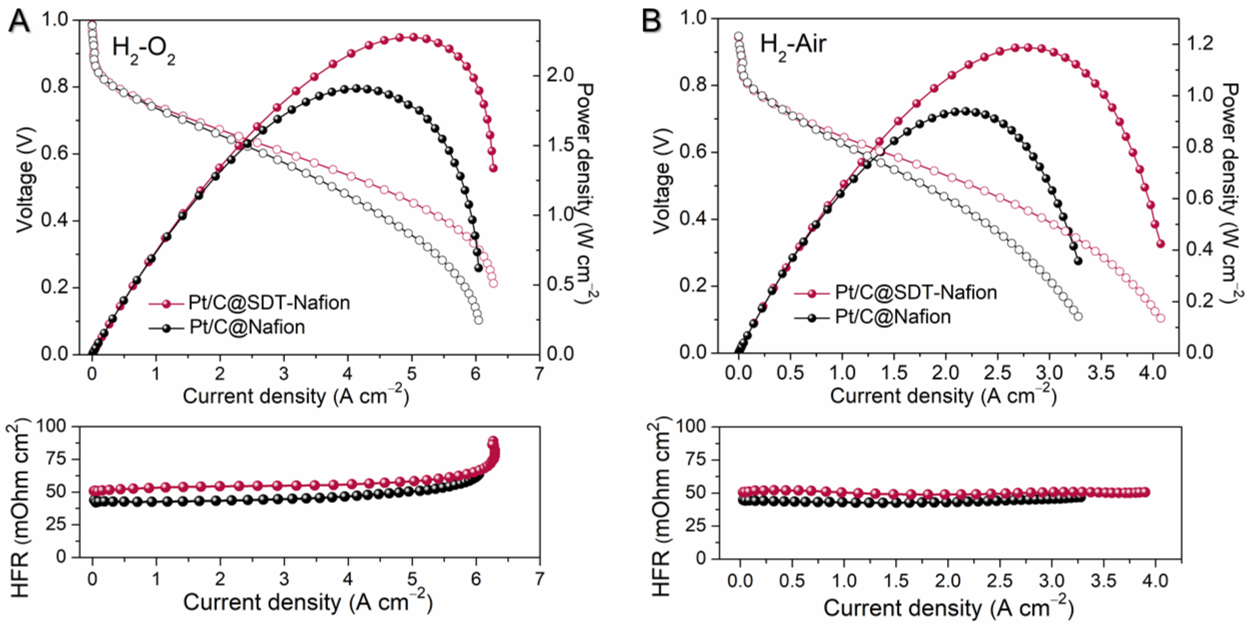


Fig. S26. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@Nafion and Pt/C@SDT-Nafion measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.3 mg_{Pt} cm^{-2}; anode, ~0.3 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

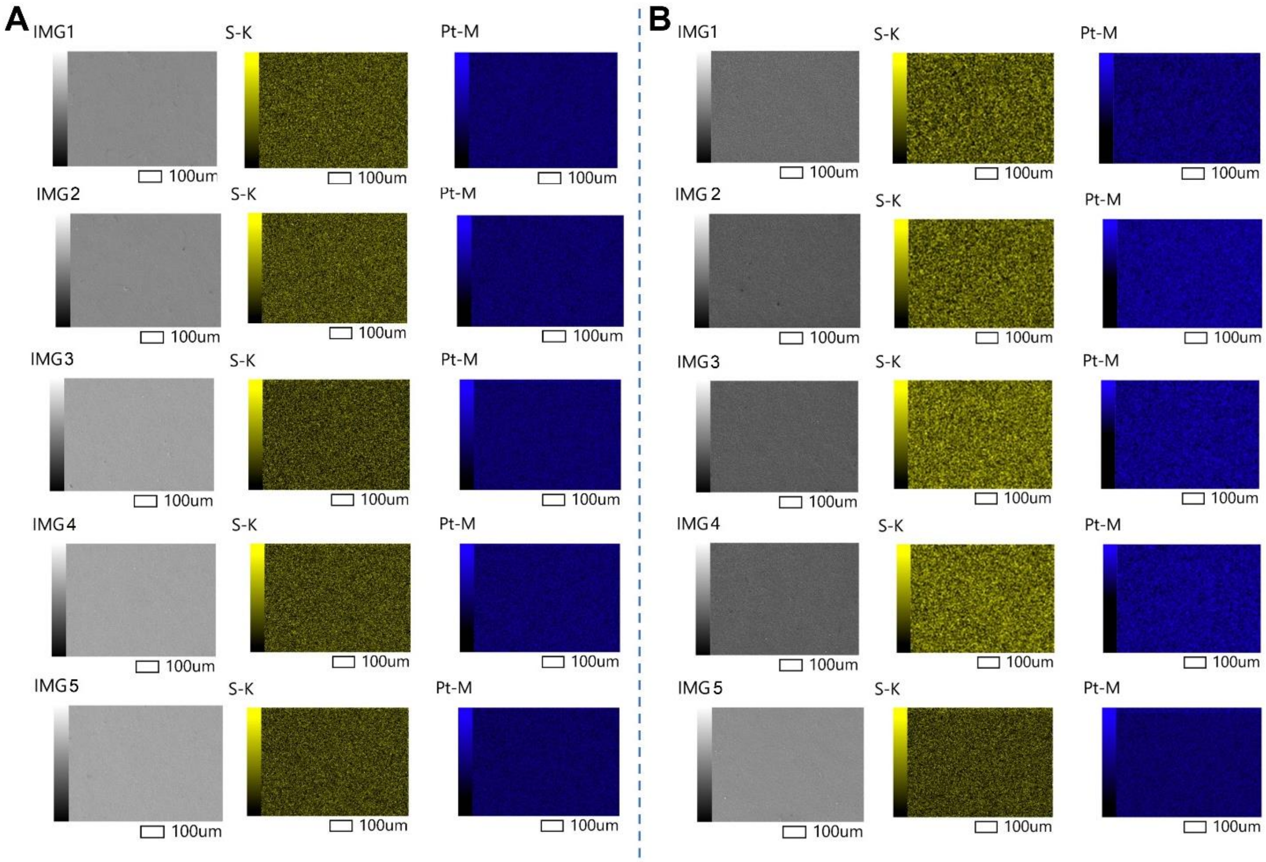


Fig. S27. The SEM and elemental mapping of MEAs with Pt-loadings at (**A**) 0.2 mg_{Pt} cm^{-2} and (**B**) 0.3 mg_{Pt} cm^{-2}. The images were captured from different areas in one CCM sample with certain Pt loading.

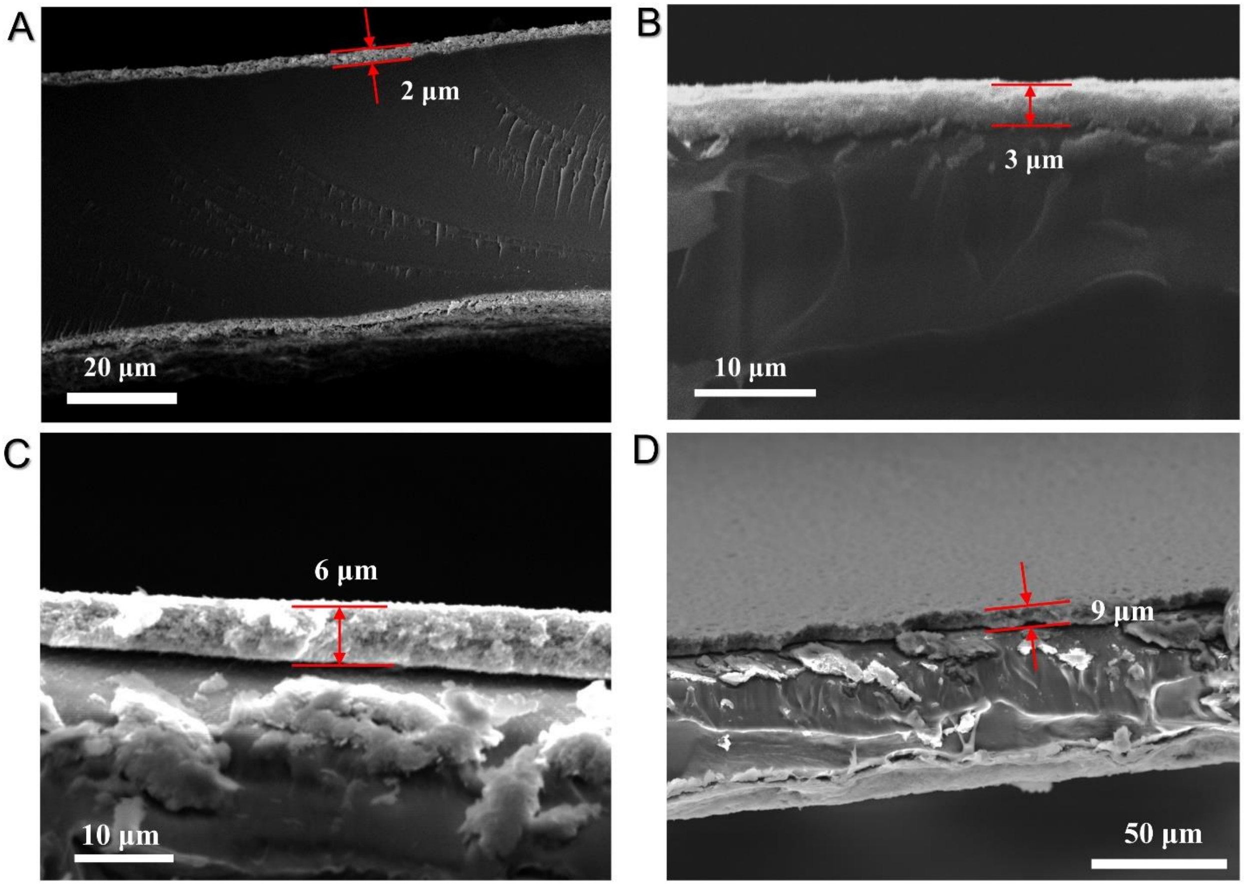


Fig. S28. (**A-D**) The cross-section SEM images of MEAs with Pt-loadings at 0.07 mg_{Pt} cm^{-2}, 0.1 mg_{Pt} cm^{-2}, 0.2 mg_{Pt} cm^{-2}, and 0.3 mg_{Pt} cm^{-2}, respectively.

For fabricating MEA with different Pt-loadings, we kept ionomer/catalyst concentration constant and increase the spray time to fabricate MEA with higher Pt-loading. Along with more Pt catalyst loaded, the thicknesses of the MEA were accordingly increased. The FE-SEM images and elemental mapping obtained from different areas showed that the catalyst was evenly distributed (Fig. S27). We speculated that the increased thickness with longer mass and electron transport path is the reason for the insignificant effect in the cell performance by increasing Pt-loadings larger than 0.1 mg_{Pt} cm^{-2}.

We noticed that other methods such as gas diffusion electrode (GDE) process or transfer printing method were also applied in MEA production, where increasing Pt loading may significantly improve the MEA performance. In our work, we focused on the influence of COF materials on MEA performance. Thus, we used the spray-coating CCM process and kept ionomer/catalyst concentration constant, which inevitably led to the increased CL thickness with a longer mass and electron transport path.

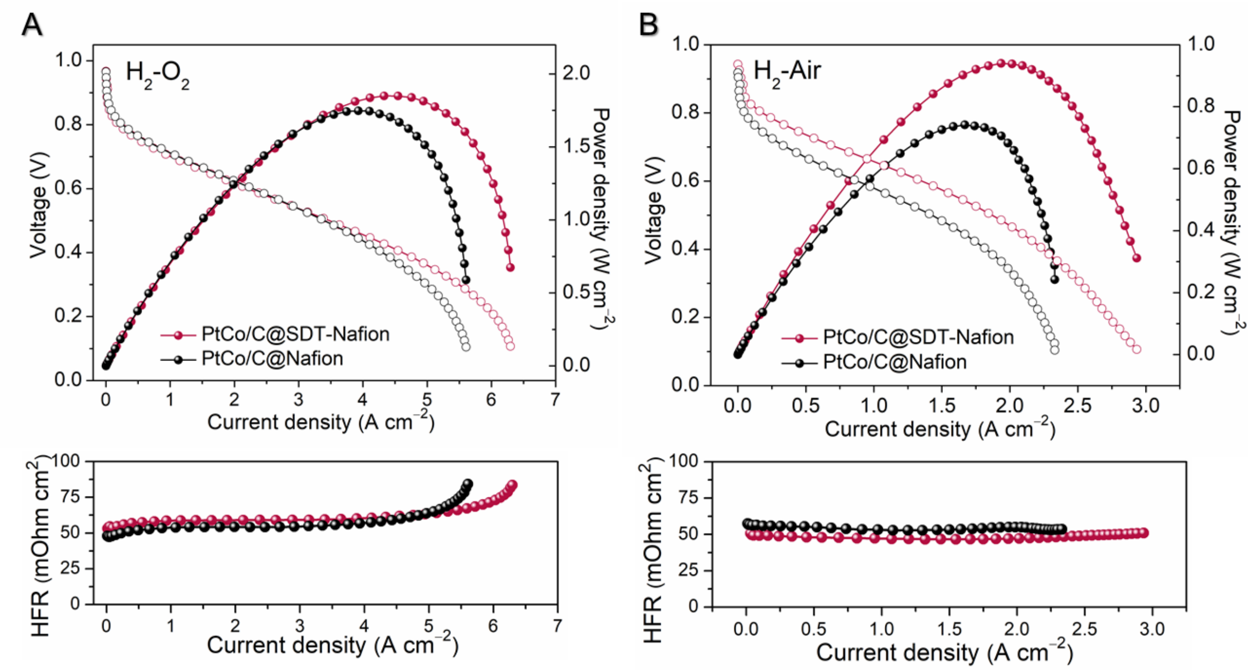


Fig. S29. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I-V* polarization (without *iR* compensation) and power density plots with PtCo/C@Nafion and PtCo/C@SDT-Nafion measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

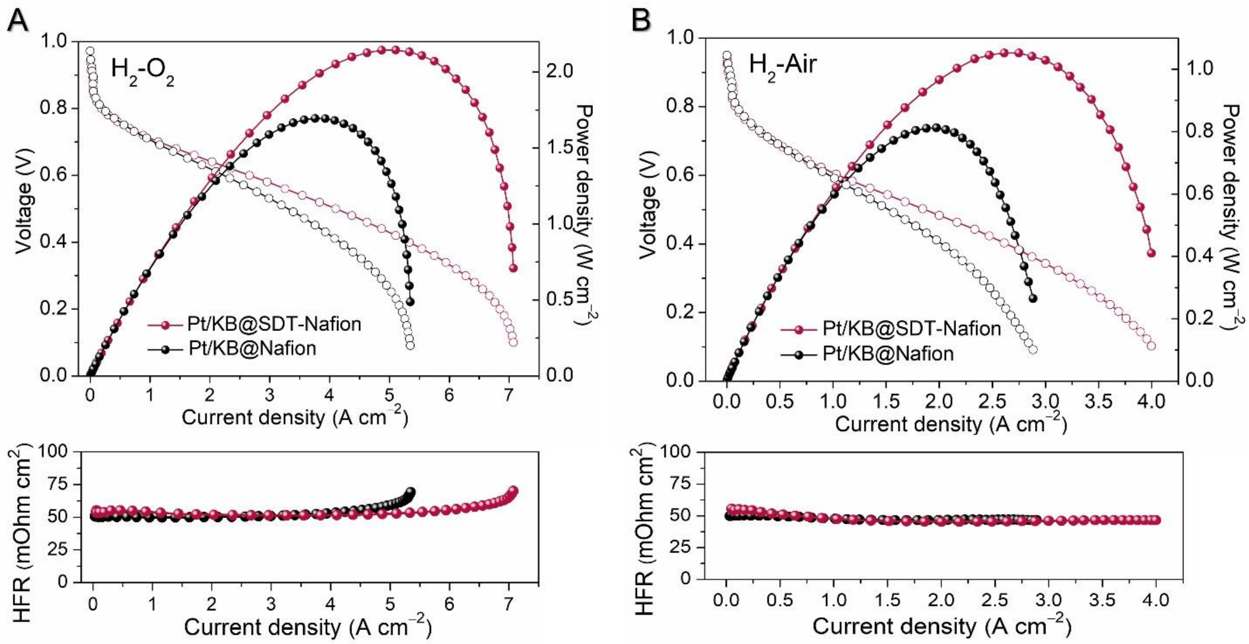


Fig. S30. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/KB@Nafion and Pt/KB@SDT-Nafion measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

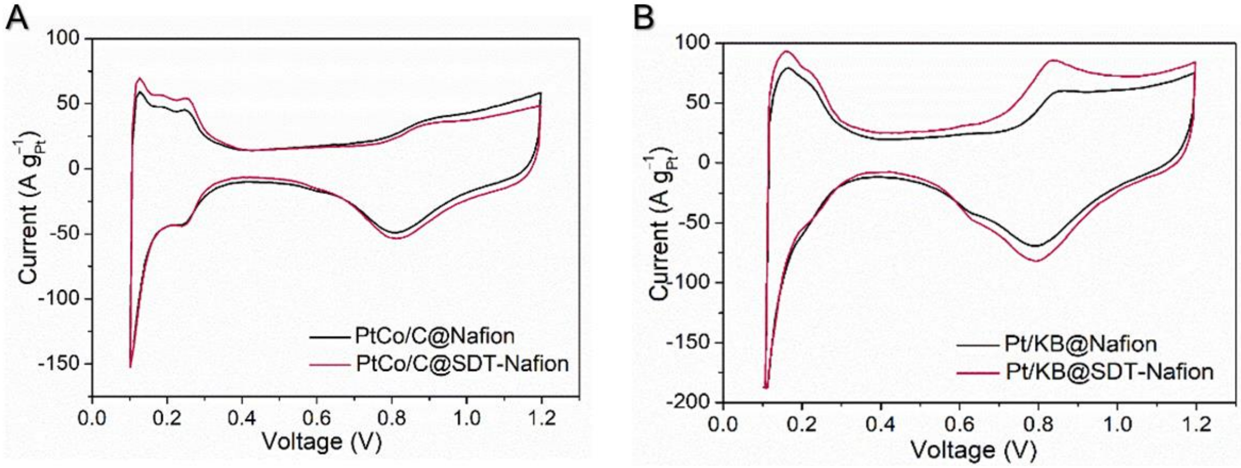


Fig. S31. The CV curves of the cells with (**A**) PtCo/C@Nafion and PtCo/C@SDT-Nafion catalyst, and (**B**) Pt/KB@Nafion and Pt/KB@SDT-Nafion catalyst with nitrogen on the cathode side. The scan rate was 50 mV s^{-1}.

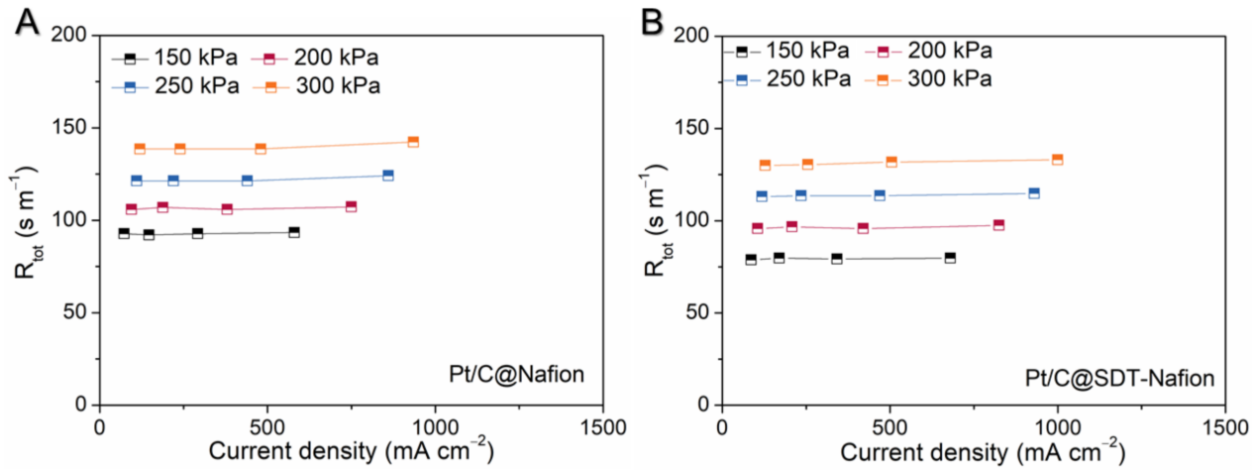


Fig. S32. R_{tot} versus limiting currents at four different gas pressures (abs), the order of X_{O2} from left to right in one line-plot is 0.5, 1, 2, and 4%.

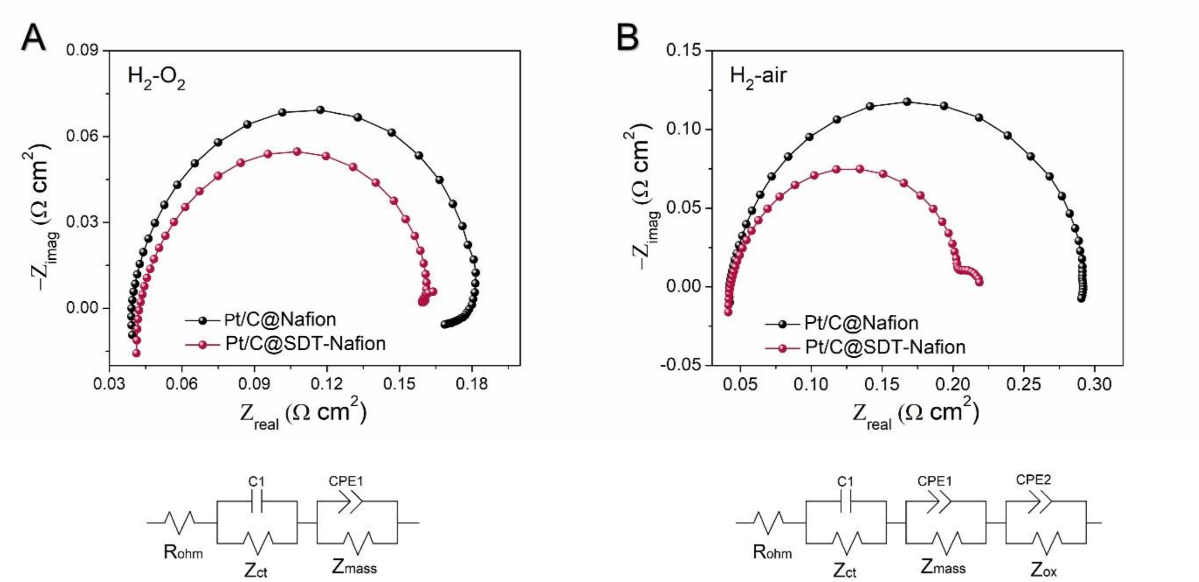


Fig. S33. Nyquist plots of Pt/C@Nafion and Pt/C@SDT-Nafion in (**A**) H_2-O_2 and (**B**) H_2-air fuel cells, respectively. The below is the equivalent circuit model. Test conditions are the same as those for the polarization curve test.

Electrochemical impedance spectroscopy (EIS) can assess the individual contributions of the transport and kinetic processes to the PEMFCs. After introducing SDT-COF, the total resistance decreases from 0.18 Ω cm^2 to 0.16 Ω cm^2 in H_2-O_2 tests, while decreasing from 0.29 Ω cm^2 to 0.22 Ω cm^2 in H_2-air tests.

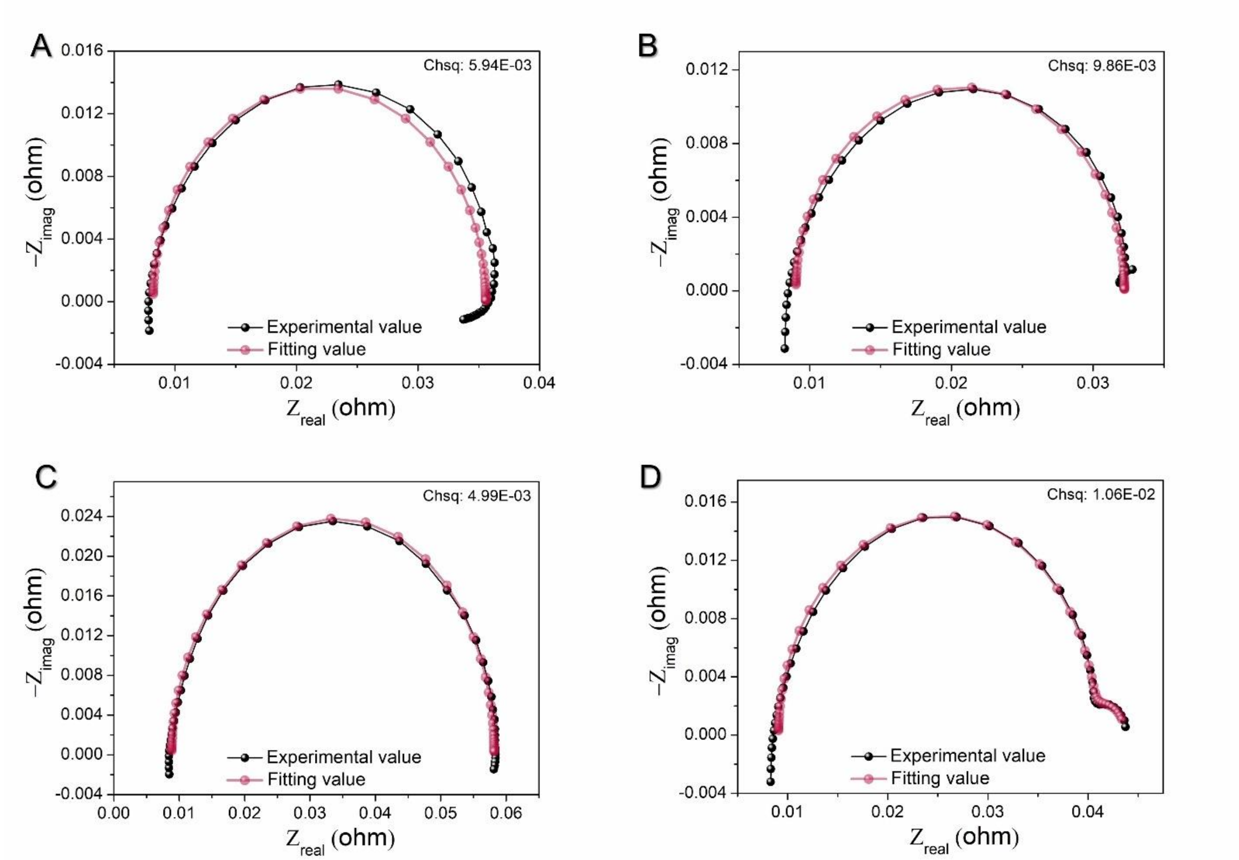


Fig. S34. Nyquist plots and fitting data of (**A**) H_2-O_2 fuel cell with Pt/C@Nafion, (**B**) H_2-O_2 fuel cell with Pt/C@SDT-Nafion, (**C**) H_2-air fuel cell with Pt/C@Nafion and (**D**) H_2-air fuel cell with Pt/C@SDT-Nafion.

The EIS data were fitted to a modified Randles cell equivalent circuit model (Table S4) to monitor the ohmic resistance (R_{ohm}), polarization resistances (Z_{ct}, Z_{ox}, Z_{mass}), and capacitances (C1, CPE1, CPE2). Z_{ct} is the charge transfer resistance related to the energy barrier of the electrochemical reaction that must be overcome. The kinetic loss at the cathode generally dominates Z_{ct}. Z_{ox} is the O_2 transfer resistance, which is related to the transmission resistance of oxygen to the active sites of the catalyst layer. For the H_2-air test, the insufficient O_2 supplement and/or N_2 occupation leads to extra mass transfer resistance (Z_{mass}). The total mass transfer resistance of O_2 reduced significantly after adding SDT-COF for both H_2-O_2 and H_2-air cells (Pt/C@SDT-Nafion: 0.68×10^{-2} Ω in H_2/O_2, 3.2×10^{-2} Ω in H_2-air; Pt/C@Nafion: 2.7×10^{-2} Ω in H_2-O_2, 5.5×10^{-2} Ω in H_2-air).

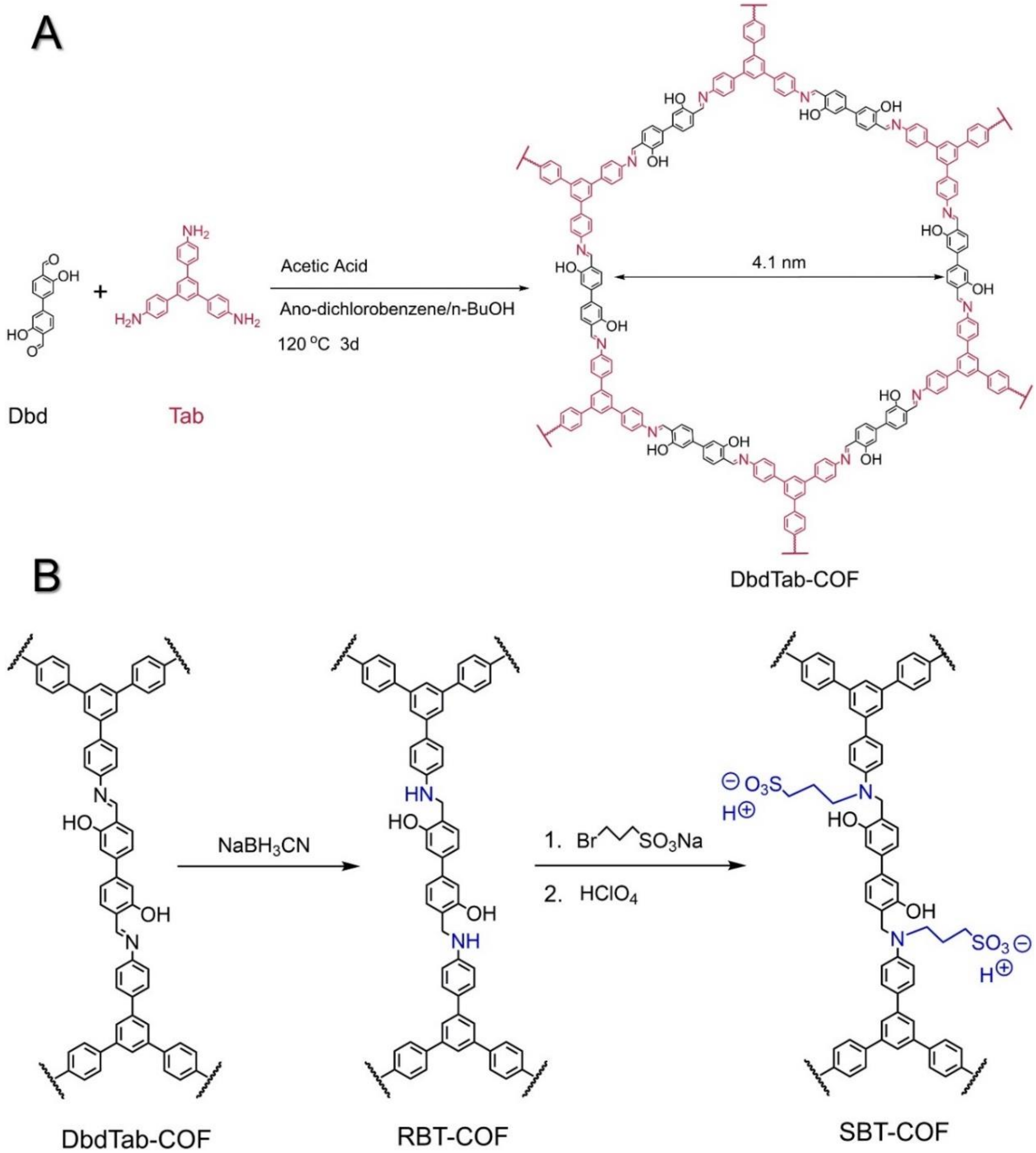


Fig. S35. (**A**) Reaction scheme of DbdTab-COF synthesis. (**B**) The synthetic route of SBT-COF.

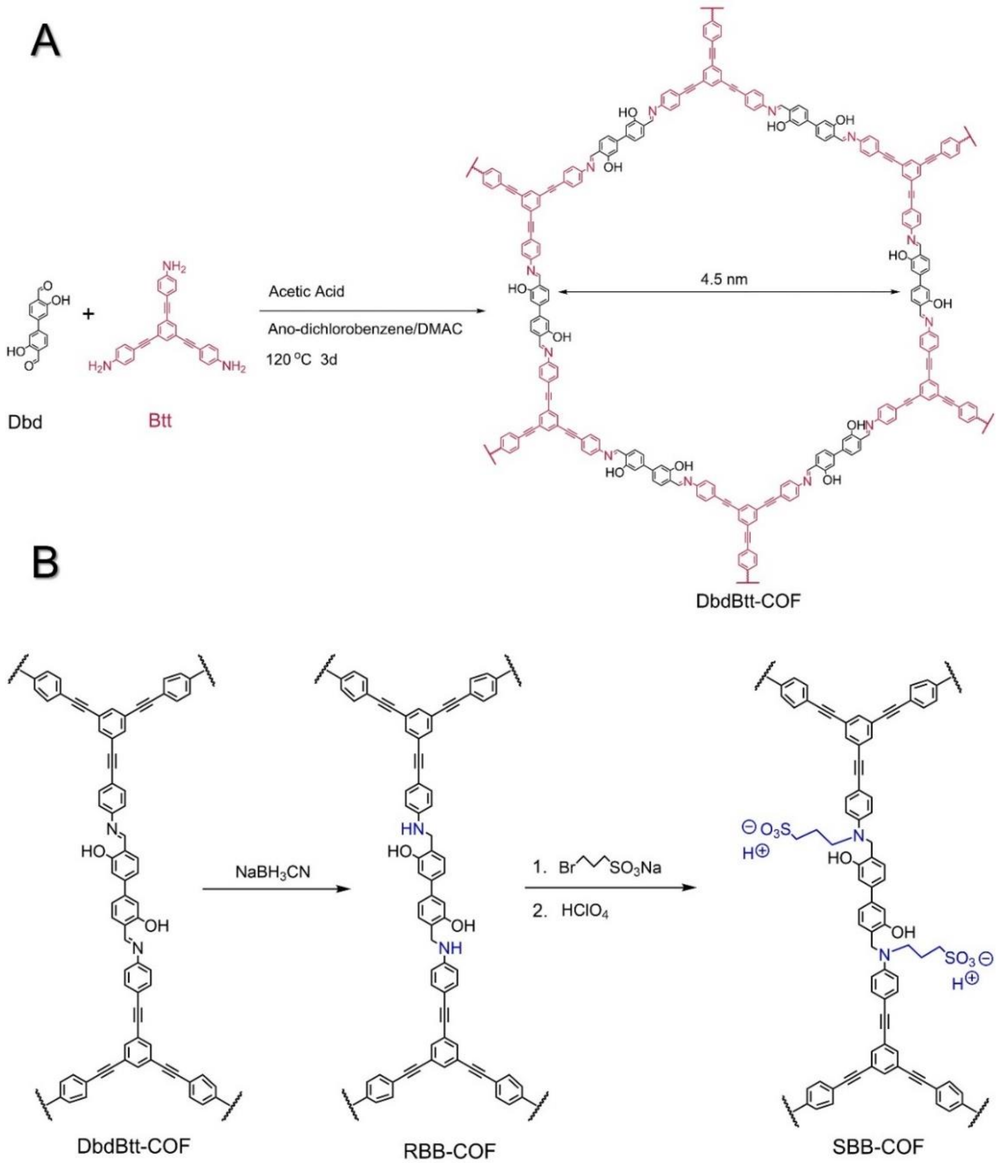


Fig. S36. (**A**) Reaction scheme of DbdBtt-COF synthesis. (**B**) The synthetic route of SBB-COF.

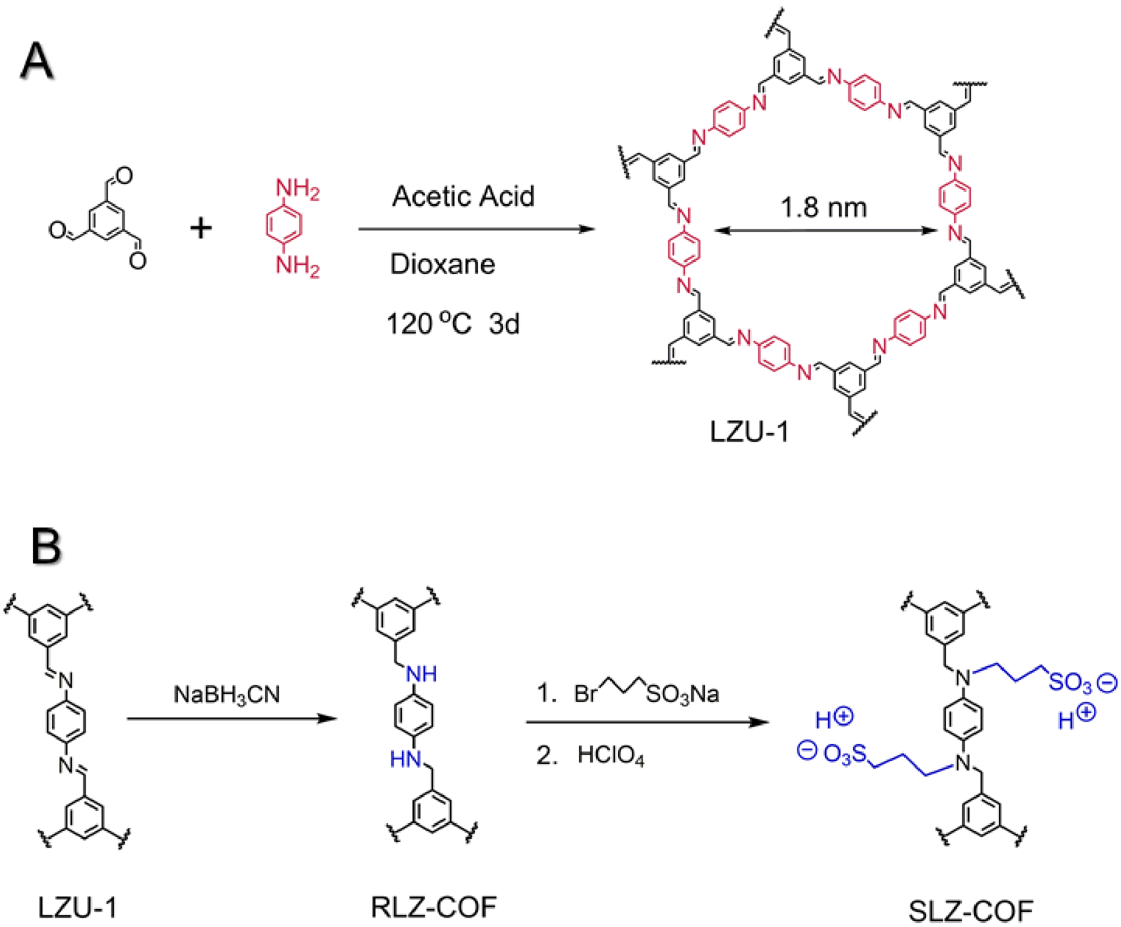


Fig. S37. (**A**) Reaction scheme of LZU-1 synthesis. (**B**) The synthetic route of SLZ-COF.

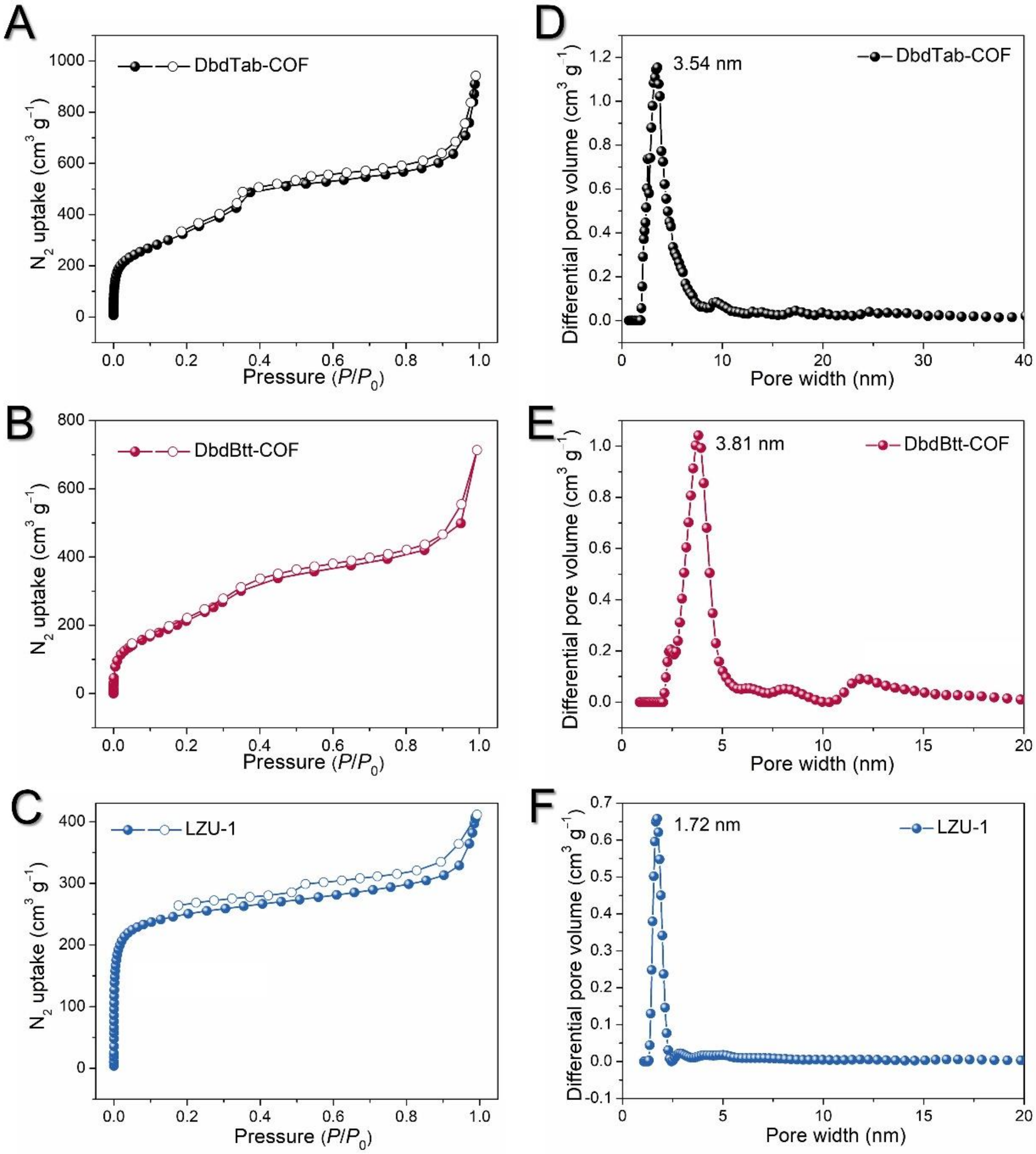


Fig. S38. The nitrogen sorption isotherm profiles of (**A**) DbdTab-COF, (**B**) DbdBtt-COF, and (**C**) LZU-1 measured at 77 K. The pore-size distribution profiles for (**D**) DbdTab-COF, (**E**) DbdBtt-COF, and (**F**) LZU-1. The obtained pore size distributions are lower than the edge-to-edge distance in the hexagonal pores, suggesting the offset stacking of the planar layers. The BET specific surface areas are: DbdTab-COF: 1255 $m^2\ g^{-1}$, DbdBtt-COF: 1081 $m^2\ g^{-1}$, LZU-1: 1308 $m^2\ g^{-1}$.

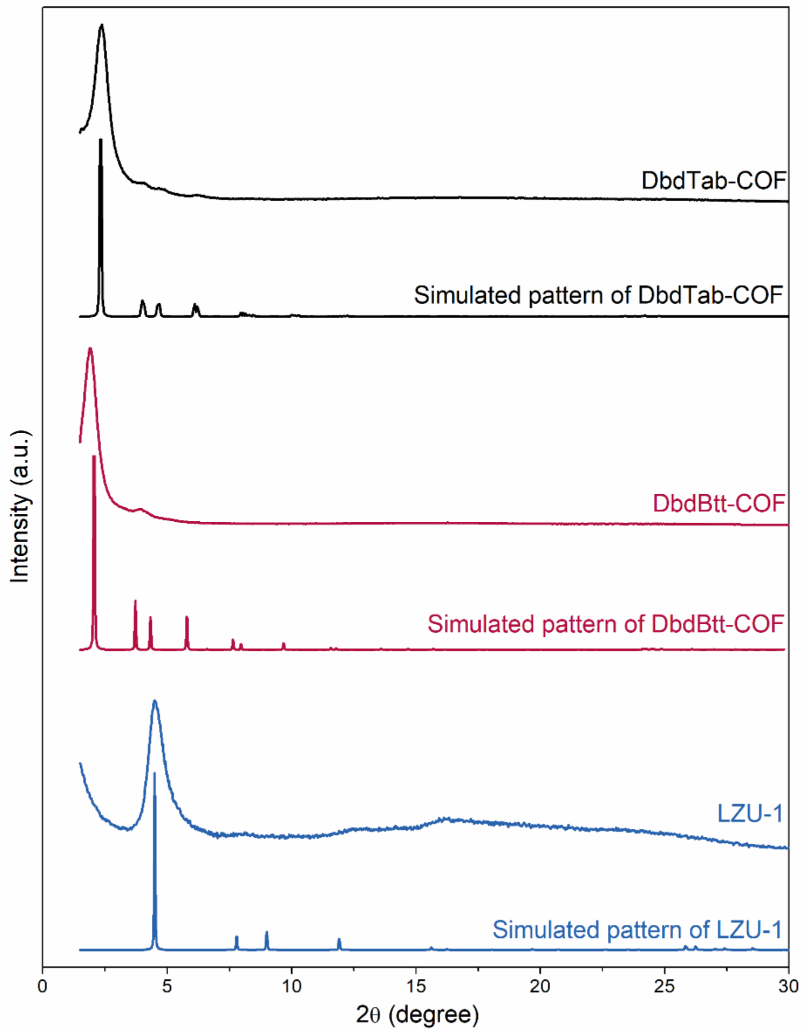


Fig. S39. The experimental and simulated (AA stacking) PXRD patterns of DbdTab-COF, DbdBtt-COF, and LZU-1.

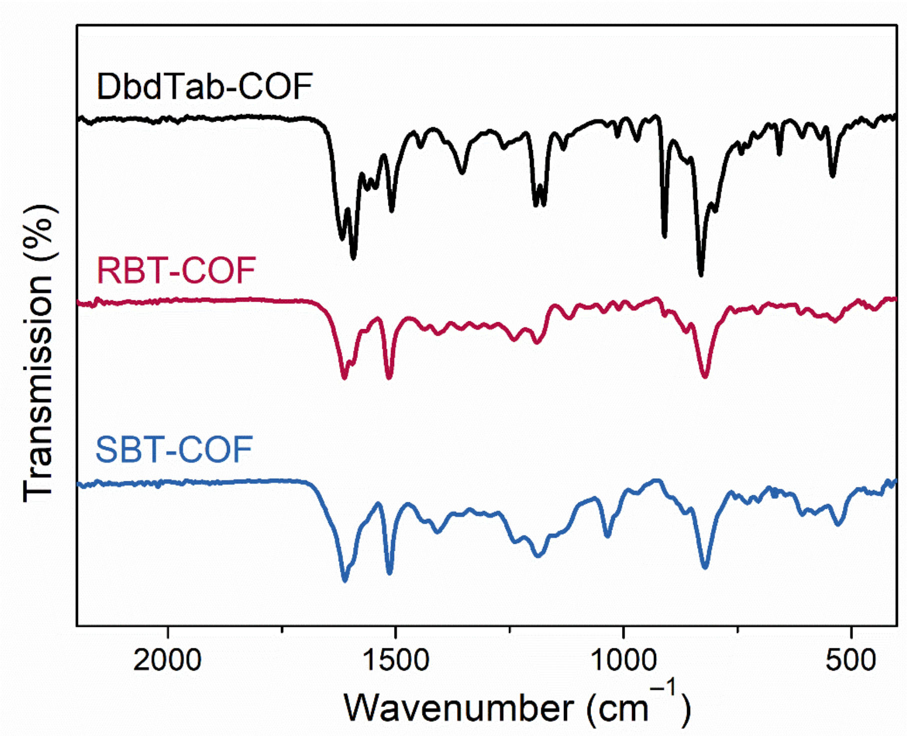


Fig. S40. The FTIR spectra of DbdTab-COF, RBT-COF, and SBT-COF.

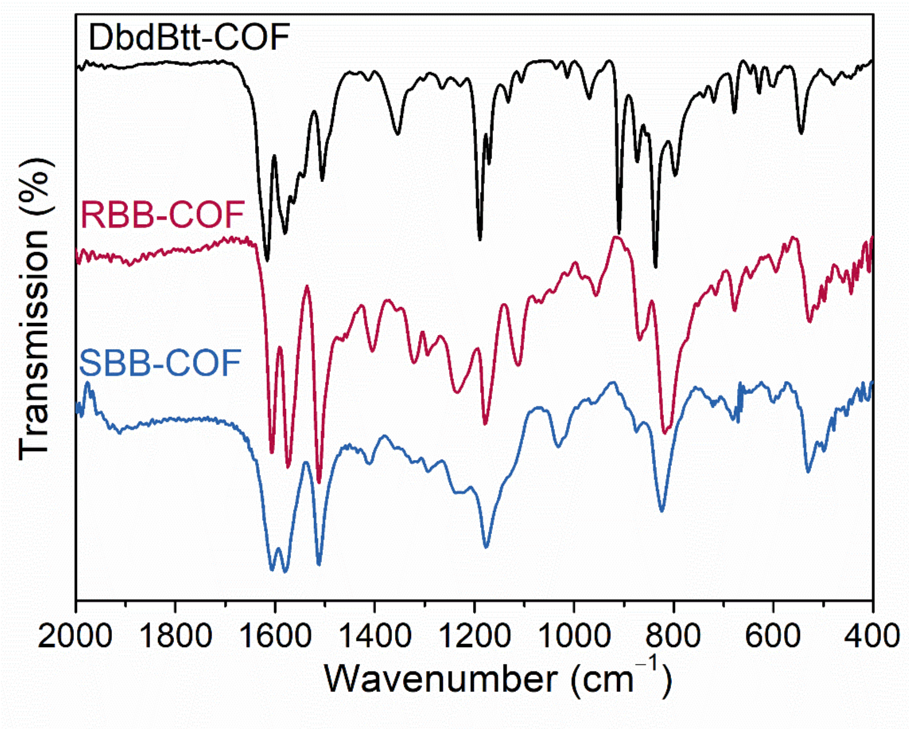


Fig. S41. The FTIR spectra of DbdBtt-COF, RBB-COF, and SBB-COF.

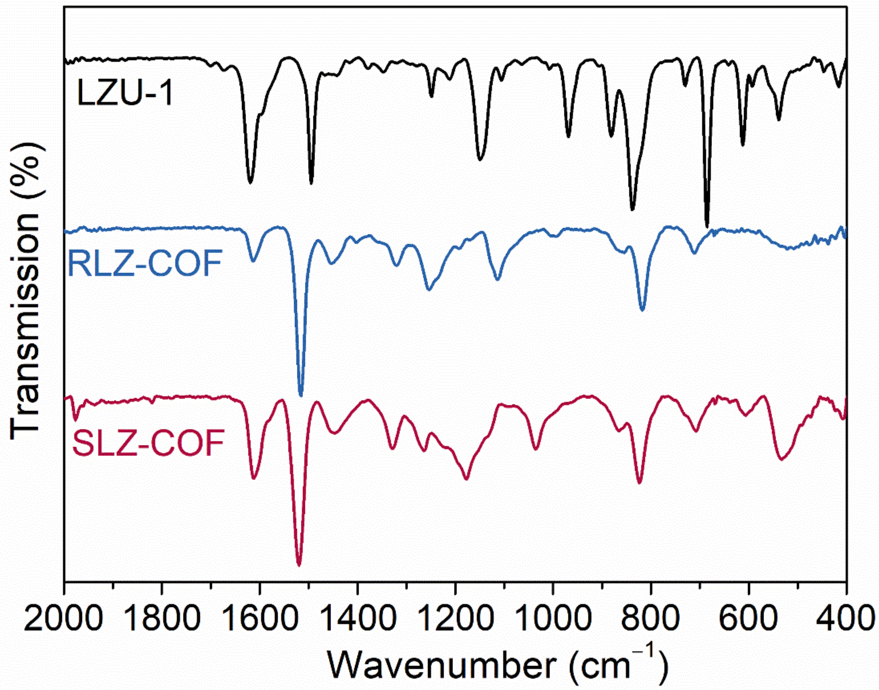


Fig. S42. The FTIR spectra of LZU-1, RLZ-COF, and SLZ-COF.

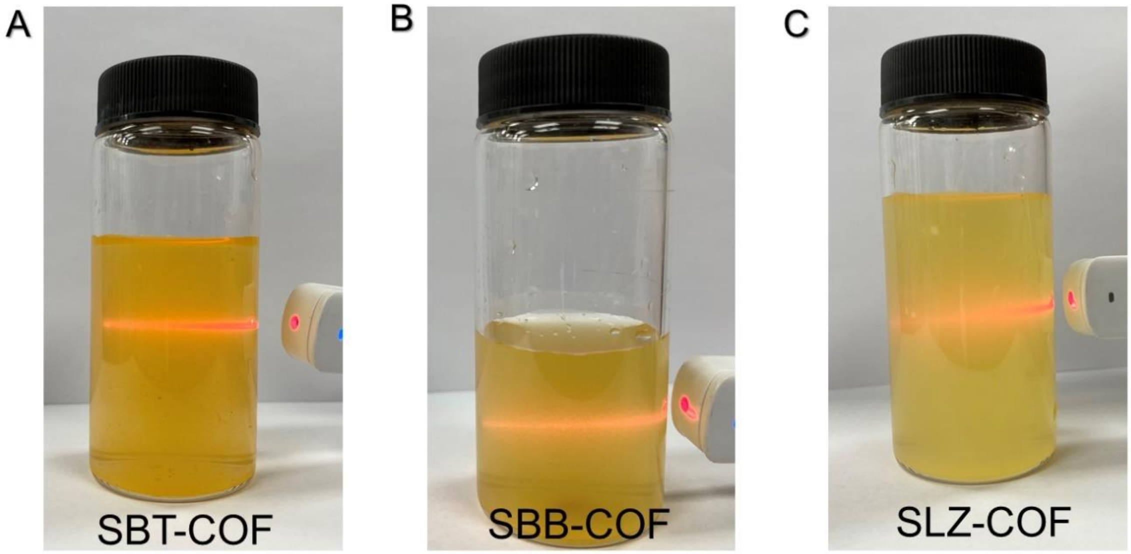


Fig. S43. The Tyndall phenomenon produced by (**A**) SBT-COF, (**B**) SBB-COF, and (**C**) SLZ-COF. The Tyndall phenomena were tested after the solution system was kept standing of 12 hours.

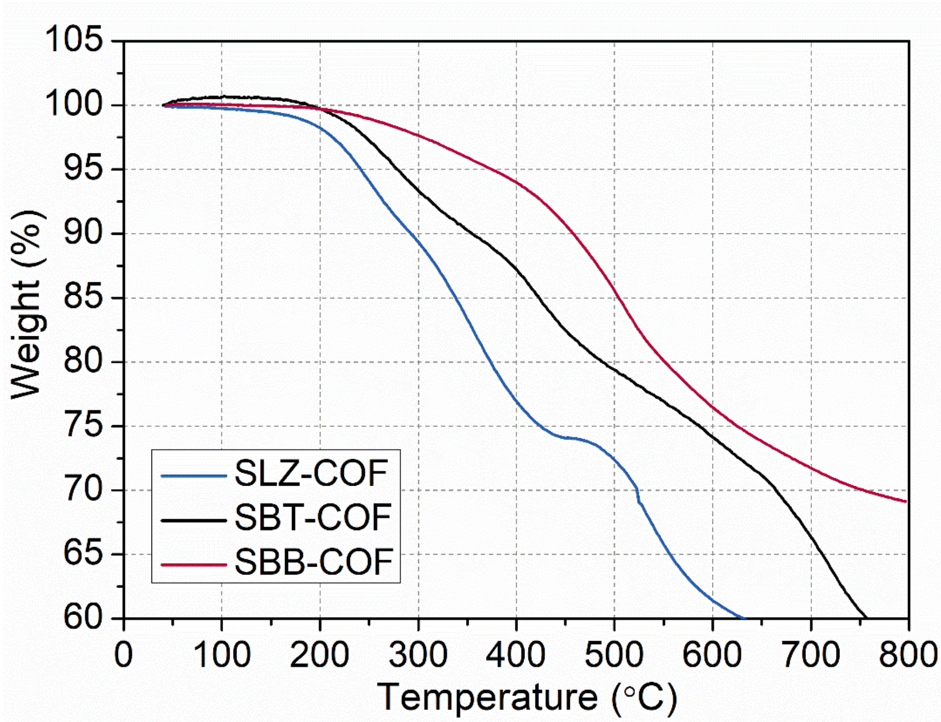


Fig. S44. TGA profiles of SLZ-COF (blue), SBT-COF (black), and SBB-COF (red).

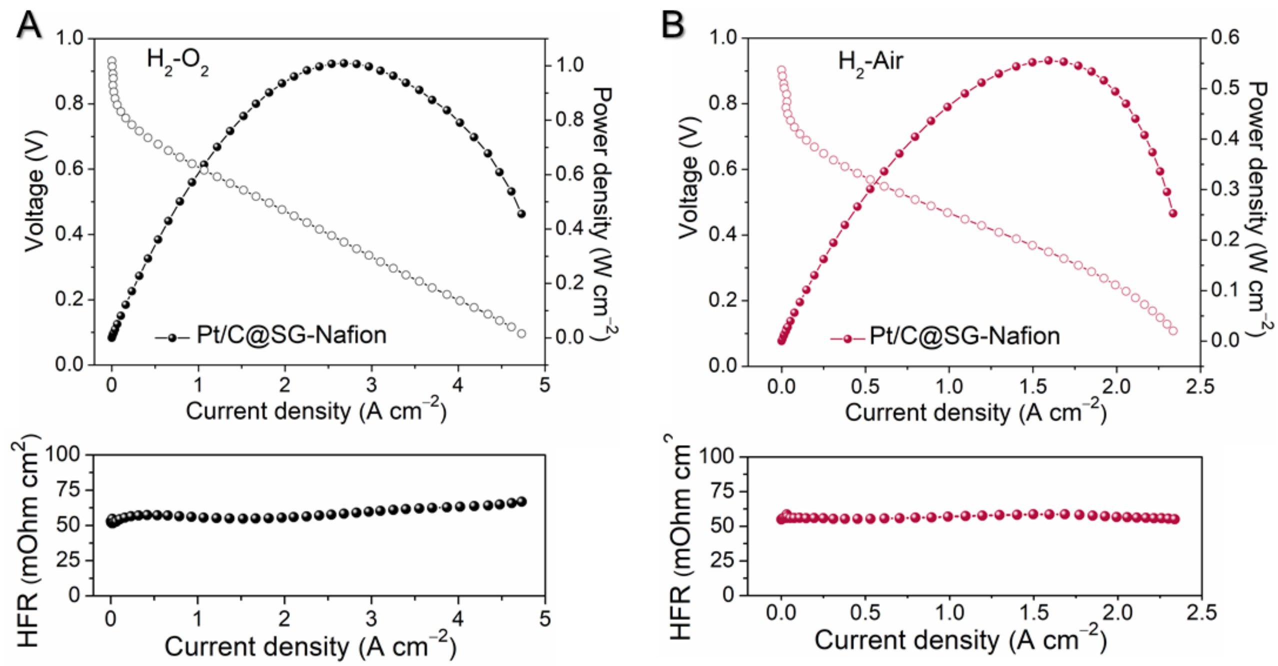


Fig. S45. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@SG-Nafion (mass ratio, Sulfonated Graphene: Nafion=1:1) measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

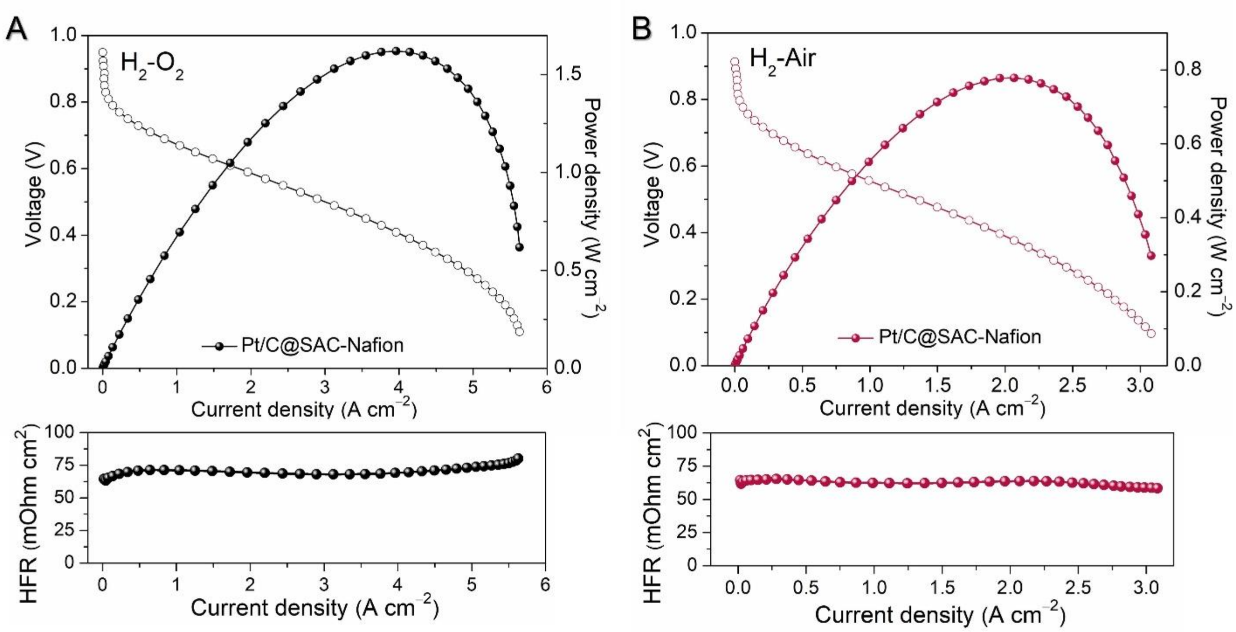


Fig. S46. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I-V* polarization (without *iR* compensation) and power density plots with Pt/C@SAC-Nafion (mass ratio, Sulfonated Activated Carbon: Nafion=1:1) measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

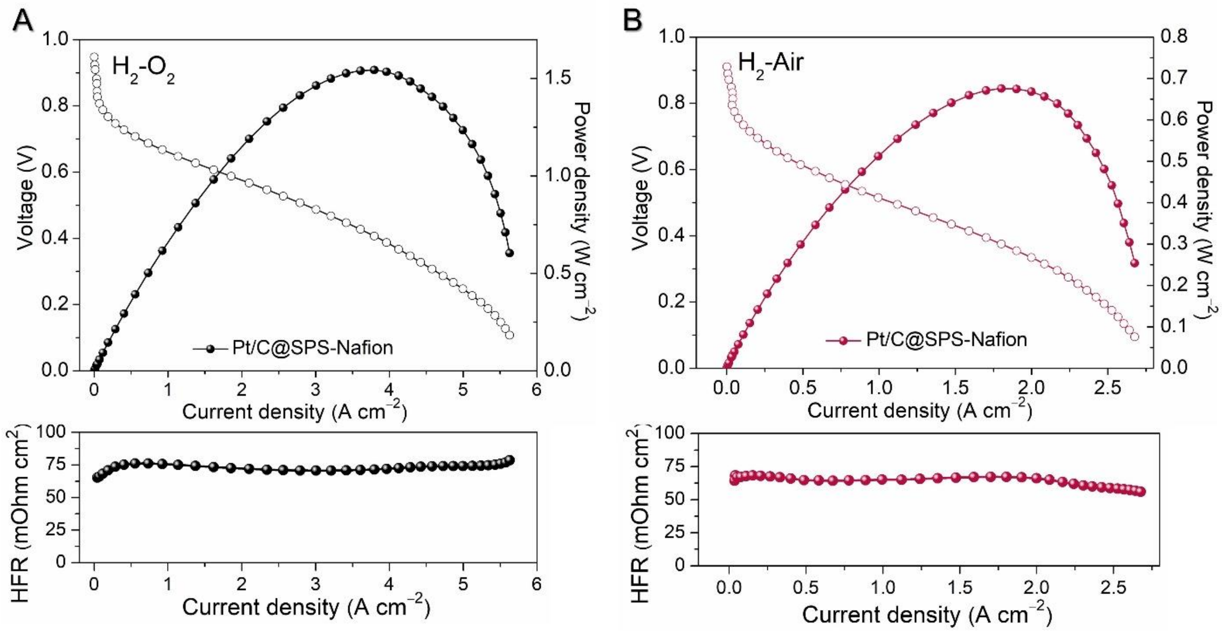


Fig. S47. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@SPS-Nafion (1:1) measured at 80 ºC and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

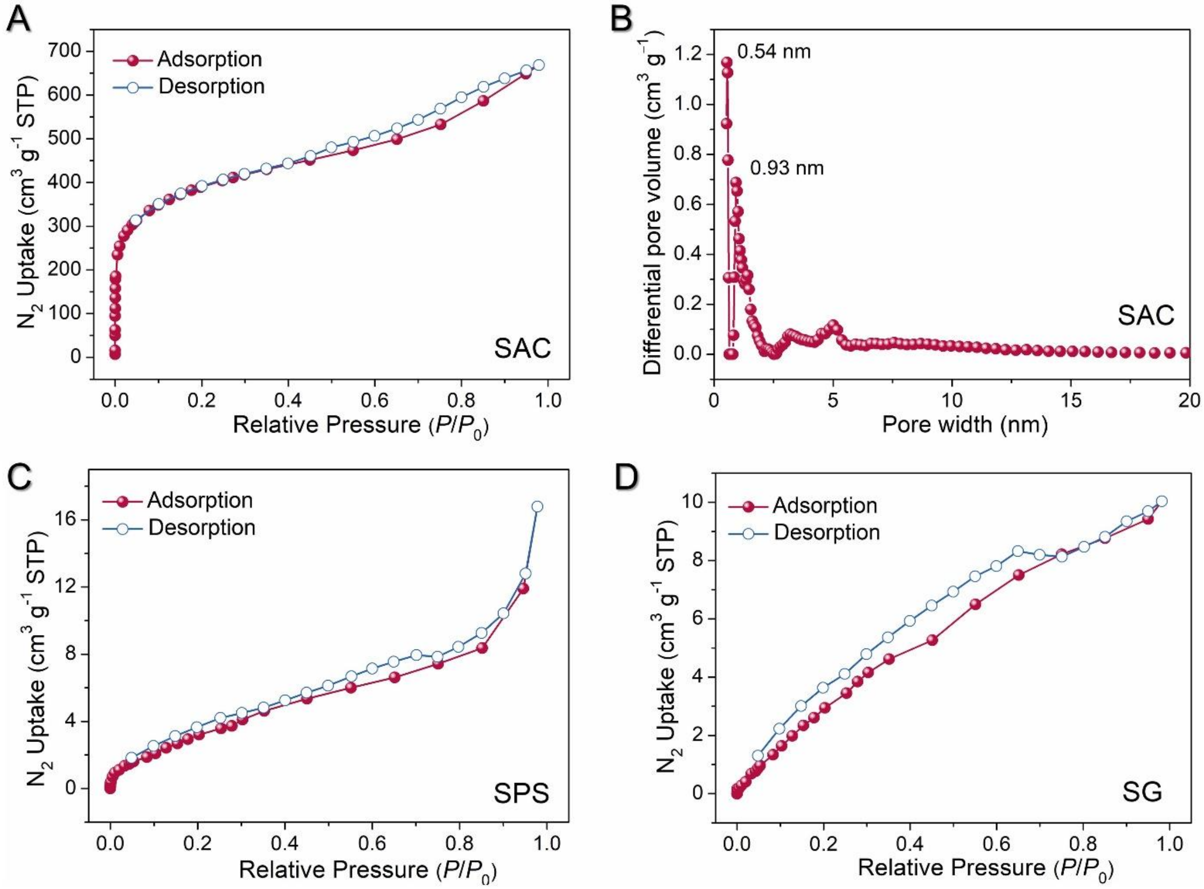


Fig. S48. (**A**) The nitrogen sorption isotherm profile of SAC measured at 77 K. The BET specific surface area of SAC is 1418 $m^2\ g^{-1}$. (**B**) The pore-size distribution profile of SAC. (**C**) The nitrogen sorption isotherm profile of SPS measured at 77 K. The BET specific surface area of SPS is 14 $m^2\ g^{-1}$. (**D**) The nitrogen sorption isotherm profile of SG measured at 77 K. The BET specific surface area of SG is 19 $m^2\ g^{-1}$.

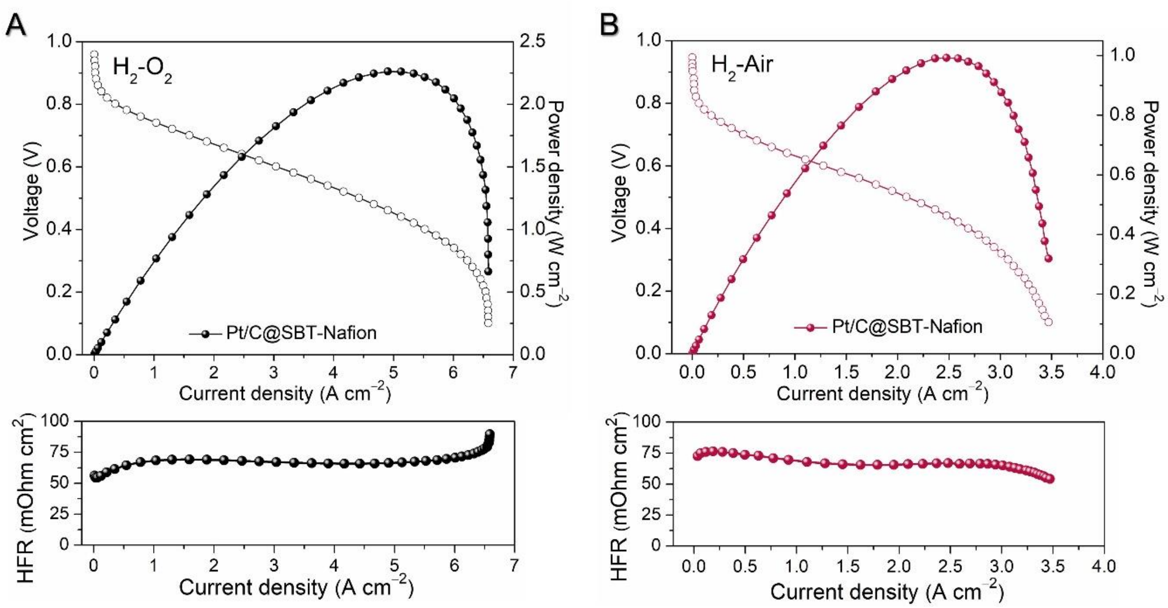


Fig. S49. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@SBT-Nafion measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

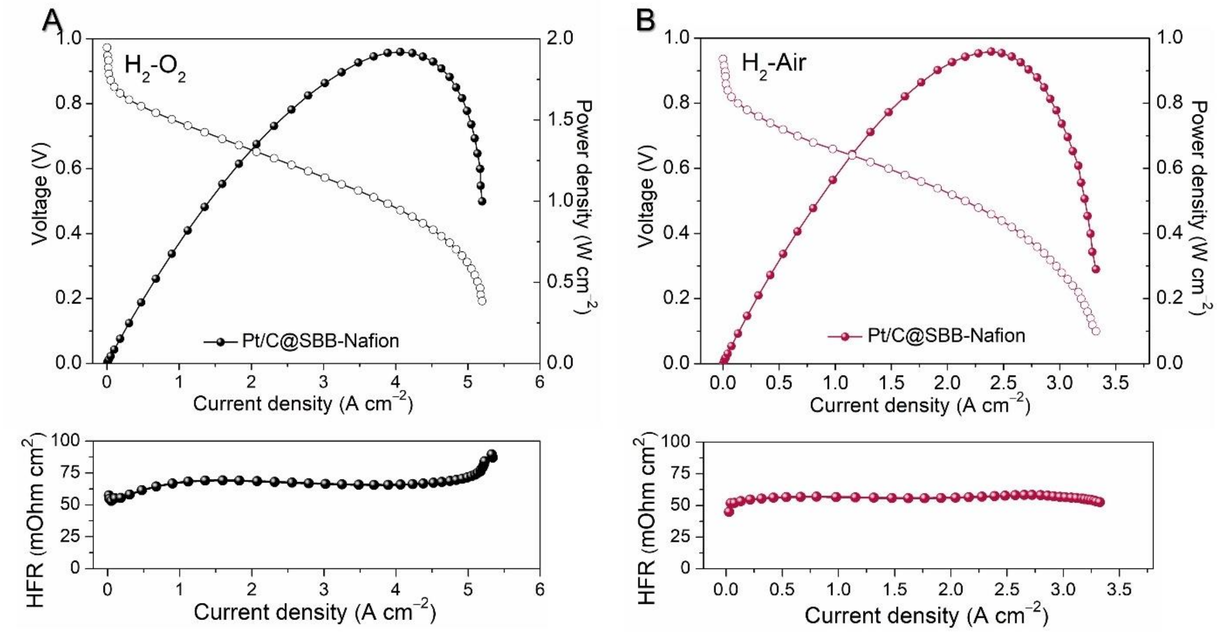


Fig. S50. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I-V* polarization (without *iR* compensation) and power density plots with Pt/C@SBB-Nafion measured at 80 °C and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

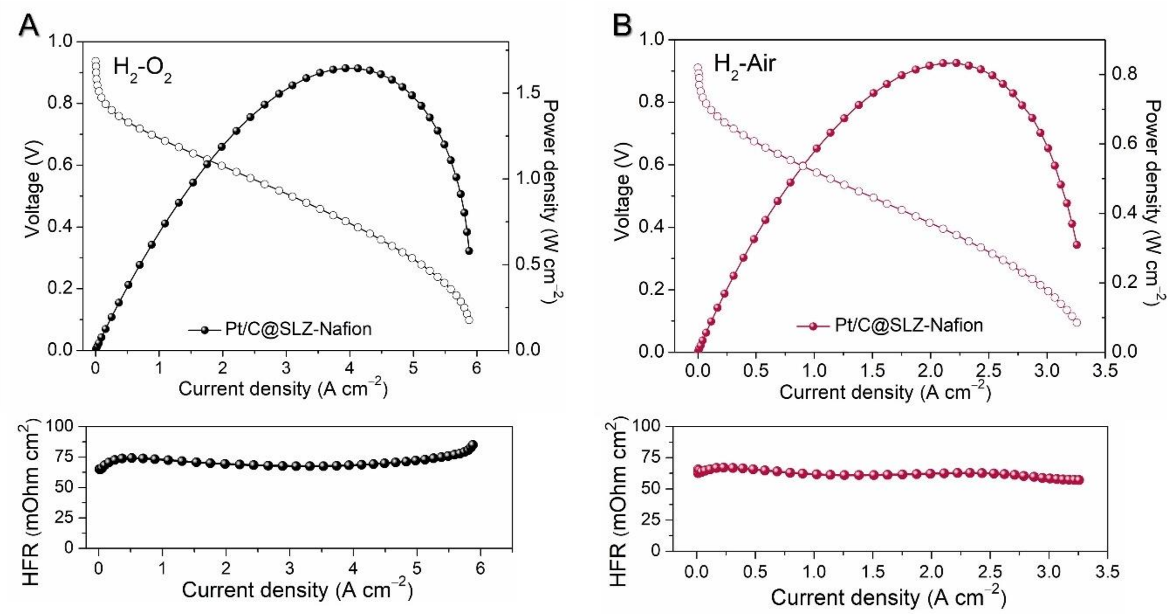


Fig. S51. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@SLZ-Nafion measured at 80 ^oC and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

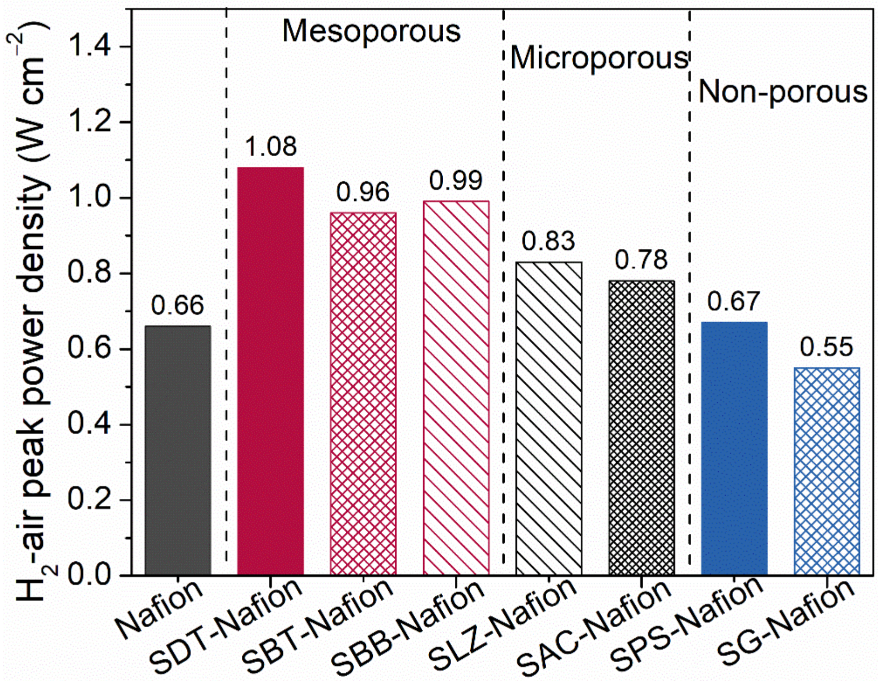


Fig. S52. Comparison of the peak power density of MEAs with different ionomers in CLs under H_2-air condition.

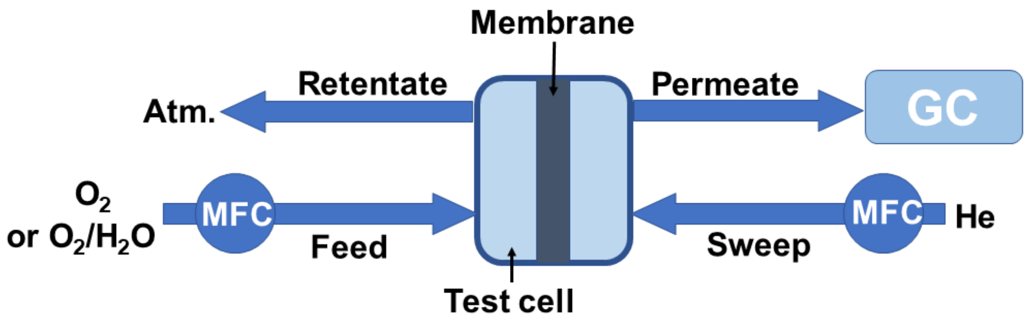


Fig. S53. Schematic view of the experimental setup according to Wicke Kallenbach used for oxygen permeation measurements for ionomer membranes.

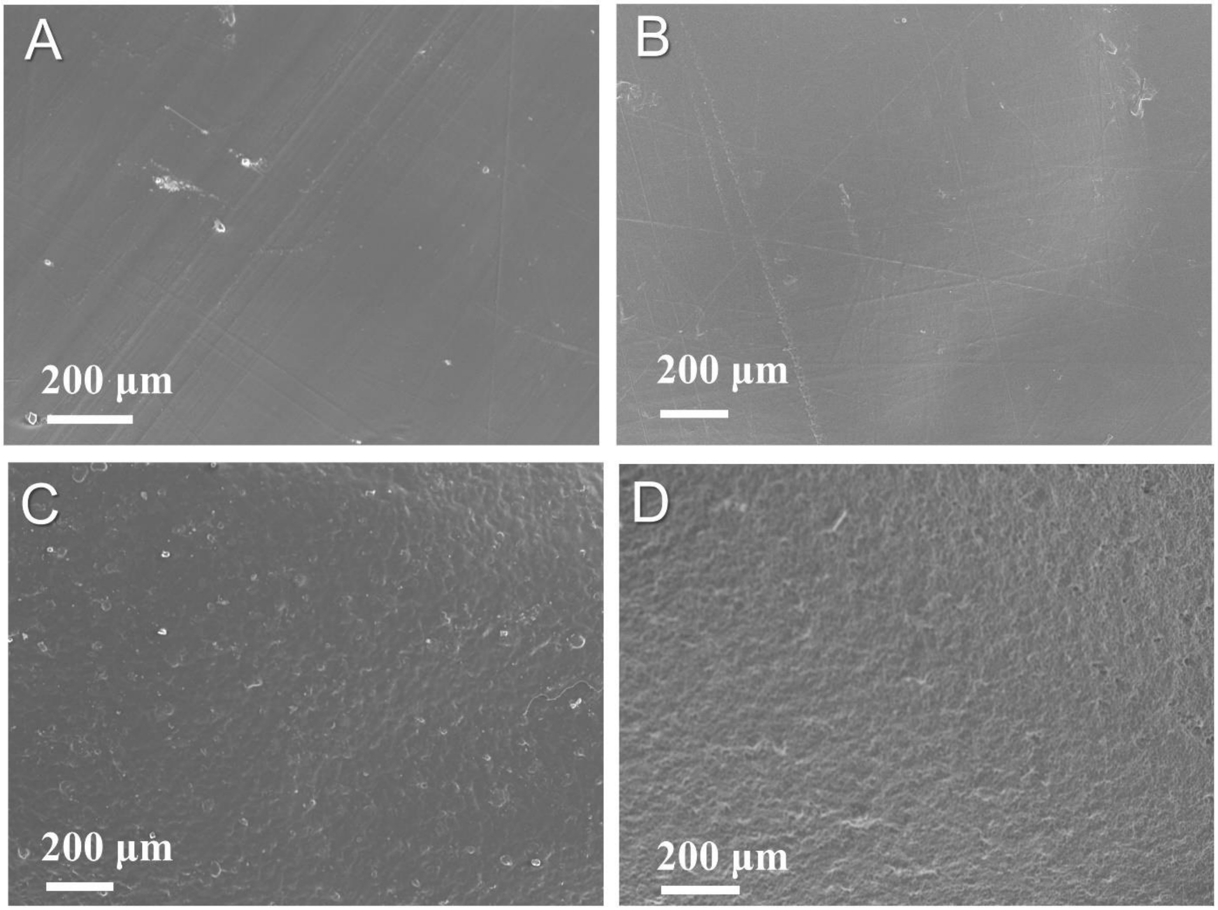
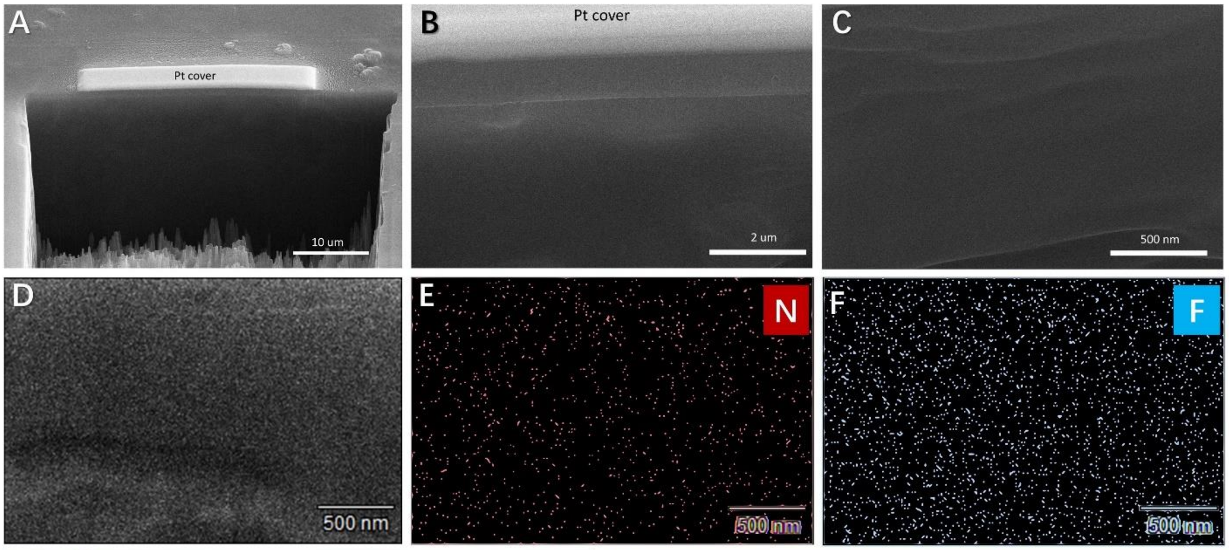
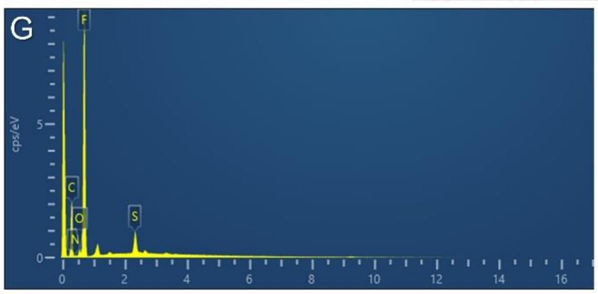


Fig. S54. (**A**) The SEM image of Nafion film for O_2 permeability test; (**B-D**) The SEM images of SDT-COF/Nafion films with the mixture ratios of 1:9 (B), 2:8 (C), and 3:7 (D), respectively.





H

Element	C	N	O	F	S	Total
wt%	33.17	0.88	1.29	59.16	5.51	100
wt% Sigma	0.29	0.20	0.09	0.30	0.11	
At%	44.61	1.02	1.30	50.30	2.77	100

Fig. S55. (**A**-**C**) The cross-section SEM images of SDT-COF/Nafion (2:8) film for oxygen permeability test. The protective platinum (Pt) was deposited on the sample prior to FIB milling. (**D**-**H**) EDS mapping of cross-section of SDT-COF/Nafion film and the element content analysis.

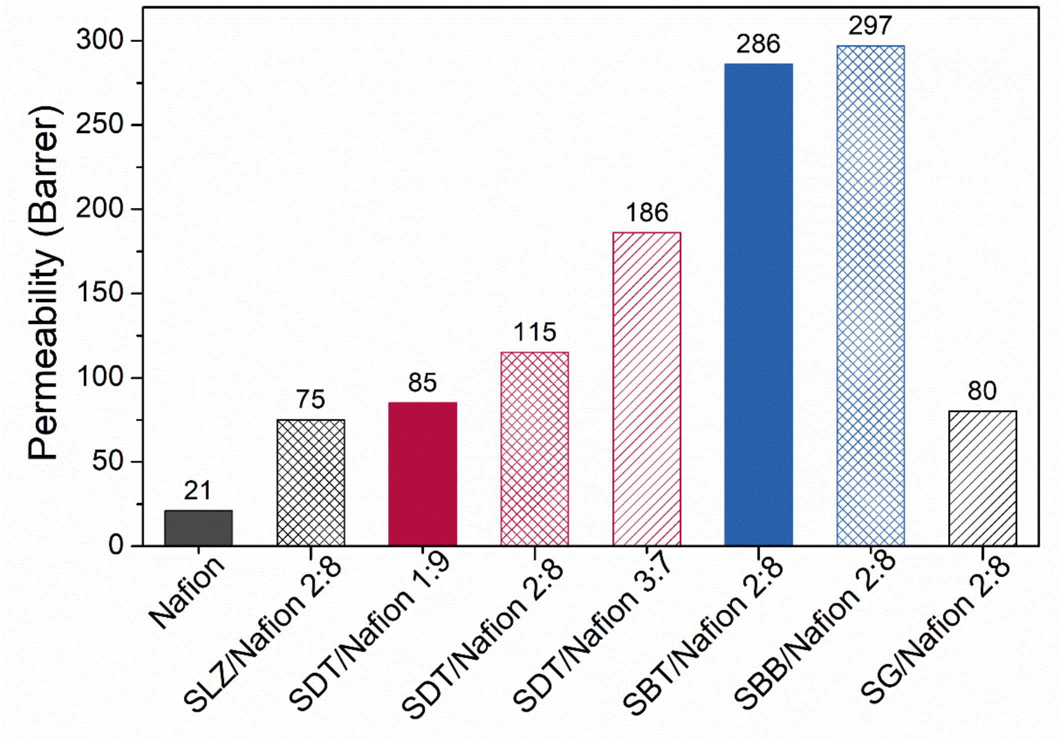


Fig. S56. The oxygen permeability comparison of different ionomer membranes.

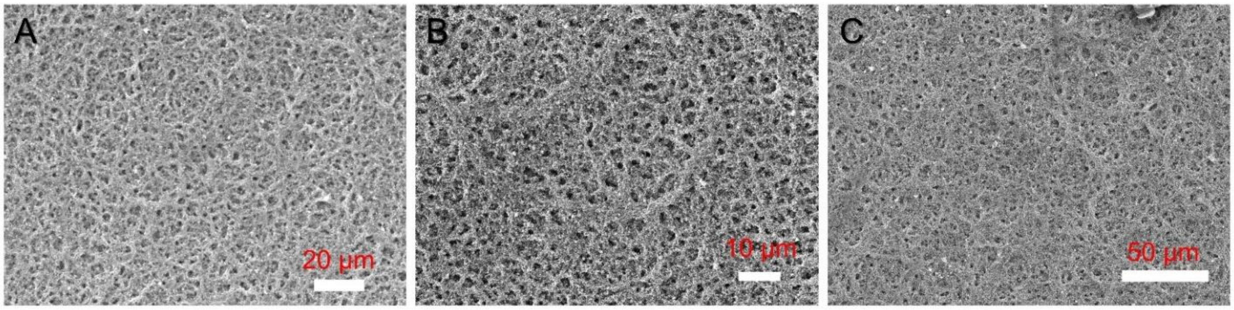


Fig. S57. The SEM images of (**A**) Pt/C@Nafion, (**B**) Pt/C@SDT-Nafion, and (**C**) Pt/C@SG-Nafion.

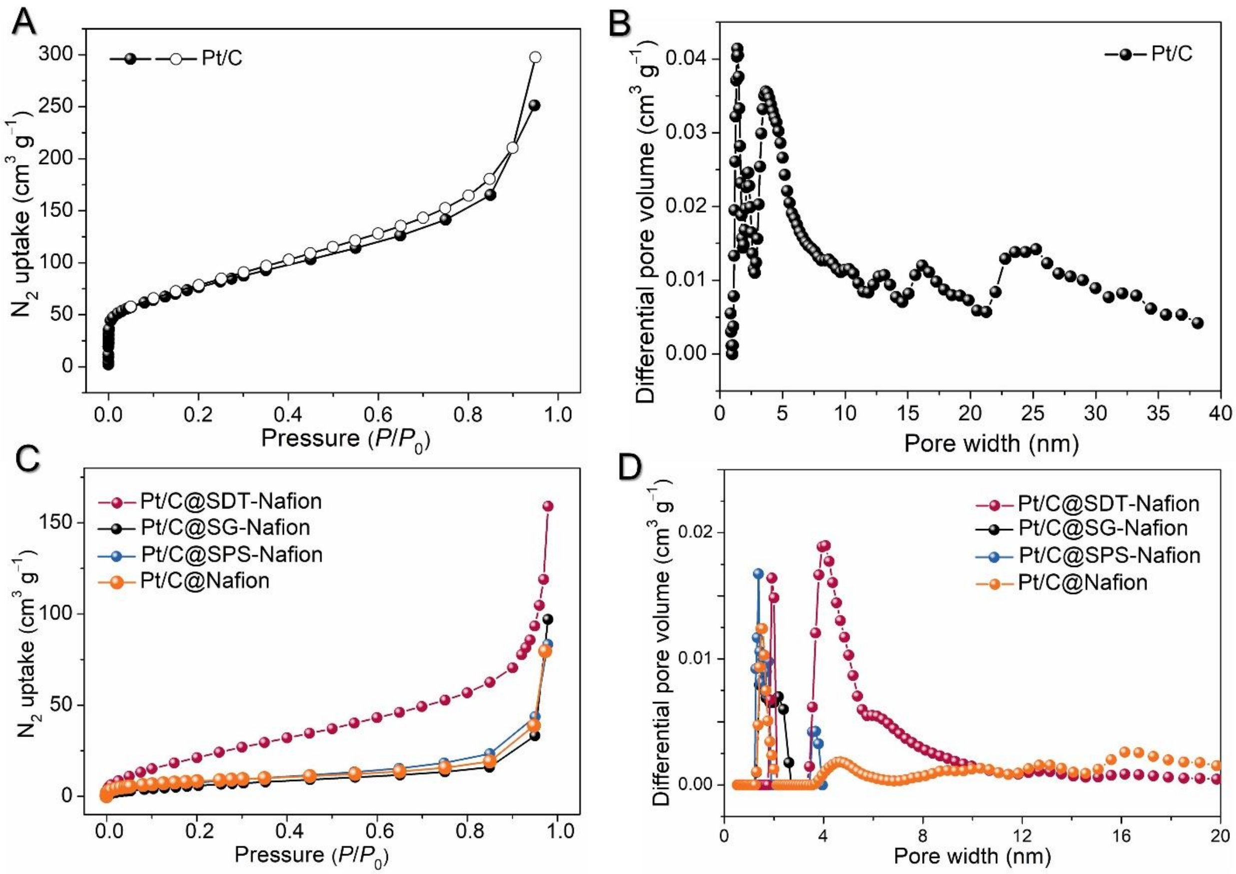


Fig. S58. (**A**) The nitrogen sorption isotherm profile of Pt/C powder measured at 77 K. (**B**) The pore-size distribution profile for Pt/C, and the BET specific surface area of Pt/C is 271 m^2 g^{-1}. (**C**) The nitrogen adsorption isotherm profiles of Pt/C@Nafion, Pt/C@SDT-Nafion, Pt/C@SG-Nafion, and Pt/C@SPS-Nafion powders measured at 77 K. (**D**) The pore-size distribution profiles for Pt/C@Nafion, Pt/C@SDT-Nafion, Pt/C@SG-Nafion, and Pt/C@SPS-Nafion, respectively. The lack of data in the pore width distribution range larger than 4 nm for Pt/C@SG-Nafion and Pt/C@SPS-Nafion is due to their low N_2 uptake. The BET specific surface areas are: Pt/C@Nafion: 31 m^2 g^{-1}; Pt/C@SDT-Nafion: 76 m^2 g^{-1}.

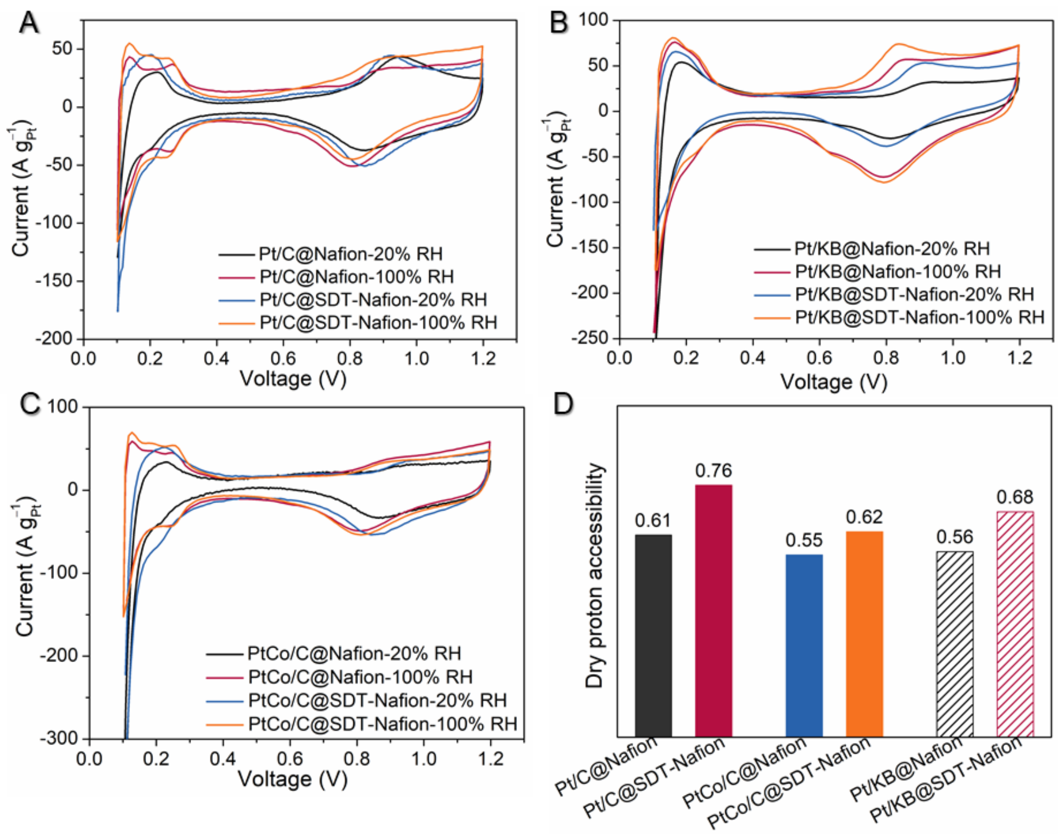


Fig. S59. The CV of the fuel cell with (**A**) Pt/C@Nafion & Pt/C@SDT-Nafion, (**B**) PtCo/C@Nafion & PtCo/C@SDT-Nafion, (**C**) Pt/KB@Nafion & Pt/KB@SDT-Nafion as CLs at 20% RH and 100% RH. The scans rate is 50 mV s^{-1}. (**D**) The comparison of dry proton accessibilities.

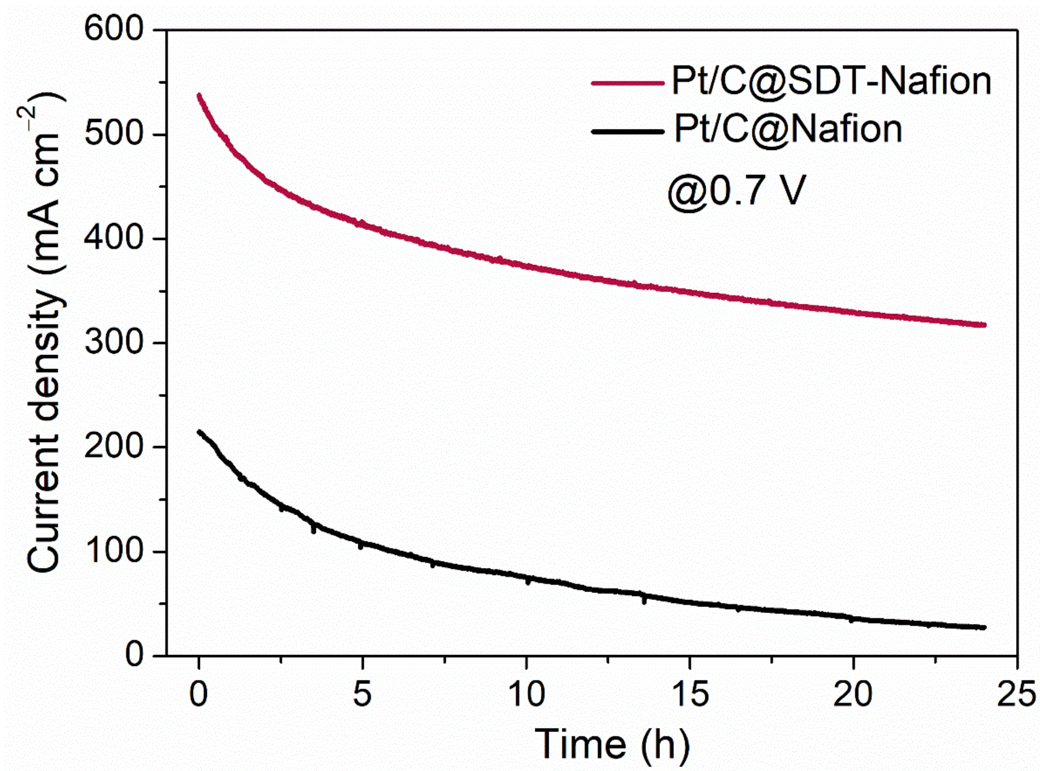


Fig. S60. The curve of current density to time for the 24-h continuous working tests for Pt/C@SDT-Nafion and Pt/C@Nafion without scheduled refreshing.

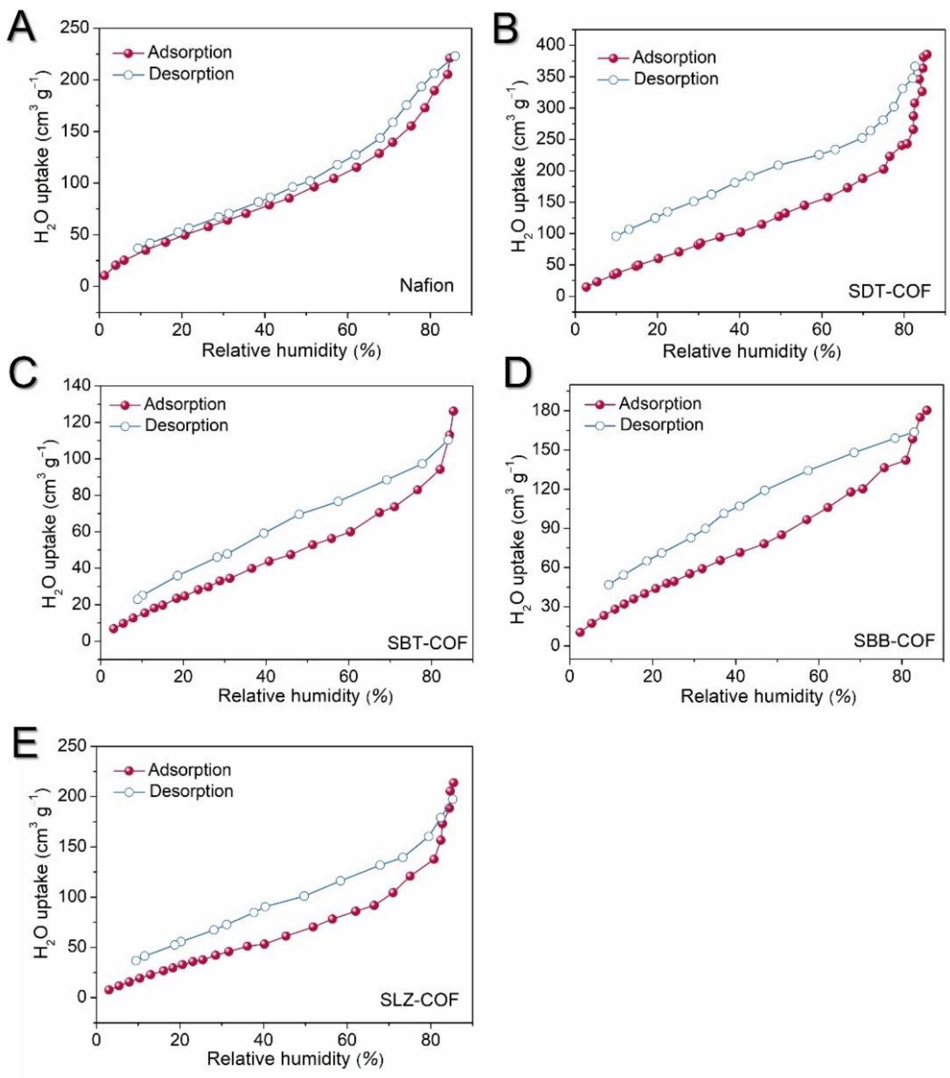


Fig. S61. Equilibrium water uptake as a function of RH in the process of water vapor adsorption and desorption at 298 K for (**A**) Nafion membrane, (**B**) SDT-COF, (**C**) SBT-COF, (**D**) SBB-COF, and (**E**) SLZ-COF in a static vapor sorption (BELSORP MAX Ⅱ Instrument). The water static uptake values of Nafion and SDT-COF are 213 and 385 $cm^3\ g^{-1}$ at 85% RH.

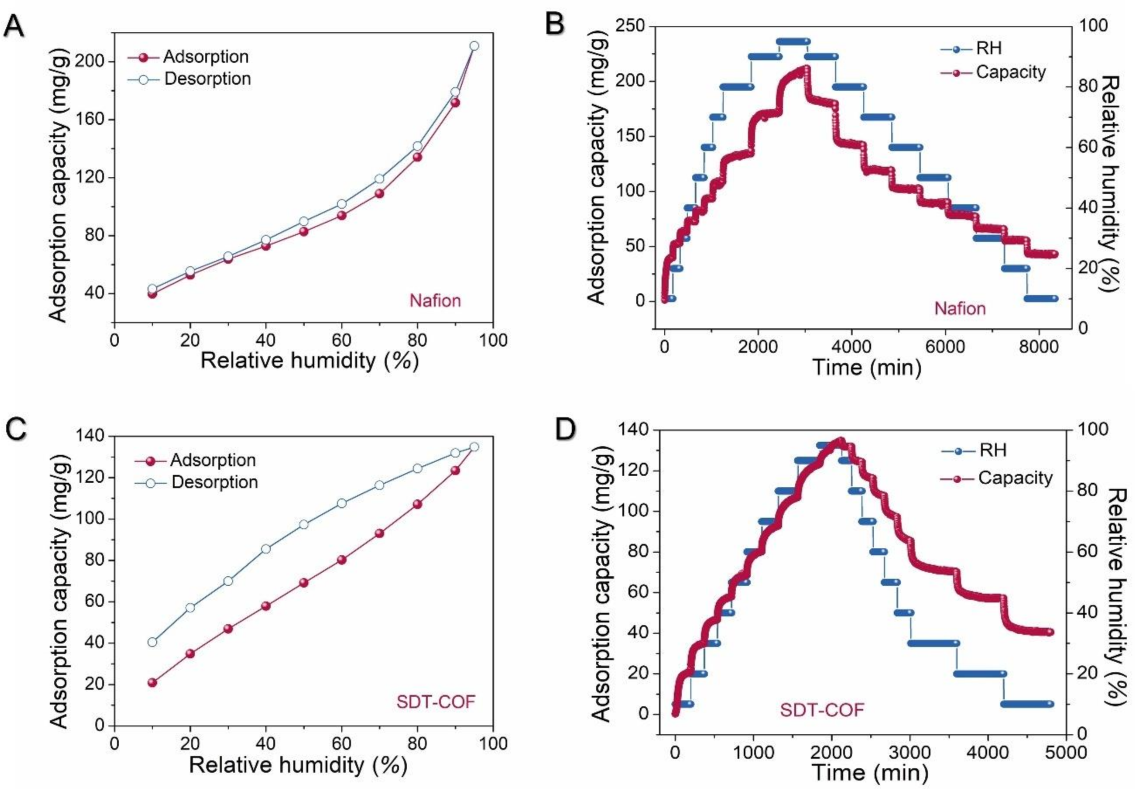


Fig. S62. Sorption kinetics curves at 298 K for (**A, B**) Nafion and (**C, D**) SDT-COF in a dynamic vapor sorption (3H-2000PW, Beishide Instrument Technology (Beijing) Co., Ltd). An ultrasensitive microbalance of resolution 0.1 μg is mounted in the thermostat heatsink with high precision temperature control. Each sample was degassed at 373 K and vacuum condition for more than 24 h. For dynamic vapor adsorption, the water uptake of SDT-COF is lower than that of Nafion. Compared with Nafion, the water desorption of SDT-COF exhibits obvious hysteresis phenomenon.

The dynamic water sorption ability of SDT-COF is much lower than its equilibrium water uptake. It demonstrates that only partial of the channel would be occupied by water molecules in a dynamic process and the oxygen gas should be able to penetrate the rest space.

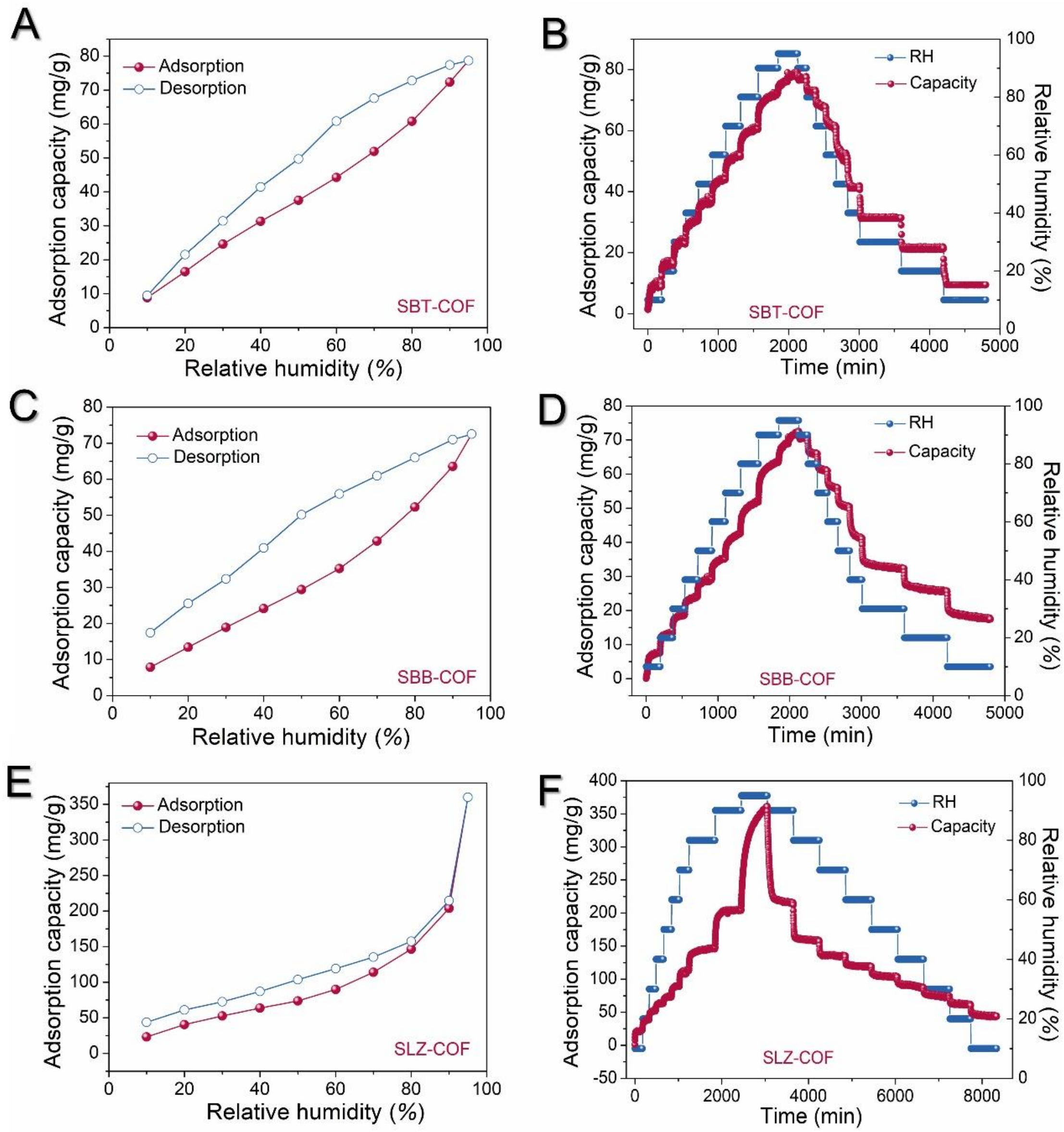


Fig. S63. Isotherm plots and sorption kinetics curves at 298 K for (**A, B**) SBT-COF, (**C, D**) SBB-COF, and (**E, F**) SLZ-COF in dynamic vapor sorption.

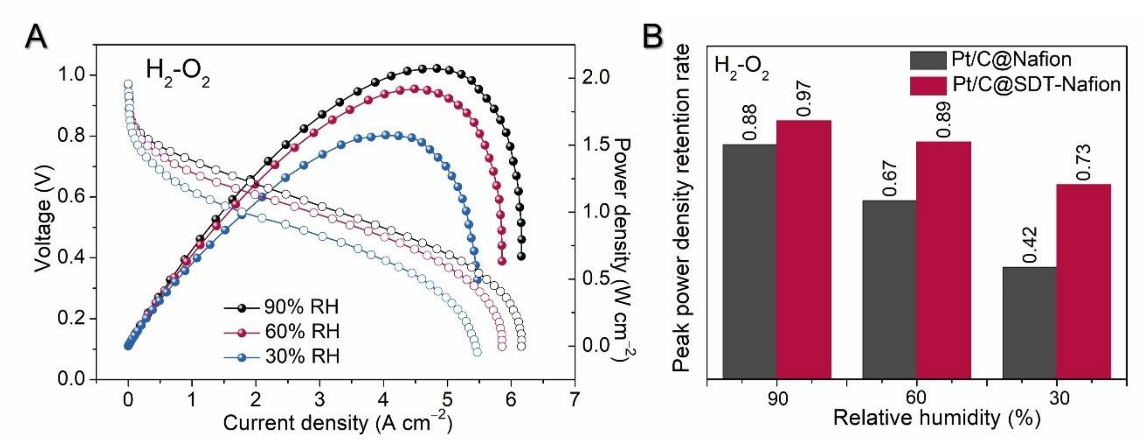


Fig. S64. (**A**) H_2-O_2 fuel cell *I-V* polarization (without *iR* compensation) and power density plots of the cells with Pt/C@Nafion and Pt/C@SDT-Nafion at 90% RH, 60% RH and 30% RH. (**B**) Comparing the peak power density retention rate at 90% RH, 60% RH and 30% RH of the cells with Pt/C@Nafion and Pt/C@SDT-Nafion.

The peak power density at 100% RH of 1.6 W cm^{-2} of the cells with Pt/C@Nafion and 2.2 W cm^{-2} of the cells with Pt/C@SDT-Nafion is set to 1. Along with the decreasing of the humidity, the decreasing degree of the cell performance significantly reduces after the addition of SDT-COF.

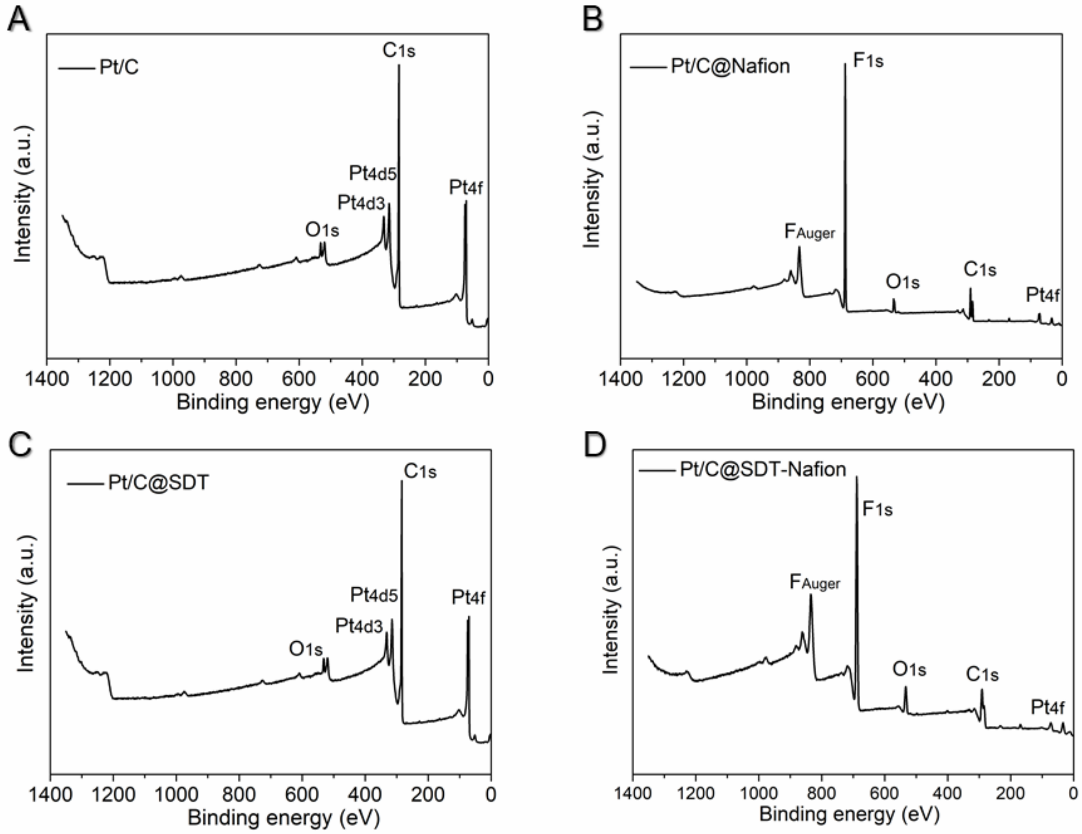


Fig. S65. Comparison of XPS spectra for CLs. (**A**) Pt/C without ionomer, (**B**) Pt/C@Nafion, (**C**) Pt/C@SDT, and (**D**) Pt/C@SDT-Nafion.

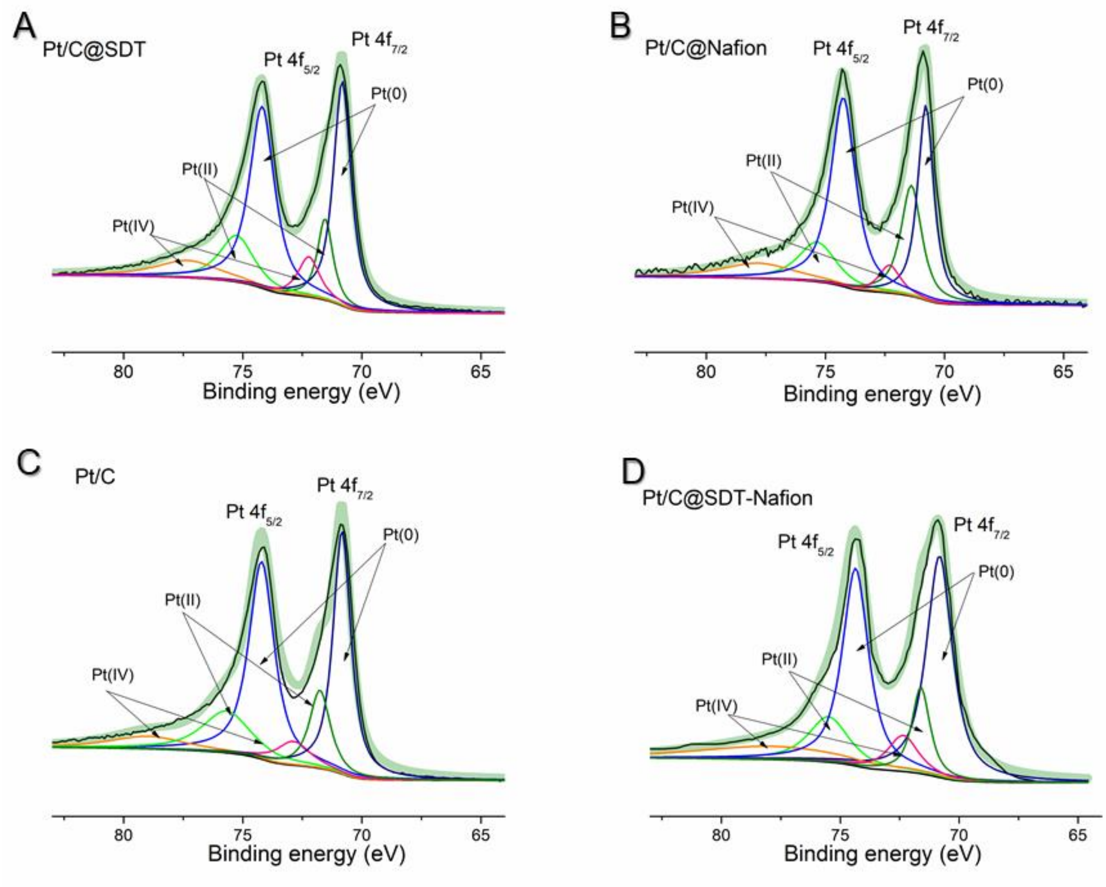


Fig. S66. Pt spectra in XPS analyses. (**A**) Pt/C without ionomer, (**B**) Pt/C@Nafion, (**C**) Pt/C@SDT, and (**D**) Pt/C@SDT-Nafion.

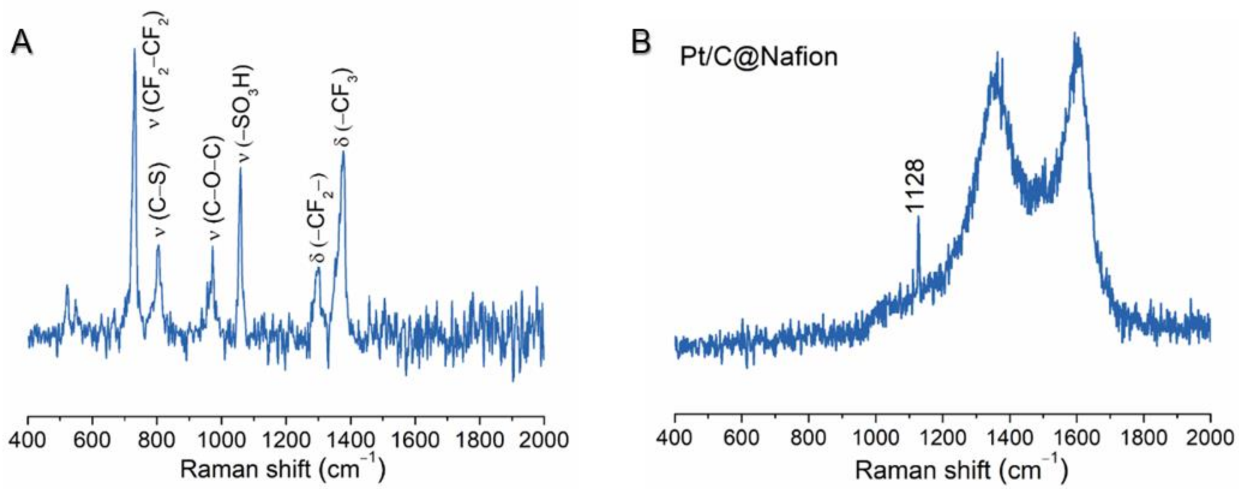


Fig. S67. (**A**) Raman spectrum of Nafion; (**B**) Raman spectrum of Pt/C@Nafion. The sharp band in the 1128 cm^{-1} represents the bending vibration, δ(Pt-SO_3). The broadband at 1300-1400 cm^{-1} range represents the D-band of carbon, while that at 1550-1650 cm^{-1} range attribute to the G-band of carbon.

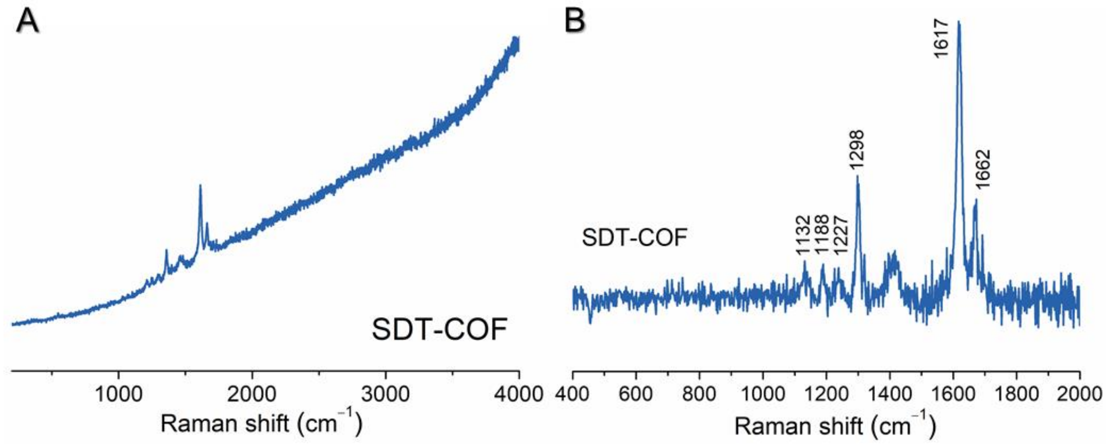


Fig. S68. Raman spectrum of SDT-COF. (**A**) SDT-COF has a noticeable fluorescence effect, and (**B**) the right chart is the baseline correction result.

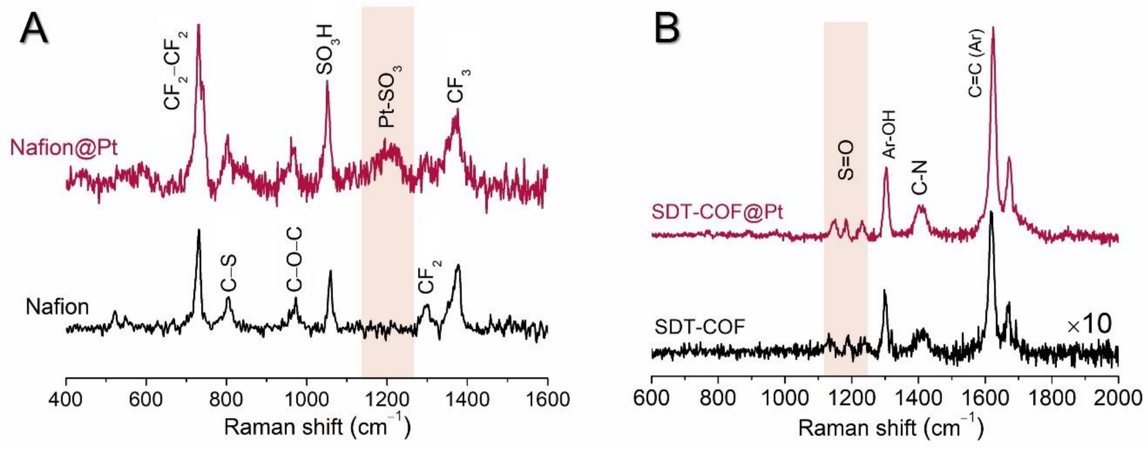


Fig. S69. Comparison of Raman spectra of (**A**) Nafion and (**B**) SDT-COF before and after mixing with nano-platinum particles. When SDT-COF is physically mixed with nano-platinum particles, a surface enhancement effect occurs. The appearance of the broad peak rather than sharp peak for δ(Pt-SO_3) is probably due to the complex interactions of Pt particles with sulfonic acid groups.

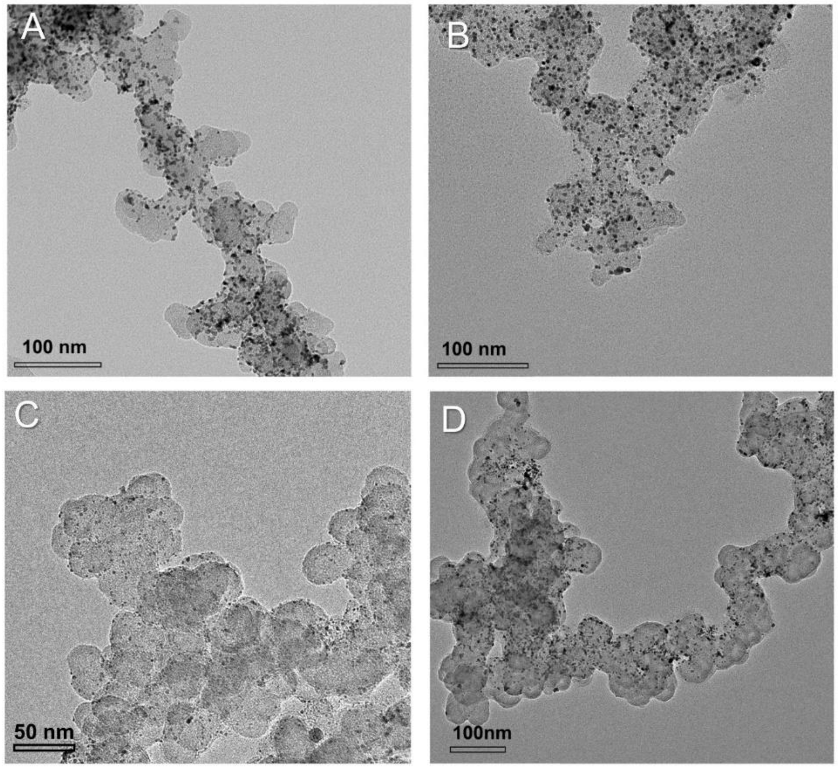


Fig. S70. (**A**) The TEM image of Pt/C without ionomer. (**B**) The TEM image of Pt/C@Nafion. (**C**-**D**) The TEM images of Pt/C@SDT-Nafion.

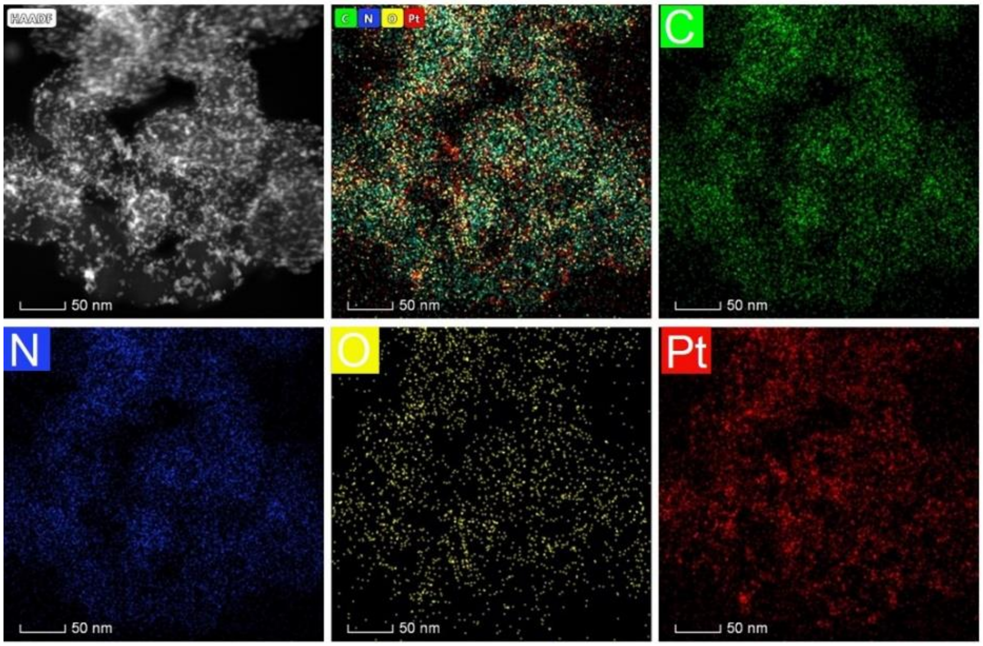


Fig. S71. EDS mapping of Pt/C@SDT-Nafion (1).

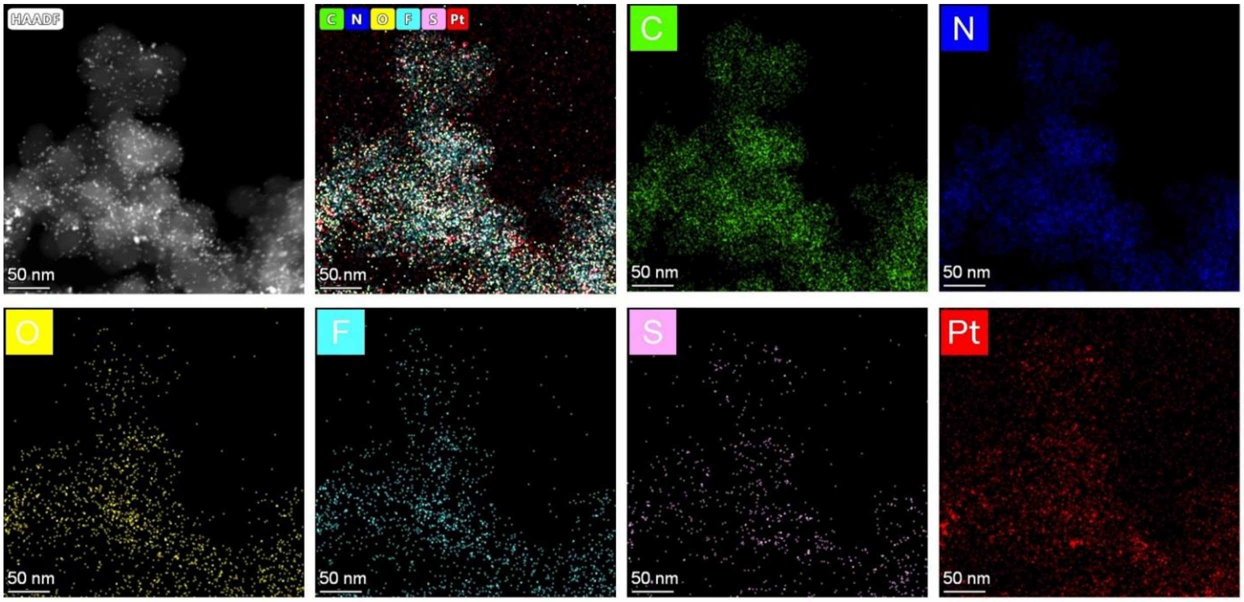


Fig. S72. EDS mapping of Pt/C@SDT-Nafion (2).

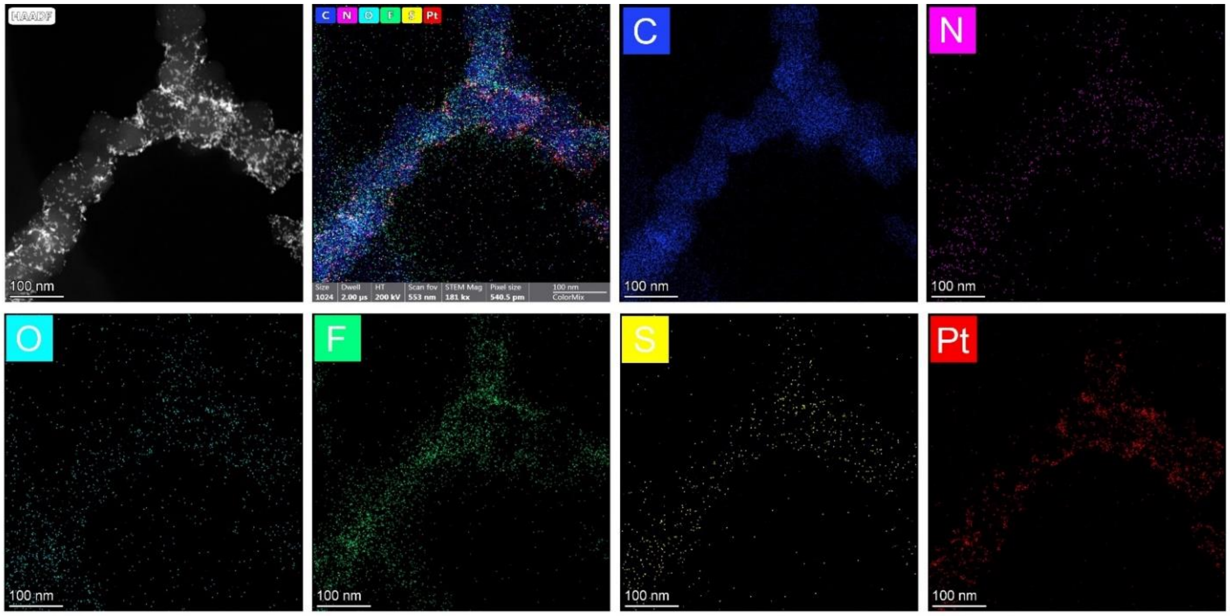


Fig. S73. EDS mapping of Pt/C@SDT-Nafion (3).

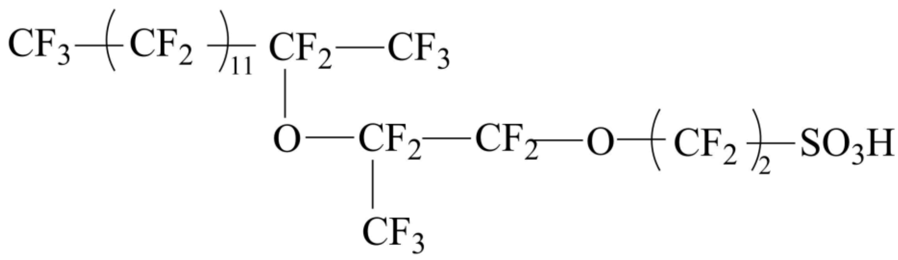


Fig. S74. Chemical structure of the simplified Nafion segment for molecular dynamics simulations.

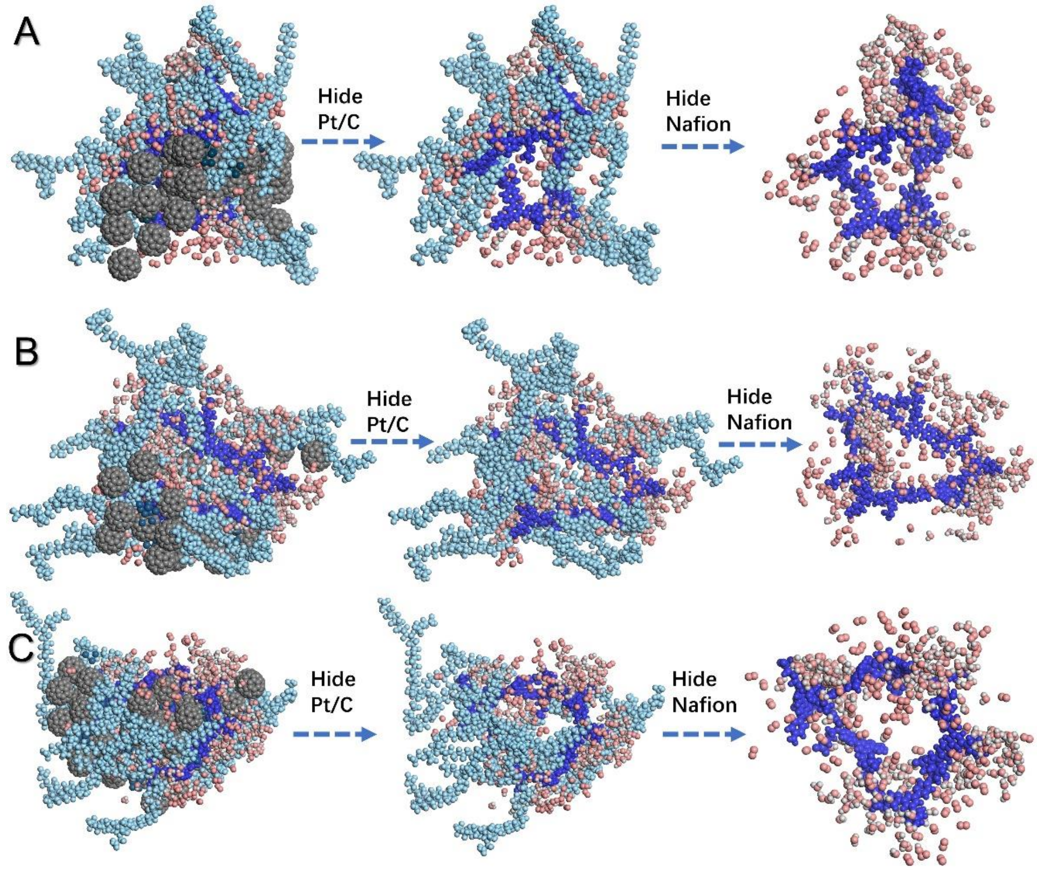


Fig. S75. (**A**-**C**) Snapshots of the molecular dynamics simulation of multi-component CL with the combination of SDT-COF and Nafion. Light blue, Nafion; bright blue, SDT-COF; indigo blue, Pt; pink, O; white, H; gray, C.

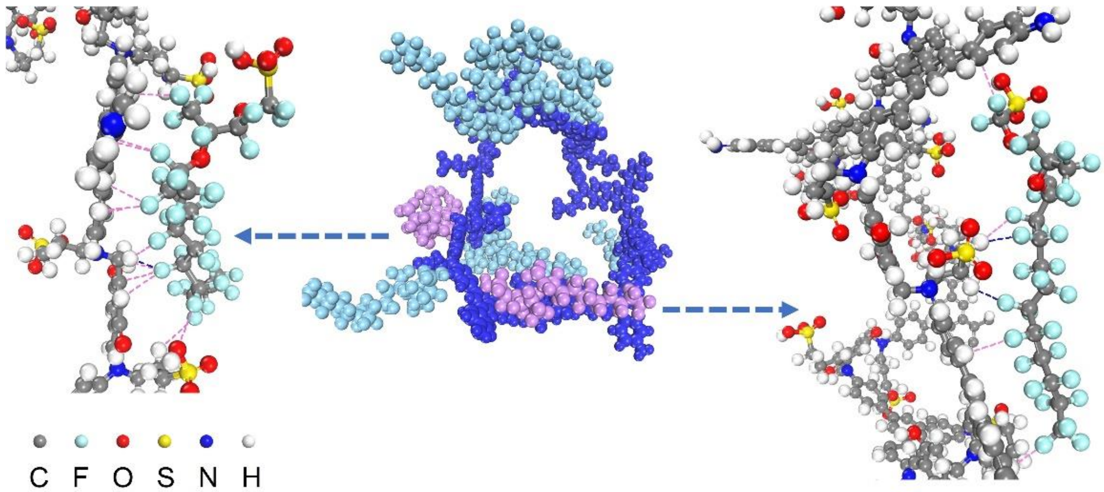


Fig. S76. Representation of the hydrogen bond and van der Waals force between SDT-COF and Nafion taken from MD simulation. Blue dashed lines refer to forces within the distance of 2.5 Å, and purple dashed lines refer to forces within the distance of 3 Å. In the middle figure, bright blue represents SDT-COF, light blue represents Nafion, and purple represents the selected Nafion.

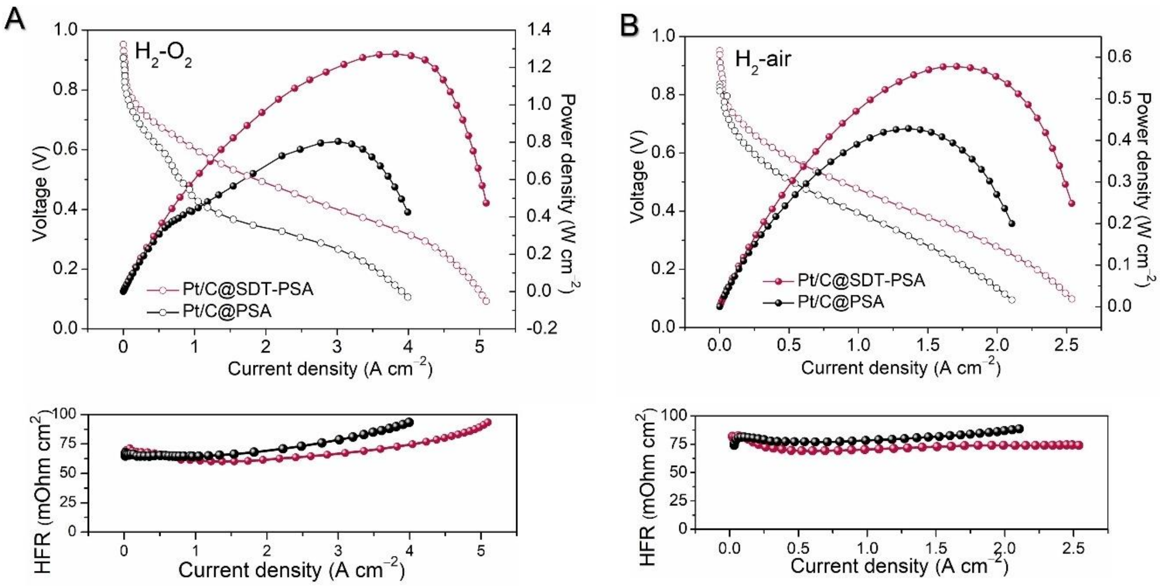


Fig. S77. (**A**) H_2-O_2 and (**B**) H_2-air fuel cell *I*-*V* polarization (without *iR* compensation) and power density plots with Pt/C@SPA and Pt/C@SDT-PSA (mass ratio, SDT-COF: PSA=1:1) measured at 80 ºC and 100% RH under 50 kPa back pressure. Cathode, ~0.07 mg_{Pt} cm^{-2}; anode, ~0.05 mg_{Pt} cm^{-2}; Nafion N212; 300 mL H_2 min^{-1} and 300 mL O_2 min^{-1} feed for H_2-O_2 test; 500 mL H_2 min^{-1} and 1,500 mL air min^{-1} feed for H_2-air test, electrode area 5 cm^2. The below is HFR.

Table S1. Peak assignments of FTIR spectra for SDT-COF, RDT-COF, and DhaTab-COF.

Peak position	**Definition**
1187 cm^{-1}	Symmetrical stretching vibration of S=O
1034 cm^{-1}	Anti-symmetric stretching vibration S=O
565 cm^{-1}	Stretching vibration of S-O
1666 cm^{-1}	Stretching vibration of C=O
1282 cm^{-1}	C-N of aromatic tertiary amines
1609 cm^{-1}	Stretching vibration of C-N
1615 cm^{-1}	Stretching vibration of C=N
1588 cm^{-1}	Stretching vibration of C=C
820 cm^{-1}	Out-of-plane bending vibration of C-H in -Ar

Table S2. Comparison of the performance of adding SDT-COF or not in the Pt/C, PtCo/C, and Pt/KB catalytic layer.

	Pt/C@ Nafion	**Pt/C@ SDT-Nafion**	**PtCo/C@ Nafion**	**PtCo/C@ SDT-Nafion**	**Pt/KB@ Nafion**	**Pt/KB@ SDT-Nafion**	**DOE 2020 target** *(40)*	**DOE 2025 target** *(32)*
ECSA ($m^2\ g_{Pt}^{-1}$)	41.4	67.4	59.8	77.4	74.3	84.5	-	-
MA@0.9 V ($A\ mg_{Pt}^{-1}$)	0.21	0.35	0.37	0.50	0.43	0.65	0.44	0.44
Rated power density (H_2-air) ($mW\ cm^{-2}$)	152@0.7 V 397@0.6 V	373@0.7 V 750@0.6 V	235@0.7 V 517@0.6 V	482@0.7 V 799@0.6 V	297@0.7 V 599@0.6 V	332@0.7 V 621@0.6 V	1,000	1,800
Peak power density (H_2-O_2) ($W\ cm^{-2}$)	1.61	2.22	1.75	1.88	1.69	2.14	-	-
Peak power density (H_2-air) ($W\ cm^{-2}$)	0.66	1.08	0.74	0.94	0.81	1.05	-	-
PGM total content （g/kW rated）	0.79@0.7 V 0.30@0.6 V	0.32@0.7 V 0.16@0.6 V	0.51@0.7 V 0.23@0.6 V	0.25@0.7 V 0.15@0.6 V	0.40@0.7 V 0.21@0.6 V	0.36@0.7 V 0.19@0.6 V	≤0.125	≤0.1

Table S3. Comparing the cathode Pt-loading, peak power density and specific parameters for MEAs in H_2-air cell based on Pt/C catalyst in this study and literatures.

Ionomer	**Cathode Pt-load (mg_{Pt} cm^{-2})**	**Peak power density (W cm^{-2})**	**Active area of MEA (cm^2)**	**Cathode flow (mL min^{-1})**	**Anode flow (mL min^{-1})**	**Pressure (kPa abs)**	**Ref.**
BPS-27k	0.4	0.69	-	1,200	400	-	*(42)*
BPS-51k	0.4	0.57	-	1,200	400	-	*(42)*
BPS-12k	0.4	0.28	-	1,200	400	-	*(42)*
SSC-1	0.4	1.20	45	9,000	4,450	135	*(43)*
SSC-2	0.4	1.10	45	9,000	4,450	135	*(43)*
Nafion	0.4	0.95	45	9,000	4,450	135	*(43)*
Aquivion	0.3	0.72	1	75	25	150	*(13)*
Nafion	0.3	0.62	1	75	25	150	*(13)*
D50	0.3	0.81	25	stoichiometry 2.0	stoichiometry 1.2	100	*(14)*
D30	0.3	0.79	25	stoichiometry 2.0	stoichiometry 1.2	100	*(14)*
D70	0.3	0.76	25	stoichiometry 2.0	stoichiometry 1.2	100	*(14)*
D100	0.3	0.73	25	stoichiometry 2.0	stoichiometry 1.2	100	*(14)*

PTFE-Nafion	0.3	0.67	5	200	50	200	*(17)*
Nafion	0.3	0.50	5	200	50	200	*(17)*
SDT-Nafion	0.3	1.19	5	1,500	500	150	This work
Nafion	0.3	0.94	5	1,500	500	150	This work
ND ionomer	0.2	0.69	$<$12.25	200	150	-	*(44)*
D521	0.2	0.65	$<$12.25	200	150	-	*(44)*
SDT-Nafion	0.2	1.23	5	1,500	500	150	This work
Nafion	0.2	0.93	5	1,500	500	150	This work
HOPI	0.12	1.14	1	2,000	500	100	*(45)*
Nafion	0.12	1.10	1	2,000	500	100	*(45)*
D72	0.1	0.92	5	800	200	100	*(15)*
D79	0.1	0.90	5	800	200	100	*(15)*
Nafion	0.1	0.77	5	800	200	100	*(15)*
e-CL	0.1	0.76	25	stoichiometry 2.0	stoichiometry 1.5	180	*(46)*
Nafion	0.1	0.60	25	stoichiometry 2.0	stoichiometry 1.5	180	*(46)*

e-CL	0.1	0.64	25	stoichiometry 1.5	stoichiometry 1.5	180	*(46)*
Nafion	0.1	0.49	25	stoichiometry 1.5	stoichiometry 1.5	180	*(46)*
SDT-Nafion	0.1	1.14	5	1,500	500	150	This work
Nafion	0.1	0.88	5	1,500	500	150	This work
SPILBCP-Nafion	0.07	0.76	5	2,500	800	150	*(47)*
SPILBCP	0.07	0.54	5	2,500	800	150	*(47)*
Nafion	0.07	0.60	5	2,500	800	150	*(47)*
SAC-Nafion	0.07	0.78	5	1,500	500	150	This work
SG-Nafion	0.07	0.55	5	1,500	500	150	This work
SPS-Nafion	0.07	0.67	5	1,500	500	150	This work
SLZ-Nafion	0.07	0.83	5	1,500	500	150	This work
SBB-Nafion	0.07	0.99	5	1,500	500	150	This work
SBT-Nafion	0.07	0.96	5	1,500	500	150	This work
SDT-Nafion	0.07	1.08	5	1,500	500	150	This work
Nafion	0.07	0.66	5	1,500	500	150	This work

SSC-1.80	0.05	0.50	29.2	Stoichiometry 2.5	stoichiometry 1.4	100	*(48)*
SSC-1.43	0.05	0.65	29.2	Stoichiometry 2.5	stoichiometry 1.4	100	*(48)*
SSC-1.02	0.05	0.62	29.2	Stoichiometry 2.5	stoichiometry 1.4	100	*(48)*
LSC-0.99	0.05	0.55	29.2	Stoichiometry 2.5	stoichiometry 1.4	100	*(48)*

Note: The most data of peak power densities of references are not given, but excavated from the H_2-air fuel cell polarization curves.

Table S4. Fitting parameters of Nyquist plots.

	Pt/C@Nafion H_2-O_2	**Pt/C@SDT-Nafion H_2-O_2**	**Pt/C@Nafion H_2-air**	**Pt/C@SDT-Nafion H_2-air**
R_{ohm}	8.20×10^{-3}	9.02×10^{-3}	1.00×10^{-2}	9.09×10^{-3}
C_1	43.9	4.32×10^{-2}	1.00×10^{-3}	18.3
Z_{ct}	1.67×10^{-2}	1.64×10^{-1}	1.00×10^{-1}	1.00×10^{-2}
CPE1	1.64×10^{-2}	5.22×10^{-2}	8.24×10^{-2}	3.88×10^{-2}
Z_{mass}	2.74×10^{-2}	6.79×10^{-3}	1.00×10^{-2}	2.62×10^{-2}
CPE2	-	-	2.43×10^{-2}	7.27×10^{-2}
Z'_{mass}	-	-	4.50×10^{-2}	5.41×10^{-3}

Table S5. The parameters and results of gas permeation.

RH	Sample	Flux (mL min^{-1})	Permeance (GPU)	Thickness (μm)	Permeability (Barrer)
0%	Nafion	3.28×10^{-4}	0.66	32.2	21
	SLZ/Nafion 2:8	4.67×10^{-4}	0.94	80.0	75
	SDT/Nafion 1:9	1.06×10^{-3}	2.12	40.2	85
	SDT/Nafion 2:8	1.42×10^{-3}	2.84	40.5	115
	SDT/Nafion 3:7	1.77×10^{-3}	3.55	52.3	186
	SBT/Nafion 2:8	3.40×10^{-3}	6.82	42.0	286
	SBB/Nafion 2:8	2.46×10^{-3}	4.93	60.2	297
	SG/Nafion 2:8	1.01×10^{-3}	2.02	39.6	80
100%	Nafion	0	0	46.6	0
	SDT/Nafion 2:8	3.84×10^{-4}	0.77	46.9	36

Table S6. Parameters for simulation.

	Experimental group Pt/C@SDT-Nafion	**Control group** Pt/C@Nafion
Particle number of Pt/C	6	6
Molecule number of Nafion	23	47
Molecule number of SDT-COF	5	0
I/C ratio	0.8	0.8
Molecule number of H_2O	500	500
Molecule number of O_2	600	600
Temperature (K)	353	353
Pressure (kPa abs)	150	150
Density (g cm^{-3})	0.37	0.50

References and Notes

1. L. Chong, J. Wen, J. Kubal, F. G. Sen, J. Zou, J. Greeley, M. Chan, H. Barkholtz, W. Ding, D.-J. Liu, Ultralow-loading platinum-cobalt fuel cell catalysts derived from imidazolate frameworks. *Science* **362**, 1276–1281 (2018).

2. X. X. Wang, M. T. Swihart, G. Wu, Achievements, challenges and perspectives on cathode catalysts in proton exchange membrane fuel cells for transportation. *Nat. Catal.* **2**, 578–589 (2019).

3. C. Y. Ahn, J. E. Park, S. Kim, O.-H. Kim, W. Hwang, M. Her, S. Y. Kang, S. Park, O. J. Kwon, H. S. Park, Y.-H. Cho, Y.-E. Sung, Differences in the electrochemical performance of Pt-based catalysts used for polymer electrolyte membrane fuel cells in liquid half- and full-cells. *Chem. Rev.* **121**, 15075–15140 (2021).

4. X. Tian, X. Zhao, Y.-Q. Su, L. Wang, H. Wang, D. Dang, B. Chi, H. Liu, E. J. M. Hensen, X. W. D. Lou, B. Y. Xia, Engineering bunched Pt-Ni alloy nanocages for efficient oxygen reduction in practical fuel cells. *Science* **366**, 850–856 (2019).

5. A. Kusoglu, A. Z. Weber, New insights into perfluorinated sulfonic-acid ionomers. *Chem. Rev.* **117**, 987–1104 (2017).

6. S. H. Kwon, H. Kang, Y.-J. Sohn, J. Lee, S. Shim, S. G. Lee, Molecular dynamics simulation study on the effect of perfluorosulfonic acid side chains on oxygen permeation in hydrated ionomers of PEMFCs. *Sci. Rep.* **11**, 8702 (2021).

7. S. Salari, M. Tam, C. McCague, J. Stumper, M. Bahrami, The ex-situ and in-situ gas diffusivities of polymer electrolyte membrane fuel cell catalyst layer and contribution of primary pores, secondary pores, ionomer and water to the total oxygen diffusion resistance. *J. Power Sources* **449**, 227479 (2020).

8. K. Shinozaki, Y. Morimoto, B. S. Pivovar, S. S. Kocha, Suppression of oxygen reduction reaction activity on Pt-based electrocatalysts from ionomer incorporation. *J. Power Sources* **325**, 745–751 (2016).

9. S. Ott, A. Orfanidi, H. Schmies, B. Anke, H. N. Nong, J. Hübner, U. Gernert, M. Gliech, M. Lerch, P. Strasser, Ionomer distribution control in porous carbon-supported catalyst layers for high-power and low Pt-loaded proton exchange membrane fuel cells. *Nat. Mater.* **19**, 77–85 (2020).

10. Y. C. Park, H. Tokiwa, K. Kakinuma, M. Watanabe, M. Uchida, Effects of carbon supports on Pt distribution, ionomer coverage and cathode performance for polymer electrolyte fuel cells. *J. Power Sources* **315**, 179–191 (2016).

11. V. Yarlagadda, M. K. Carpenter, T. E. Moylan, R. S. Kukreja, R. Koestner, W. Gu, L. Thompson, A. Kongkanand, Boosting fuel cell performance with accessible carbon mesopores. *ACS Energy Lett.* **3**, 618–621 (2018).

12. K. Kodama, K. Motobayashi, A. Shinohara, N. Hasegawa, K. Kudo, R. Jinnouchi, M. Osawa, Y. Morimoto, Effect of the side-chain structure of perfluoro-sulfonic acid ionomers on the oxygen reduction reaction on the surface of Pt. *ACS Catal.* **8**, 694–700 (2018).

13. K. Talukdar, P. Gazdzicki, K. A. Friedrich, Comparative investigation into the performance and durability of long and short side chain ionomers in polymer electrolyte membrane fuel cells. *J. Power Sources* **439**, 227078 (2019).

14. S. So, H. Kang, D. Choi, K. H. Oh, Tunable aggregation of short-side-chain perfluorinated sulfonic acid ionomers for the catalyst layer in polymer electrolyte membrane fuel cells. *Int. J. Hydrogen Energy* **45**, 19891–19899 (2020).

15. H. Ren, Y. Teng, X. Meng, D. Fang, H. Huang, J. Geng, Z. Shao, Ionomer network of catalyst layers for proton exchange membrane fuel cell. *J. Power Sources* **506**, 230186 (2021).

16. S. Wang, X. Li, Z. Wan, Y. Chen, J. Tan, M. Pan, Effect of hydrophobic additive on oxygen transport in catalyst layer of proton exchange membrane fuel cells. *J. Power Sources* **379**, 338–343 (2018).

17. Z. Q. Tian, X. L. Wang, H. M. Zhang, B. L. Yi, S. P. Jiang, Microwave-assisted synthesis of PTFE/C nanocomposite for polymer electrolyte fuel cells. *Electrochem. Commun.* **8**, 1158–1162 (2006).

18. A. Fischer, J. Jindra, H. Wendt, Porosity and catalyst utilization of thin layer cathodes in air operated PEM-fuel cells. *J. Appl. Electrochem.* **28**, 277–282 (1998).

19. K. Gu, E. J. Kim, S. K. Sharma, P. R. Sharma, S. Bliznakov, B. S. Hsiao, M. H. Rafailovich, Mesoporous carbon aerogel with tunable porosity as the catalyst support for enhanced proton-exchange membrane fuel cell performance. *Mater. Today Energy* **19**, 100560 (2021).

20. E. Şengül, S. Erkan, I. Eroglu, N. Bac, Effect of gas diffusion layer characteristics and addition of pore-forming agents on the performance of polymer electrolyte membrane fuel cells. *Chem. Eng. Commun.* **196**, 161–170 (2009).

21. S. Zhao, C. Jiang, J. Fan, S. Hong, P. Mei, R. Yao, Y. Liu, S. Zhang, H. Li, H. Zhang, C. Sun, Z. Guo, P. Shao, Y. Zhu, J. Zhang, L. Guo, Y. Ma, J. Zhang, X. Feng, F. Wang, H. Wu, B. Wang, Hydrophilicity gradient in covalent organic frameworks for membrane distillation. *Nat. Mater.* **20**, 1551–1558 (2021).

22. C. S. Diercks, O. M. Yaghi, The atom, the molecule, and the covalent organic framework. *Science* **355**, eaal1585 (2017).

23. J. Li, S. Zhao, B. Wang, X. Feng, Strategies in constructing covalent organic framework membranes for molecular sieving. *Sci. China Chem.* **65**, 836–839 (2022).

24. Y. Yang, X. He, P. Zhang, Y. H. Andaloussi, H. Zhang, Z. Jiang, Y. Chen, S. Ma, P. Cheng, Z. Zhang, Combined intrinsic and extrinsic proton conduction in robust covalent organic frameworks for hydrogen fuel cell applications. *Angew. Chem. Int. Ed.* **59**, 3678–3684 (2020).

25. L. Cao, H. Wu, Y. Cao, C. Fan, R. Zhao, X. He, P. Yang, B. Shi, X. You, Z. Jiang, Weakly humidity-dependent proton-conducting COF membranes. *Adv. Mater.* **32**, e2005565 (2020).

26. S. Kandambeth, V. Venkatesh, D. B. Shinde, S. Kumari, A. Halder, S. Verma, R. Banerjee, Self-templated chemically stable hollow spherical covalent organic framework. *Nat. Commun.* **6**, 6786 (2015).

27. X. Huang, C. Sun, X. Feng, Crystallinity and stability of covalent organic frameworks. *Sci. China Chem.* **63**, 1367–1390 (2020).

28. H. Liu, J. Chu, Z. Yin, X. Cai, L. Zhuang, H. Deng, Covalent organic frameworks linked by amine bonding for concerted electrochemical reduction of CO_2. *Chem* **4**, 1696–1709 (2018).

29. S. J. Lyle, T. M. Osborn Popp, P. J. Waller, X. Pei, J. A. Reimer, O. M. Yaghi, Multistep solid-state organic synthesis of carbamate-linked covalent organic frameworks. *J. Am. Chem. Soc.* **141**, 11253–11258 (2019).

30. A. Suzuki, U. Sen, T. Hattori, R. Miura, R. Nagumo, H. Tsuboi, N. Hatakeyama, A. Endou, H. Takaba, M. C. Williams, A. Miyamoto, Ionomer content in the catalyst layer of polymer electrolyte membrane fuel cell (PEMFC): Effects on diffusion and performance. *Int. J. Hydrogen Energy* **36**, 2221–2229 (2011).

31. M. R. Lee, H.-Y. Lee, S.-D. Yim, C.-S. Kim, Y.-G. Shul, A. Kucernak, D. Shin, Effects of ionomer carbon ratio and ionomer dispersity on the performance and durability of MEAs. *Fuel Cells (Weinh.)* **18**, 129–136 (2018).

32. The US DOE Fuel Cell Technologies Office (FCTO), Fuel cell technical team roadmap (DOE, 2017); https://www.energy.gov/sites/default/files/2017/11/f46/FCTT_Roadmap_Nov_2017_FINAL.pdf.

33. Y. V. Yakovlev, Y. V. Lobko, M. Vorokhta, J. Nováková, M. Mazur, I. Matolínová, V. Matolín, Ionomer content effect on charge and gas transport in the cathode catalyst layer of proton-exchange membrane fuel cells. *J. Power Sources* **490**, 229531 (2021).

34. S. M. Alia, K. S. Reeves, J. S. Baxter, D. A. Cullen, The impact of ink and spray variables on catalyst layer properties, electrolyzer performance, and electrolyzer durability. *J. Electrochem. Soc.* **167**, 144512 (2020).

35. A. S. Rad, DFT study of hydrogen fluoride and sulfur trioxide interactions on the surface of Pt-decorated graphene. *J. Theor. App. Phys.* **10**, 307–313 (2016).

36. P. Jagadesan, G. Eder, P. L. McGrier, The excited-state intramolecular proton transfer properties of three imine-linked two-dimensional porous organic polymers. *J. Mater. Chem. C Mater. Opt. Electron. Devices* **5**, 5676–5679 (2017).

37. S. Y. Ding, J. Gao, Q. Wang, Y. Zhang, W.-G. Song, C.-Y. Su, W. Wang, Construction of covalent organic framework for catalysis: Pd/COF-LZU1 in Suzuki-Miyaura coupling reaction. *J. Am. Chem. Soc.* **133**, 19816–19822 (2011).

38. T. Navessin, M. Eikerling, Q. Wang, D. Song, Z. Liu, J. Horsfall, K. V. Lovell, S. Holdcroft, Influence of membrane ion exchange capacity on the catalyst layer performance in an operating PEM fuel cell. *J. Electrochem. Soc.* **152**, A796 (2005).

39. B. Fang, N. K. Chaudhari, M.-S. Kim, J. H. Kim, J.-S. Yu, Homogeneous deposition of platinum nanoparticles on carbon black for proton exchange membrane fuel cell. *J. Am. Chem. Soc.* **131**, 15330–15338 (2009).

40. The U.S. Department of Energy (DOE) Fuel Cell Technologies Office (FCTO), Fuel cells multi-year research development and demonstration plan (2016); https://www.energy.gov/sites/prod/files/2016/06/f32/fcto_myrdd_fuel_cells_0.pdf.

41. W. Wang, Z. Qu, X. Wang, J. Zhang, A molecular model of PEMFC catalyst layer: Simulation on reactant transport and thermal conduction. *Membranes* **11**, 148 (2021).

42. J. E. Chae, S. J. Yoo, J. Y. Kim, J. H. Jang, S. Y. Lee, K. H. Song, H.-J. Kim, Hydrocarbon-based electrode ionomer for proton exchange membrane fuel cells. *Int. J. Hydrogen Energy* **45**, 32856–32864 (2020).

43. S. Shahgaldi, I. Alaefour, X. Li, Influence of ionomer structures and ratios on performance and degradation of PEM fuel cells. *ECS Trans.* **83**, 71–78 (2018).

44. C. Y. Ahn, J. Ahn, S. Y. Kang, O.-H. Kim, D. W. Lee, J. H. Lee, J. G. Shim, C. H. Lee, Y.-H. Cho, Y.-E. Sung, Enhancement of service life of polymer electrolyte fuel cells through application of nanodispersed ionomer. *Sci. Adv.* **6**, eaaw0870 (2020).

45. R. Jinnouchi, K. Kudo, K. Kodama, N. Kitano, T. Suzuki, S. Minami, K. Shinozaki, N. Hasegawa, A. Shinohara, The role of oxygen-permeable ionomer for polymer electrolyte fuel cells. *Nat. Commun.* **12**, 4956 (2021).

46. S. Choi, J. Jeon, J. Chae, S. Yuk, D.-H. Lee, G. Doo, D. W. Lee, J. Hyun, J. Kwen, S. Q. Choi, H.-T. Kim, Single-step fabrication of a multiscale porous catalyst layer by the emulsion template method for low Pt-Loaded proton exchange membrane fuel cells. *ACS Appl. Energy Mater.* **4**, 4012–4020 (2021).

47. Y. Li, T. Van Cleve, R. Sun, R. Gawas, G. Wang, M. Tang, Y. A. Elabd, J. Snyder, K. C. Neyerlin, Modifying the electrocatalyst-ionomer interface via sulfonated poly(ionic liquid) block copolymers to enable high-performance polymer electrolyte fuel cells. *ACS Energy Lett.* **5**, 1726–1731 (2020).

48. Y. C. Park, K. Kakinuma, H. Uchida, M. Watanabe, M. Uchida, Effects of short-side-chain perfluorosulfonic acid ionomers as binders on the performance of low Pt loading fuel cell cathodes. *J. Power Sources* **275**, 384–391 (2015).